

Design of printed circuit boards using Altium Designer

1.- Software Framework

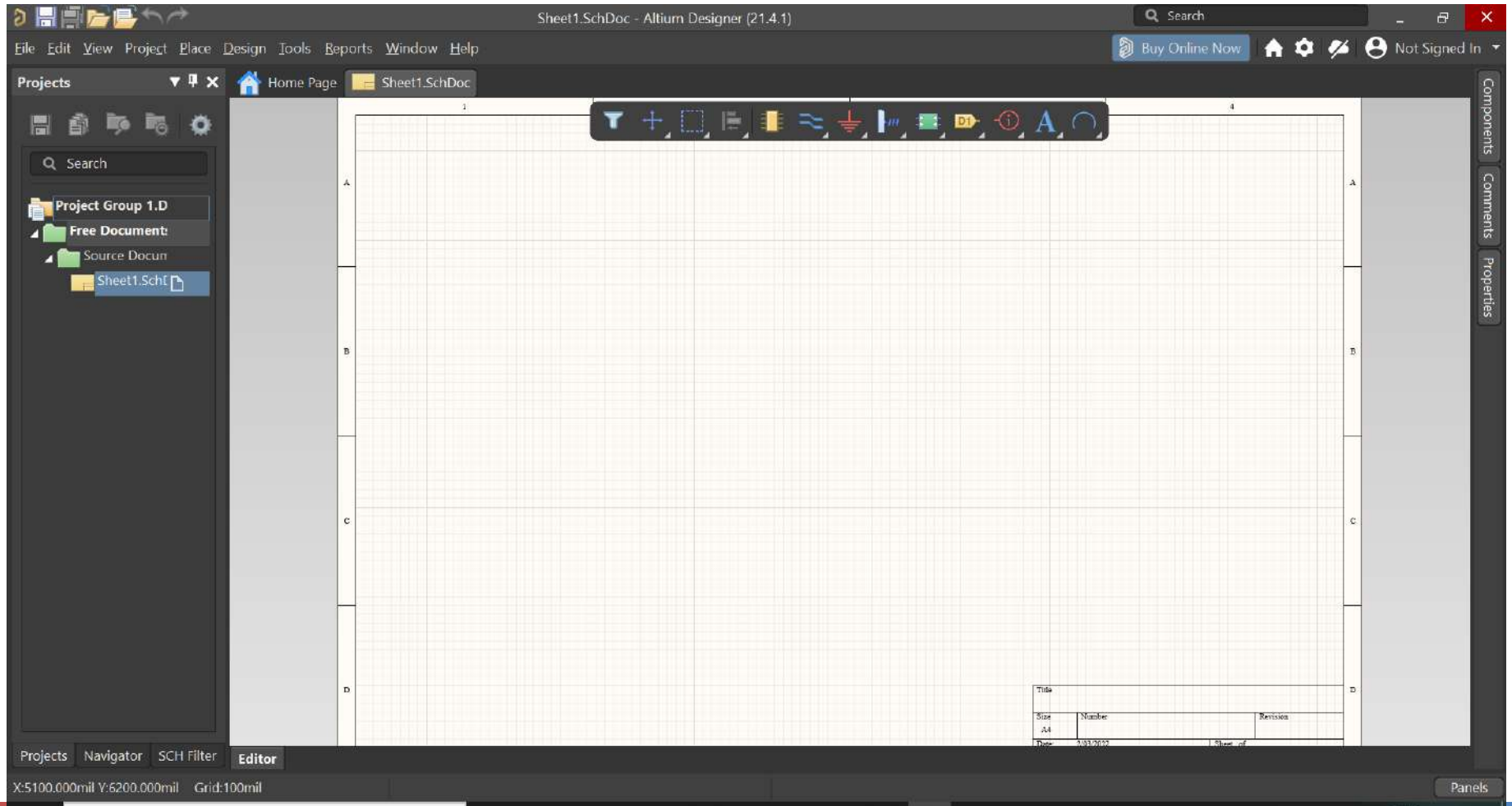
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Dr. Ignacio Bravo Muñoz
Departamento de Electrónica
Univeresidad de Granada

Creating a project

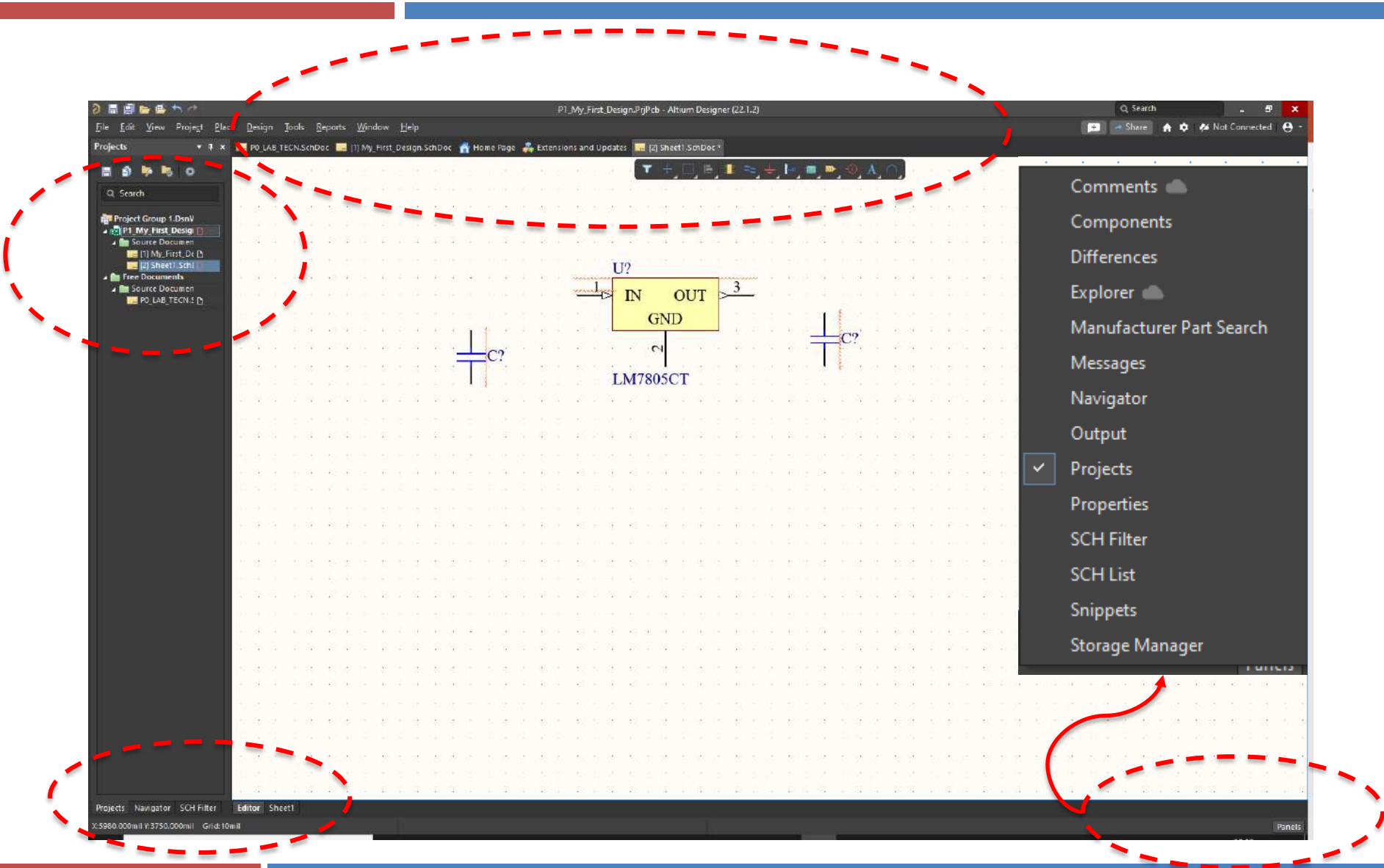
- ❑ A project can be composed by several parts:
 - ❑ Schematics
 - ❑ PCB
 - ❑ Documentation
 - ❑ Gerber files
 - ❑ Etc.
- ❑ Even we could create a Schematics (for instance) and included within a project later
- ❑ We will only address Schematics + PCB in this lab

Schematics

□ Regular view of Schematics tool



Schematics



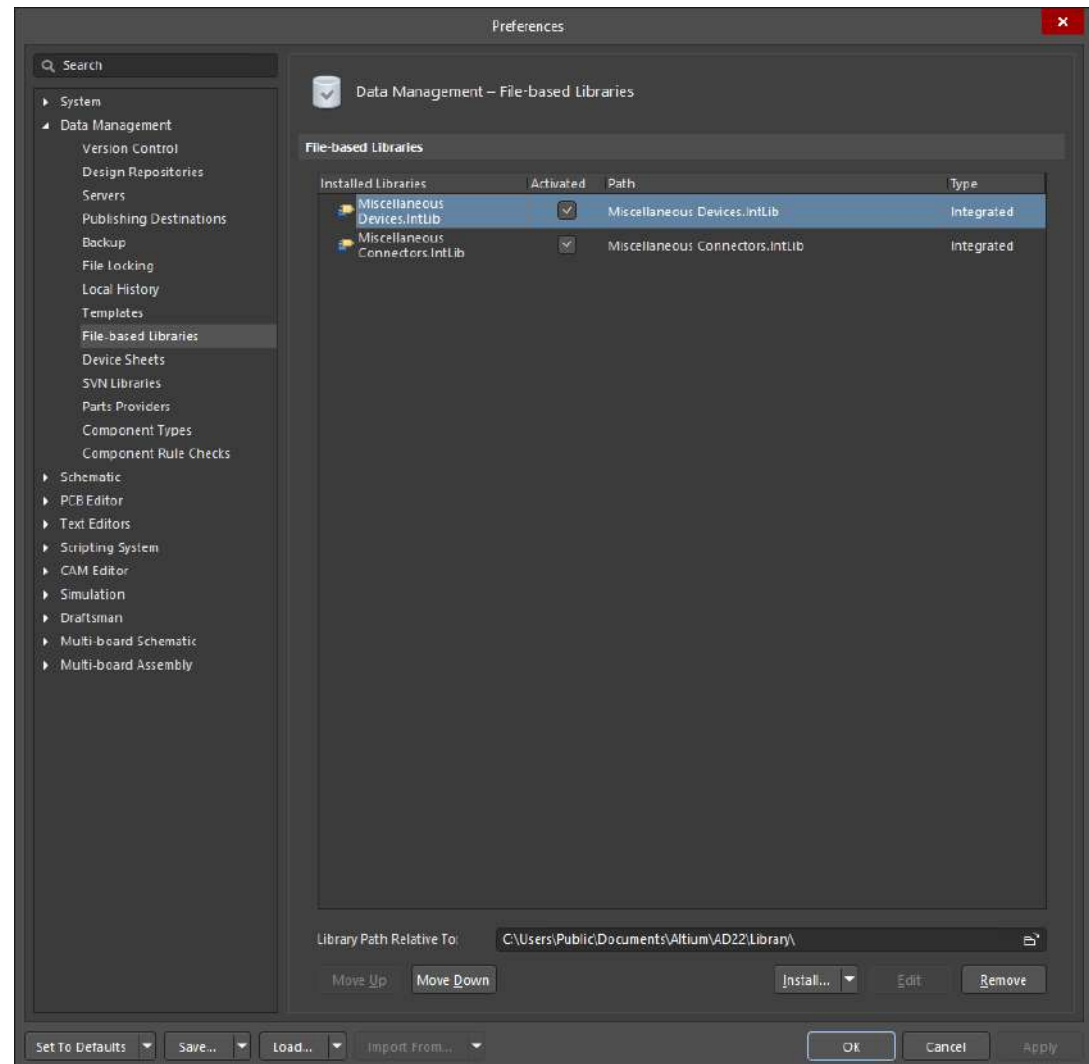
Schematics



- ❑ Schematics quick tool bar.
- ❑ It contains the most common elements to be used within a Schematics.
- ❑ We must distinguish between:
 - ❑ Physical devices (R's, C's, circuits, etc.)
 - ❑ Symbols (GND, Vcc, bus, entry bus, etc.)
 - ❑ Both are included inside components

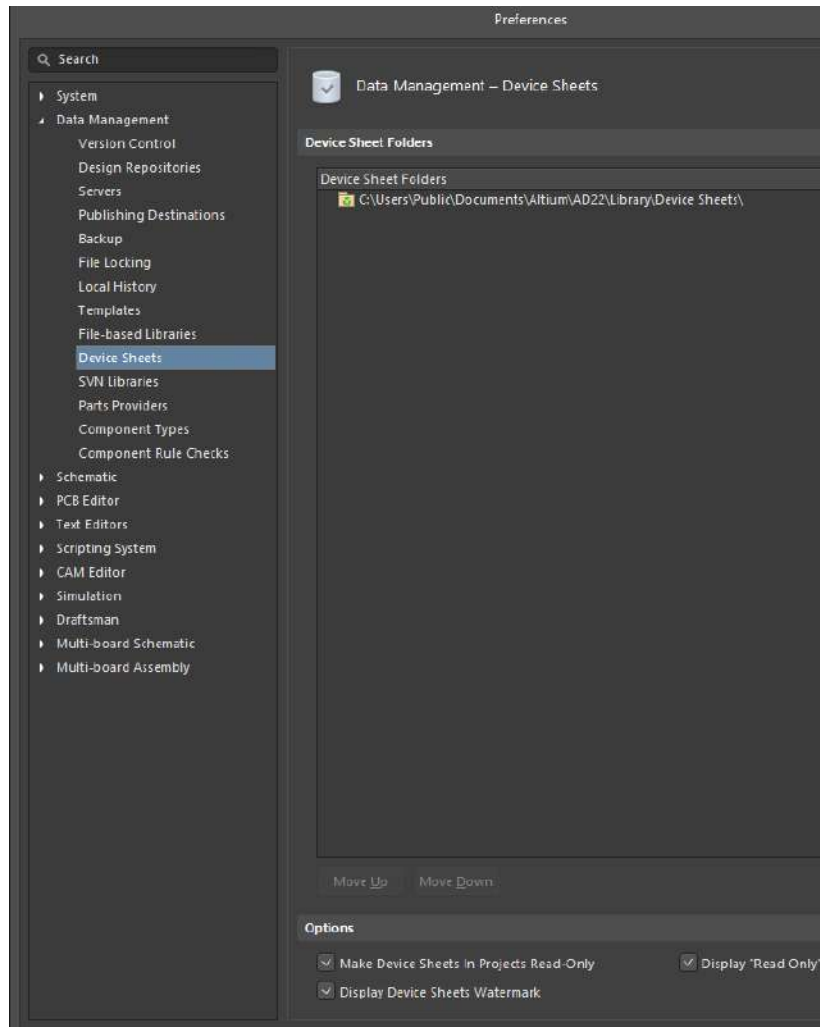
Schematics

- Tools → Preferences
- Data Management → File based libraries
- By default libraries included in panel of components.



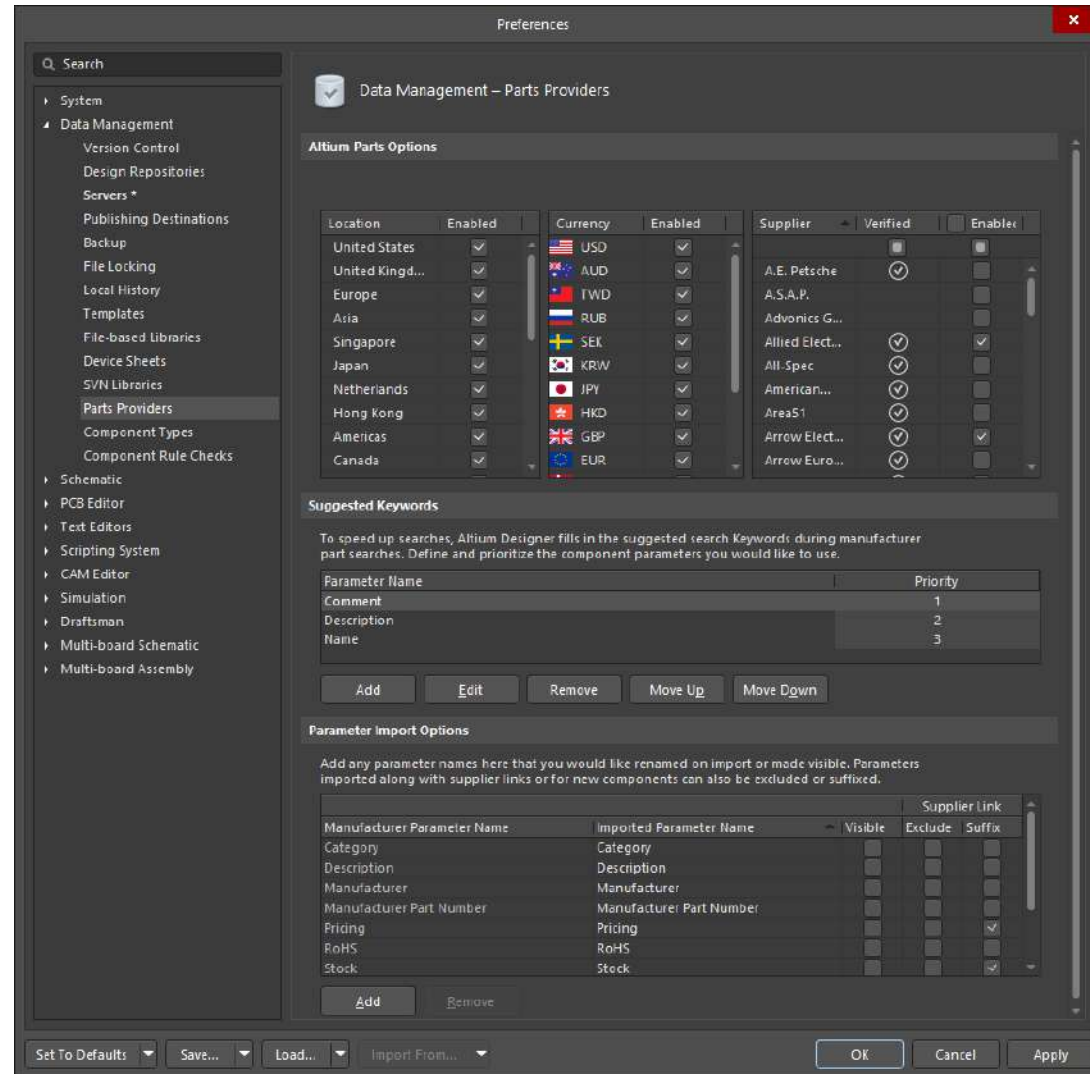
Schematics

- ❑ Data Management
 ➔ Device Sheets
- ❑ Interesting place
 for loading
 datasheets



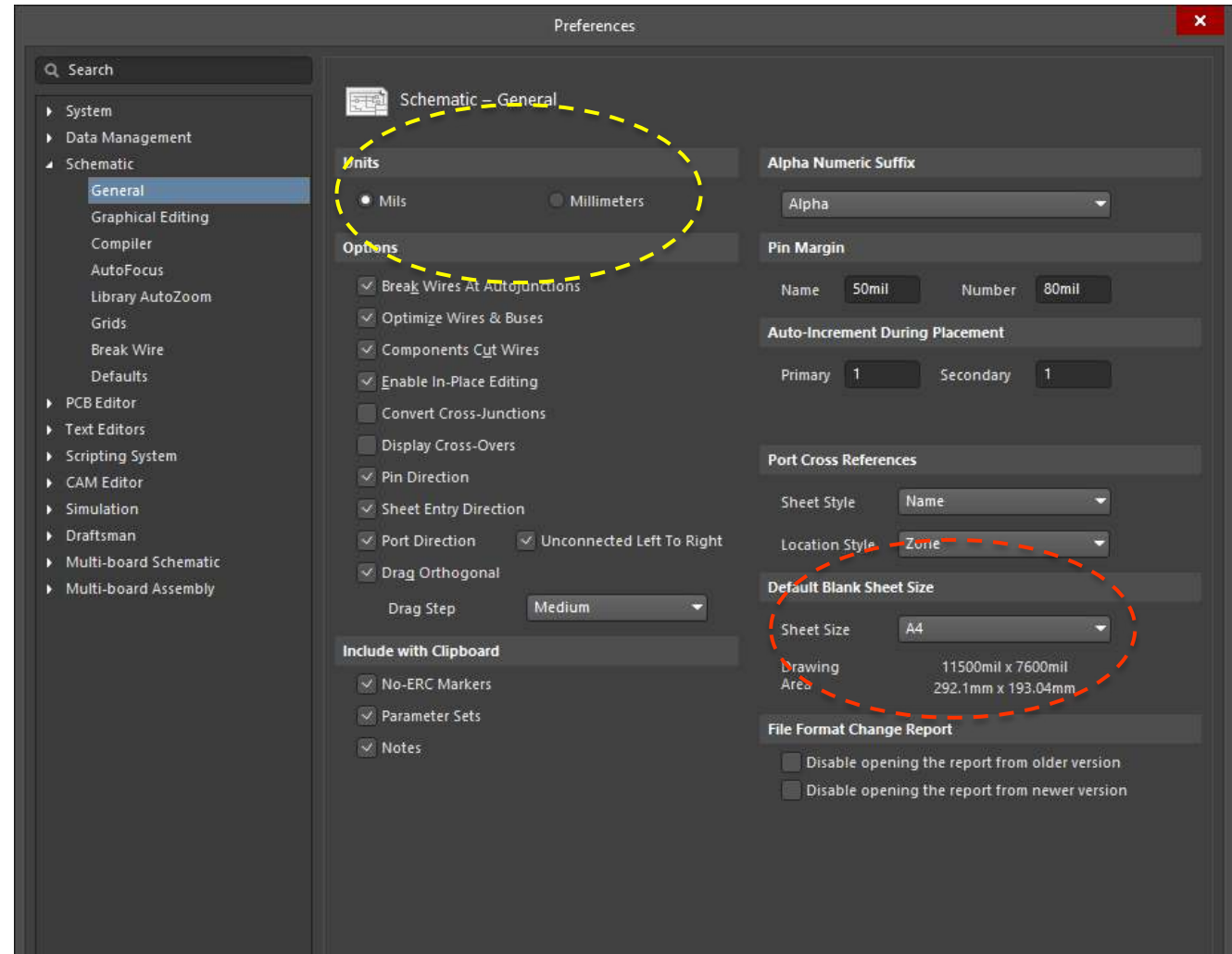
Schematics

- Path Providers:
Device suppliers
and currency.
Relevant for
BOM (Bill of
Materials)



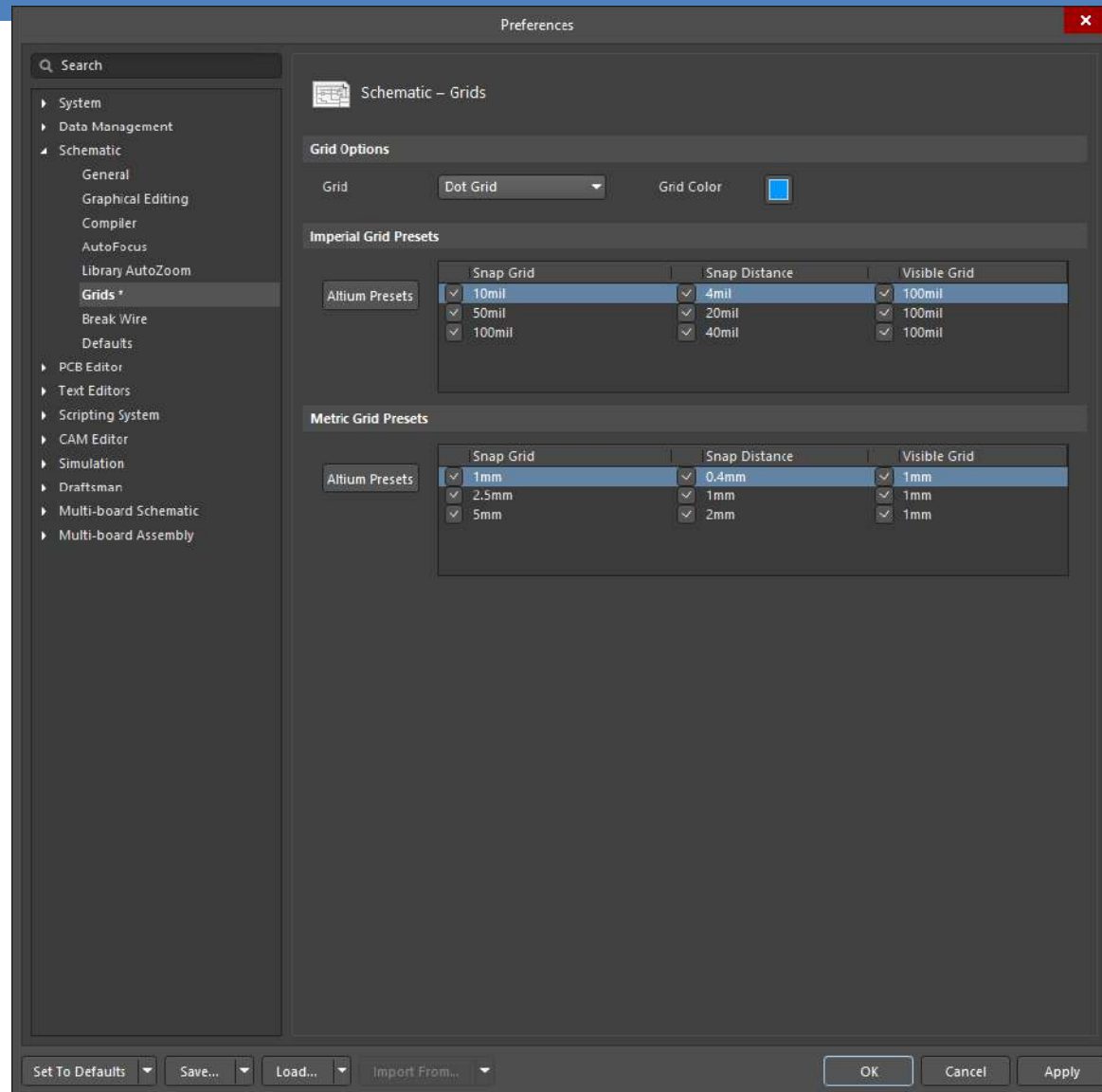
Schematics

- ❑ Schematic → General
- ❑ Units: Mils (inches) or milometers
- ❑ Sheet size



Schematics

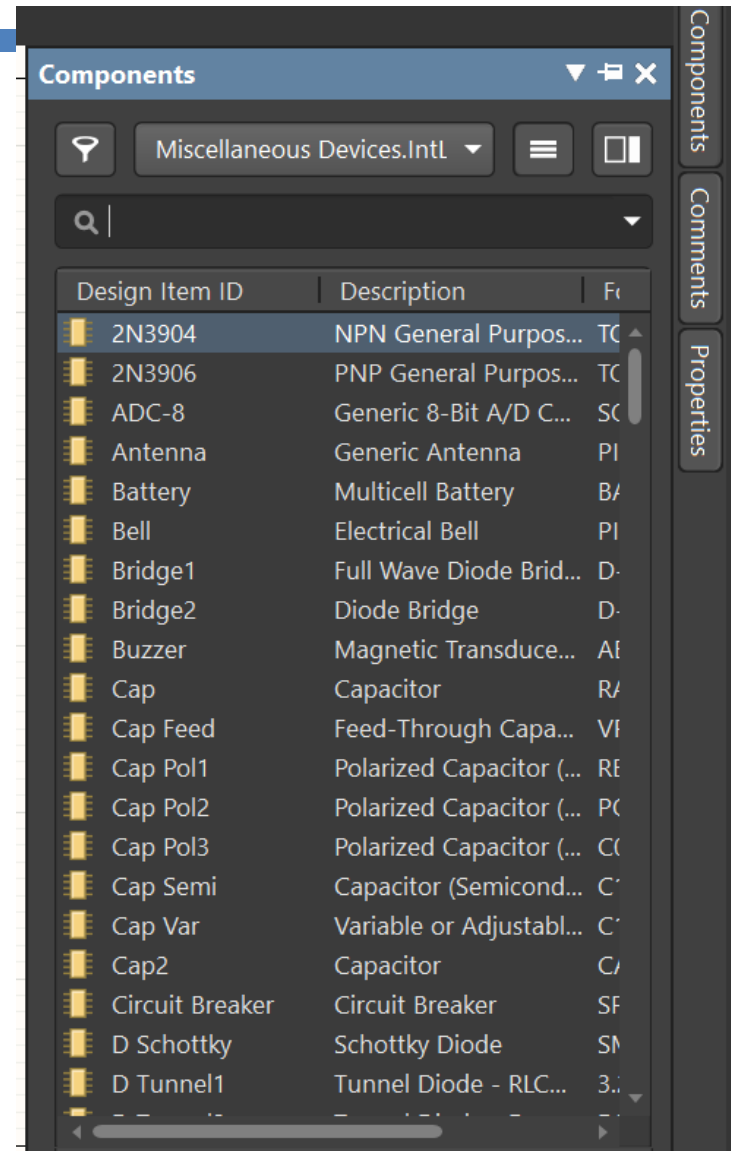
- Grid: Setup grid options.



- ❑ Some tips related to shortcuts:
 - ❑ Zoom in: CTRL + Up Wheel
 - ❑ Zoom out: CTRL + Down Wheel
 - ❑ Right click: Moving the grid / Escape (wiring)
 - ❑ Escape: Finishing mode
 - ❑ Space bar: Rotating

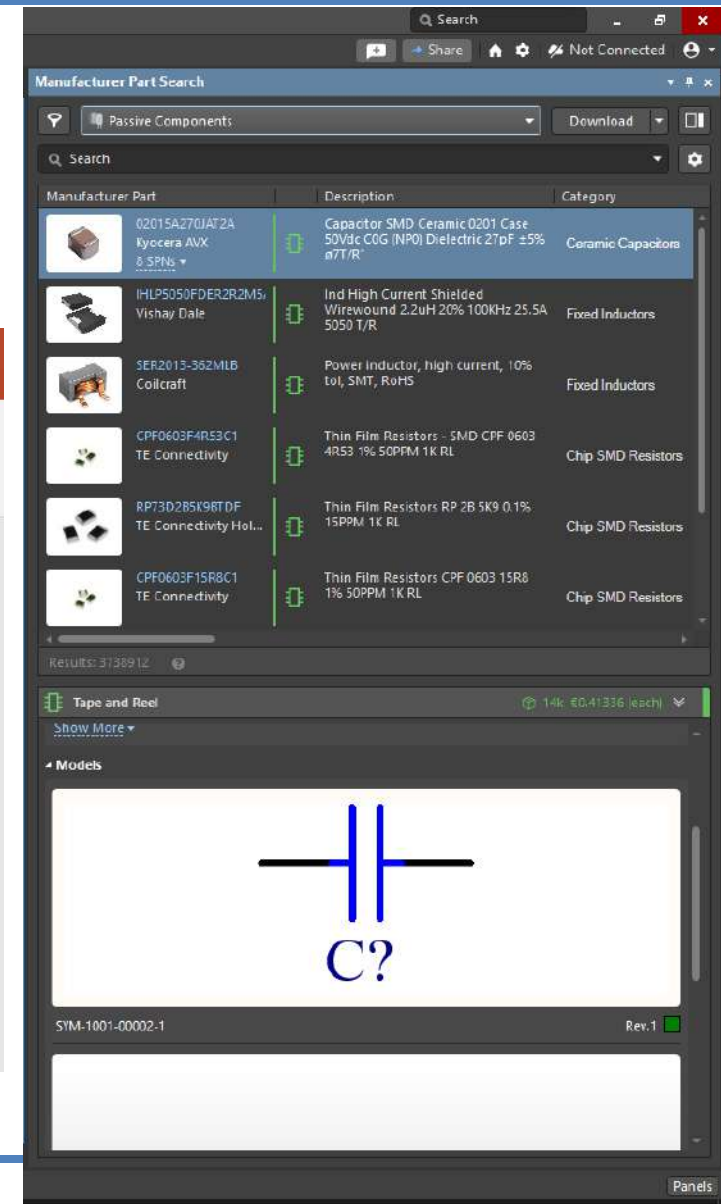
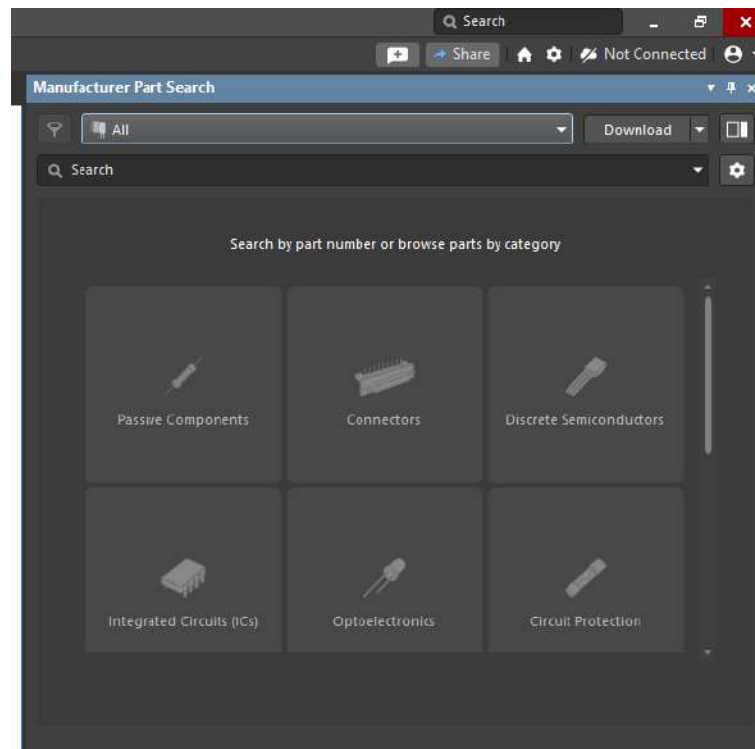
Schematics

□ Components view:



Schematics

❑ Manufacturer Part Search

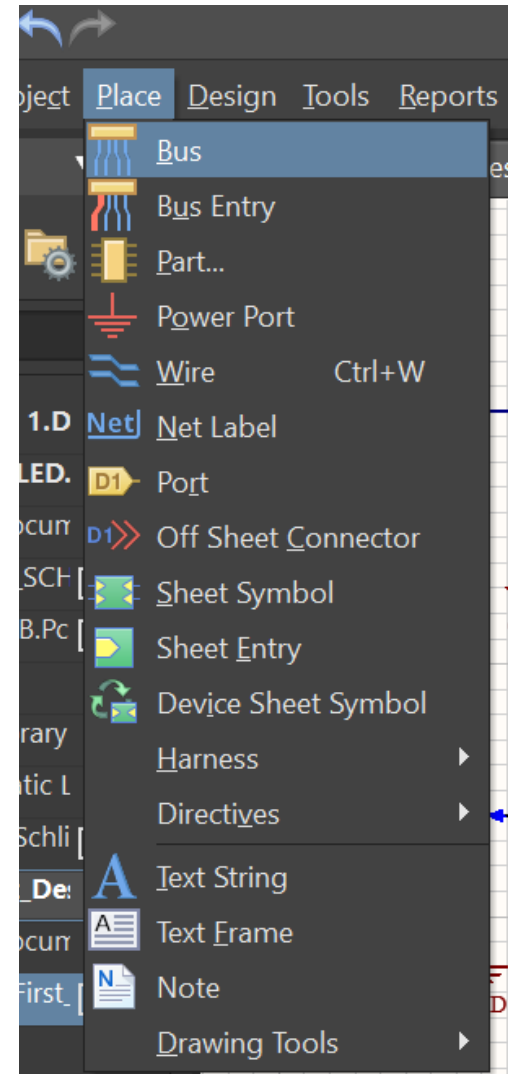


- ☐ What is a Library?
 - ☐ Place where a collection of components are included.
- ☐ May one component belong to different libraries?
 - ☐ Yes, it is. Even we can create our own library.
- ☐ What types of library can be managed?
 - ☐ Schematic Library
 - ☐ PCB Library
 - ☐ Project Library

- ❑ What happen if we need a component and it is not included in our environment?
 - ❑ Option 1: To install the library
 - ❑ Option 2: To find the library in Altium community
 - ❑ Option 3: Create our own device/component

Schematics

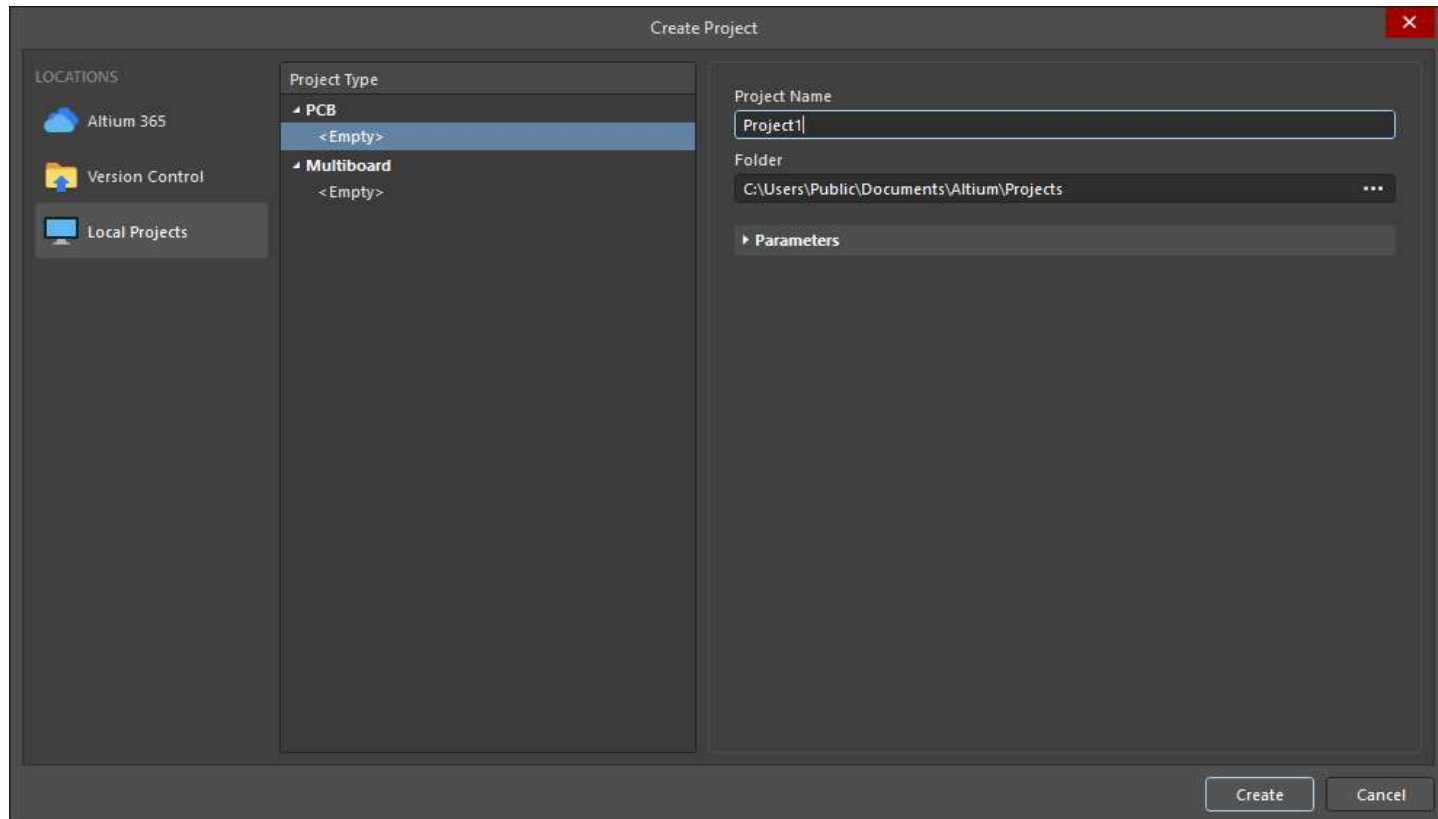
- ❑ Before doing our first design:
- ❑ Difference between wire and bus:
 - ❑ Bus is a set of wires (usually related to digital signals) joined to ease the schematic view and design.
 - ❑ The connexion between the bus and the physical pin is done by a Bus Entry.
 - ❑ Wire is a single line to create an electrical connection between two points.



- Other issues:
 - Net Labels: Beyond clarifying purposes, it works as an electrical connection between several parts instead of drawing a long wire.
 - V_{CC} : take care regarding the power supply. It will come from some battery, connector, etc. “It is not magical”
 - GND: it is used to have only one when we are designing an analog system. If we have digital and analog signals two grounds can be used.

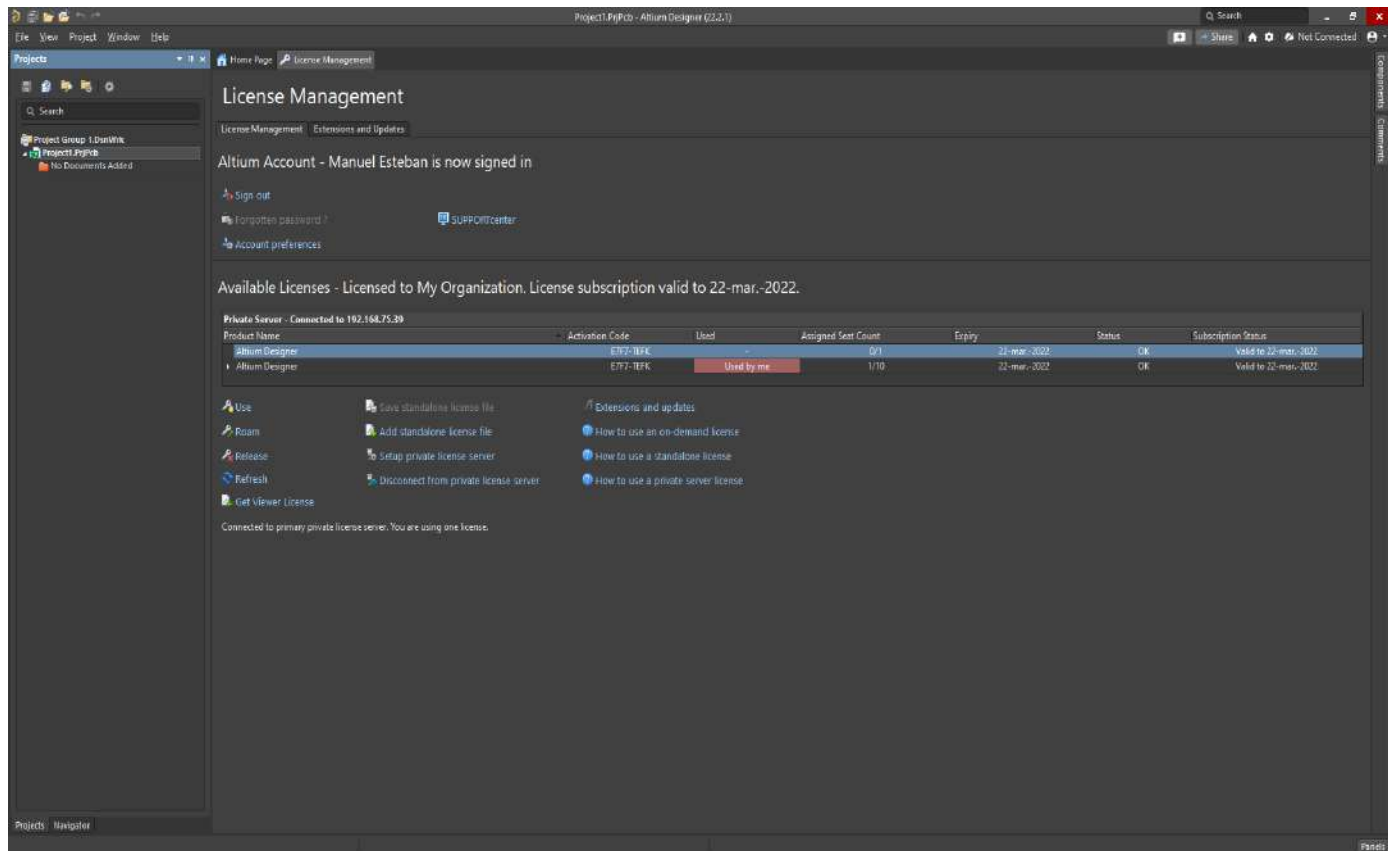
My first design

□ Creating a new project



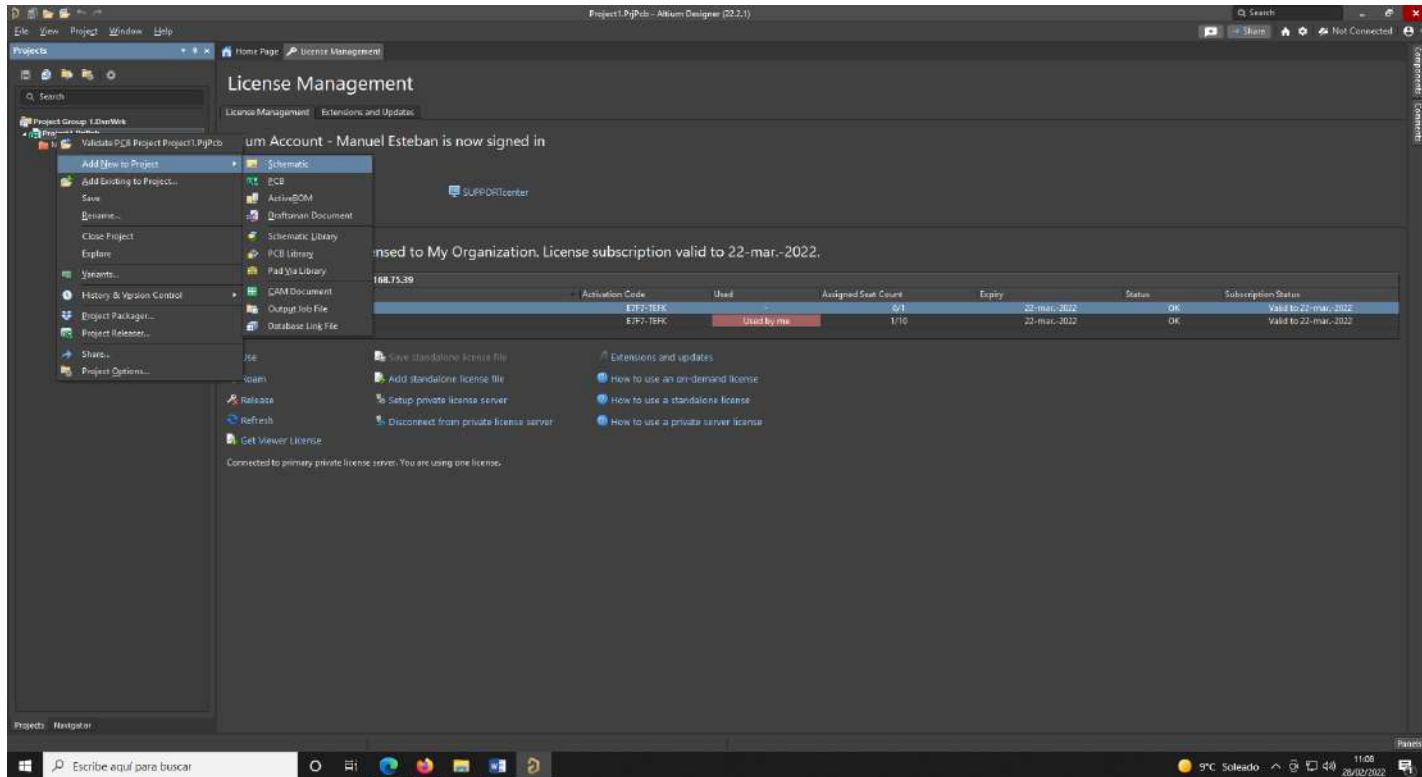
My first design

□ Reviewing the license



My first design

□ Generating an empty schematic sheet

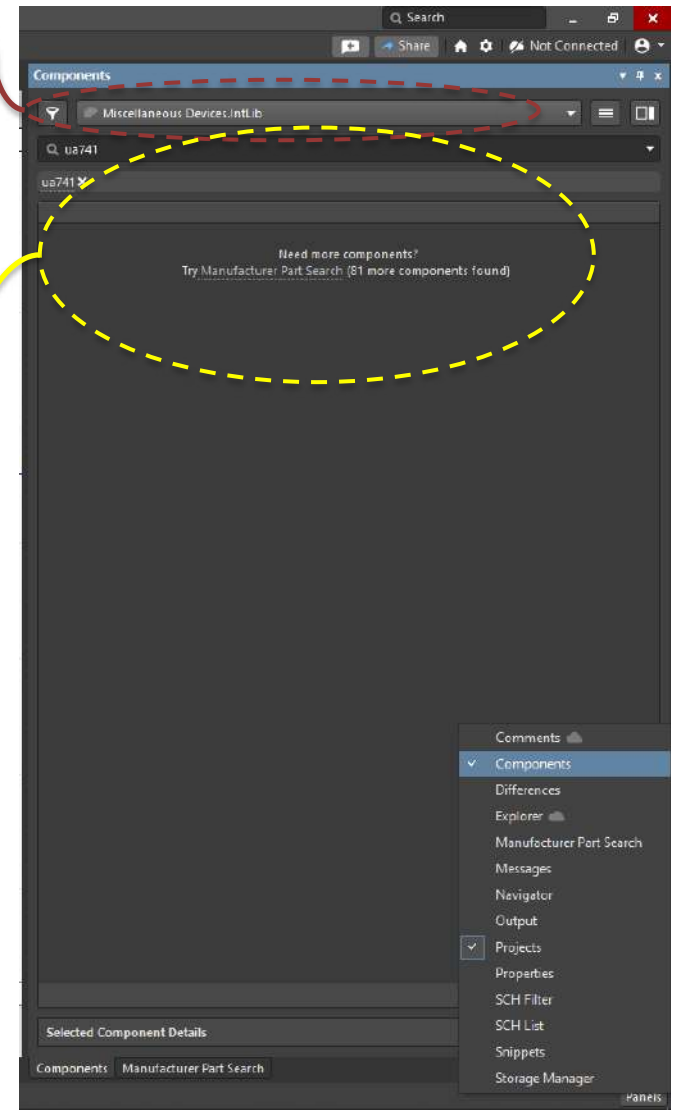


My first design

- Searching and selecting the components (Panels Tab → Components)

Installed libraries in the PC

Unlisted components can be searched through this panel or Manufacturer Part Search Tab



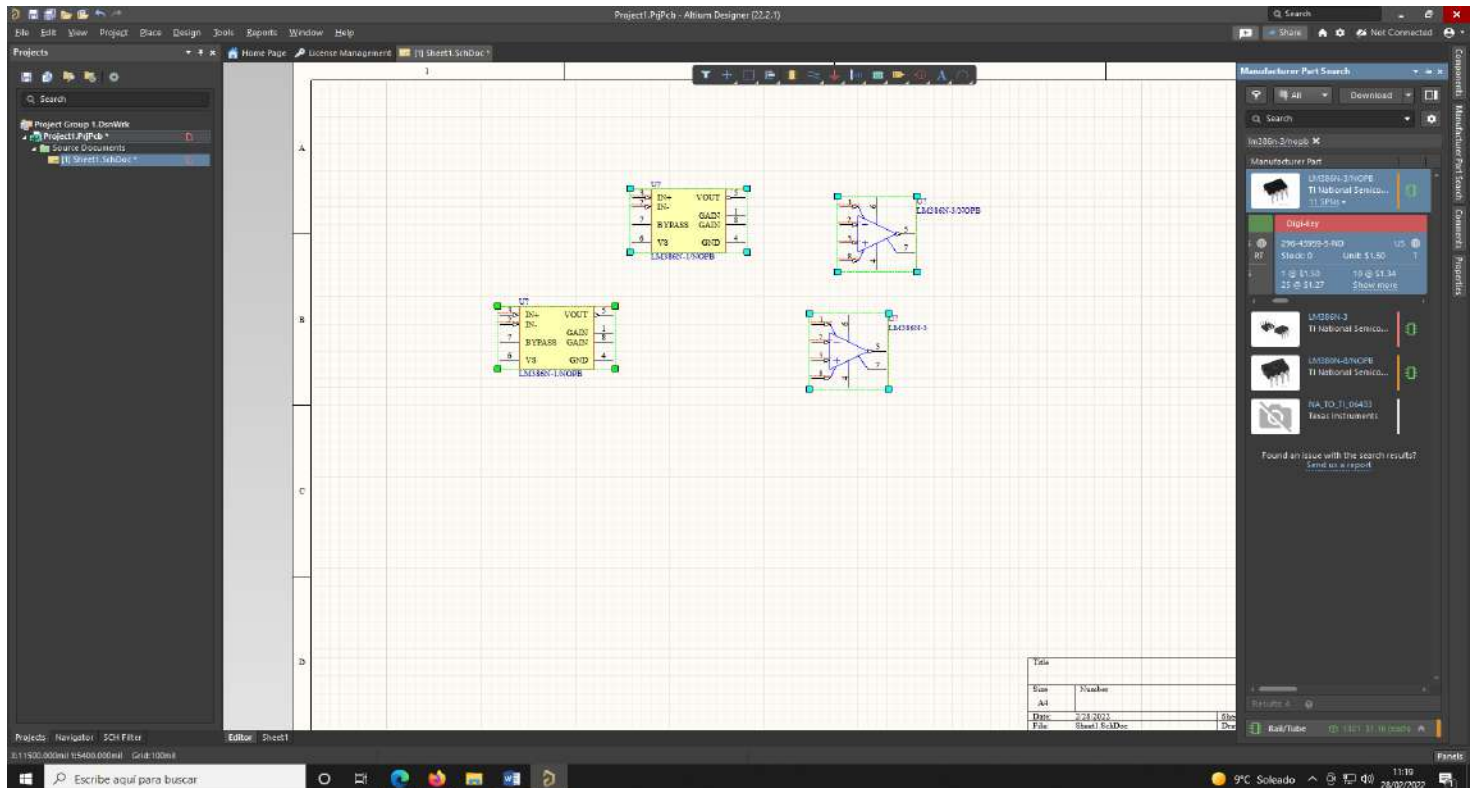
My first design

- ❑ Searching and selecting components not installed in local libraries.
- ❑ Several options can be got. Relevant issues such as availability, quantity or SCH symbol should be considered

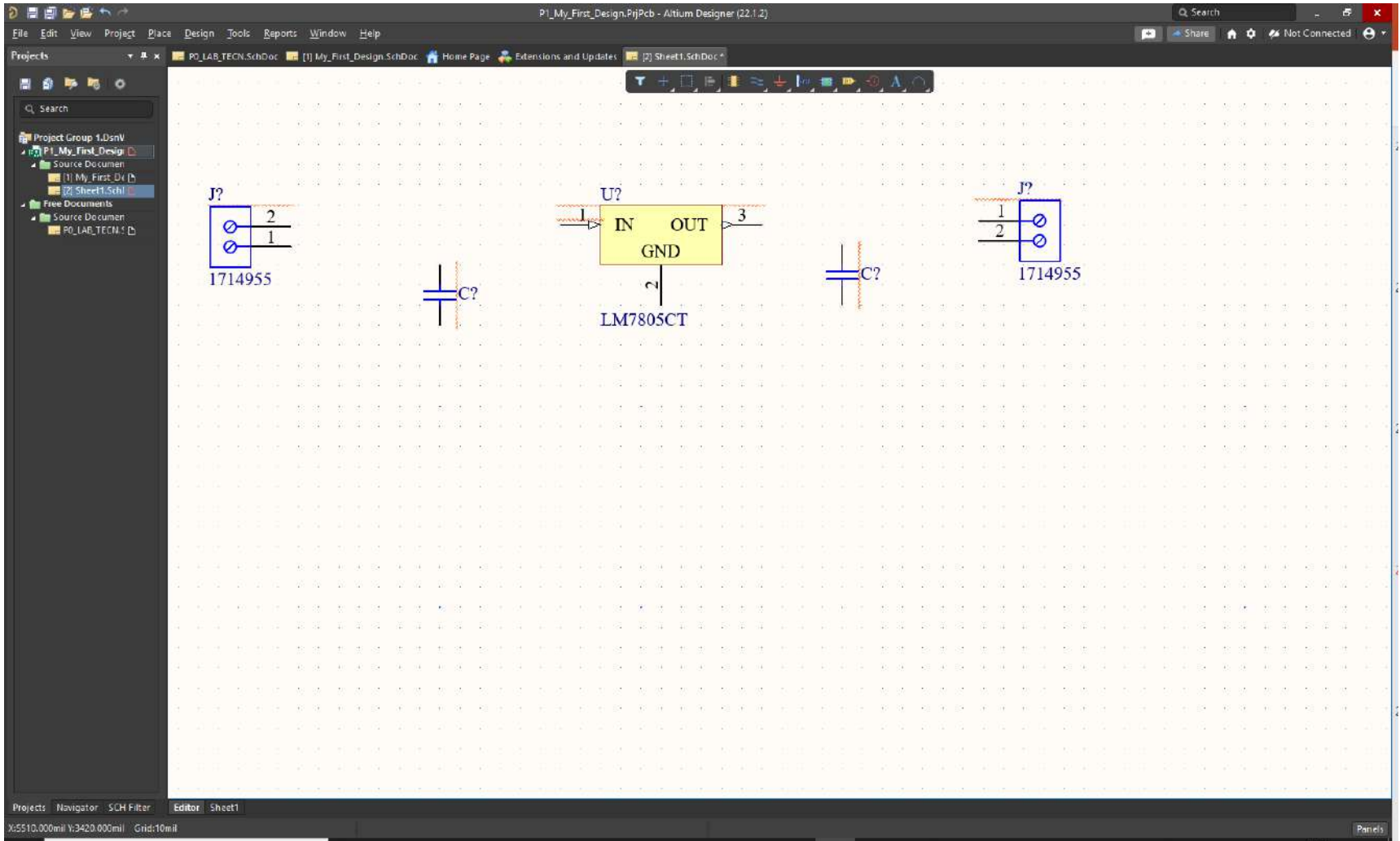


My first design

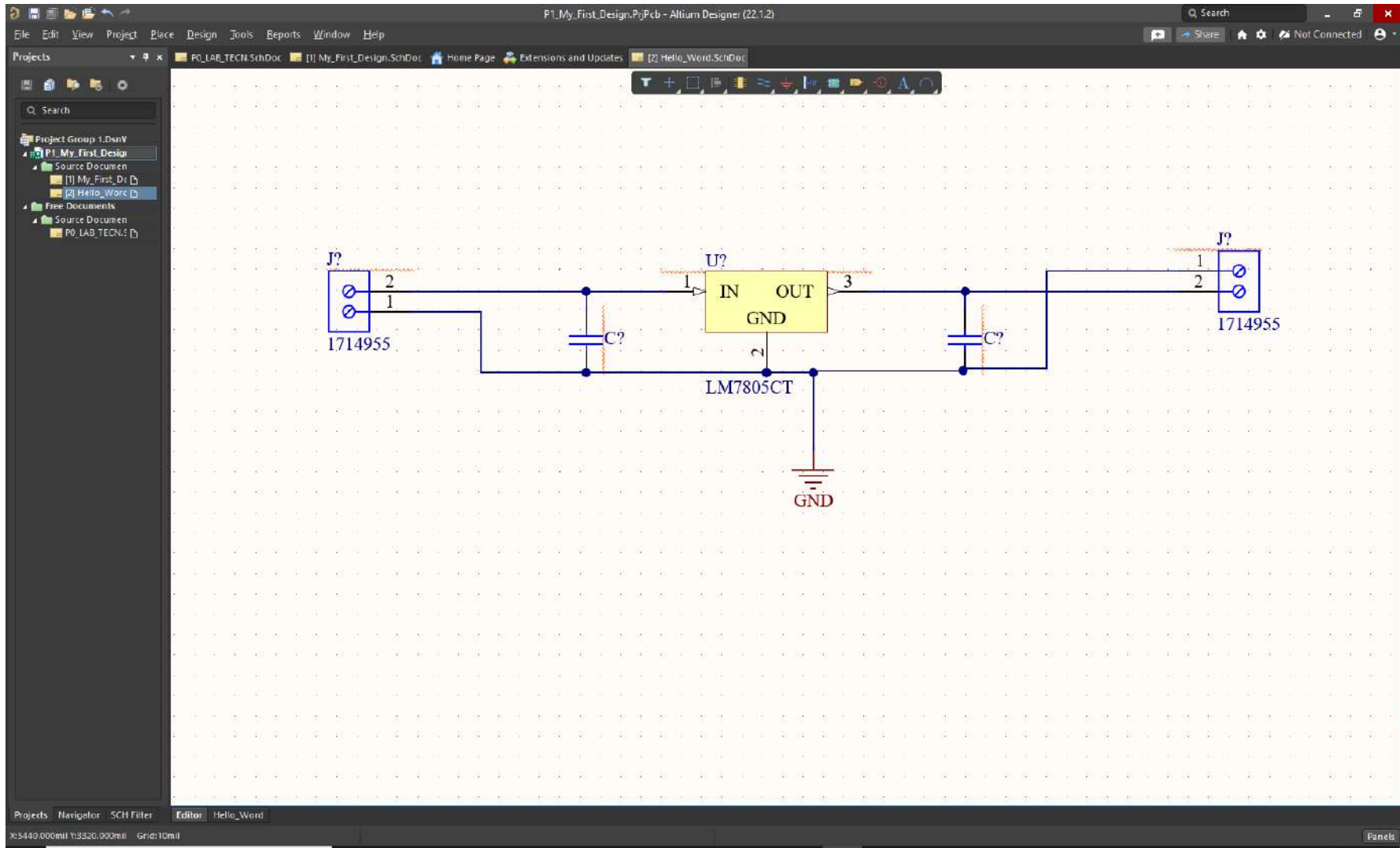
- Right click over the component and Place option must be selected to include in our design



“Hello World”

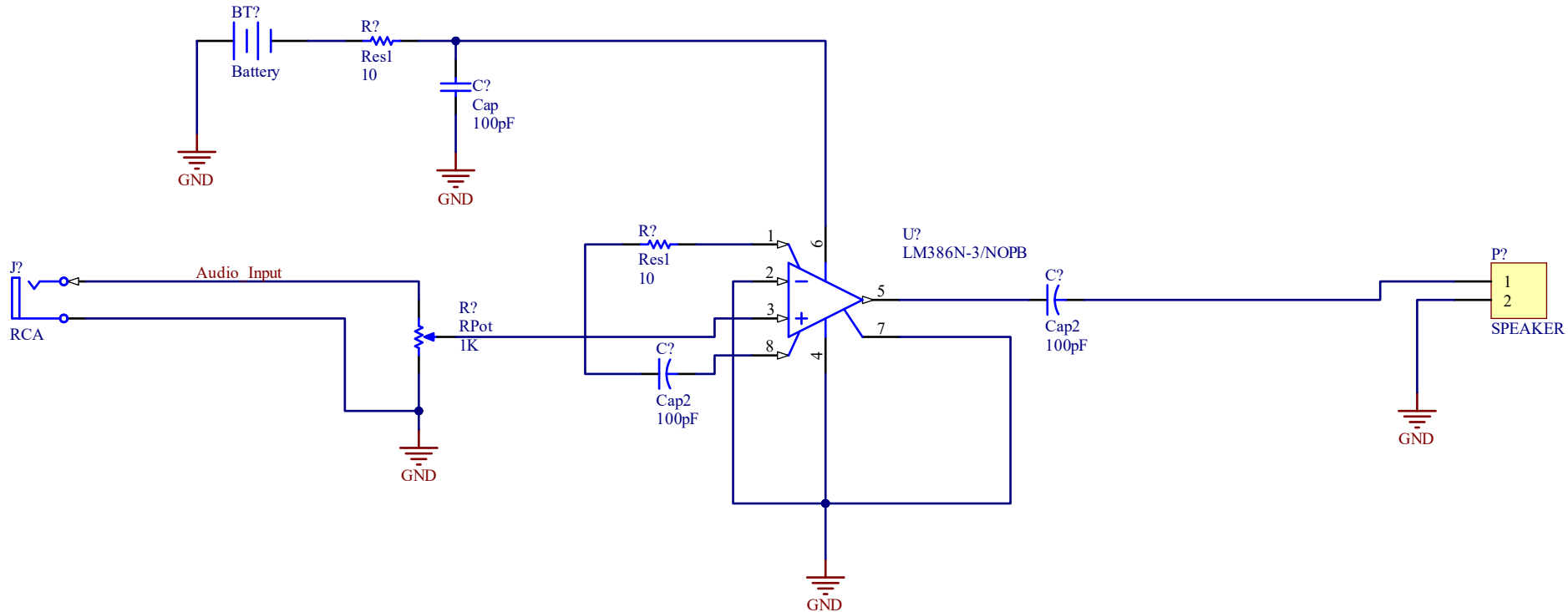


“Hello World”



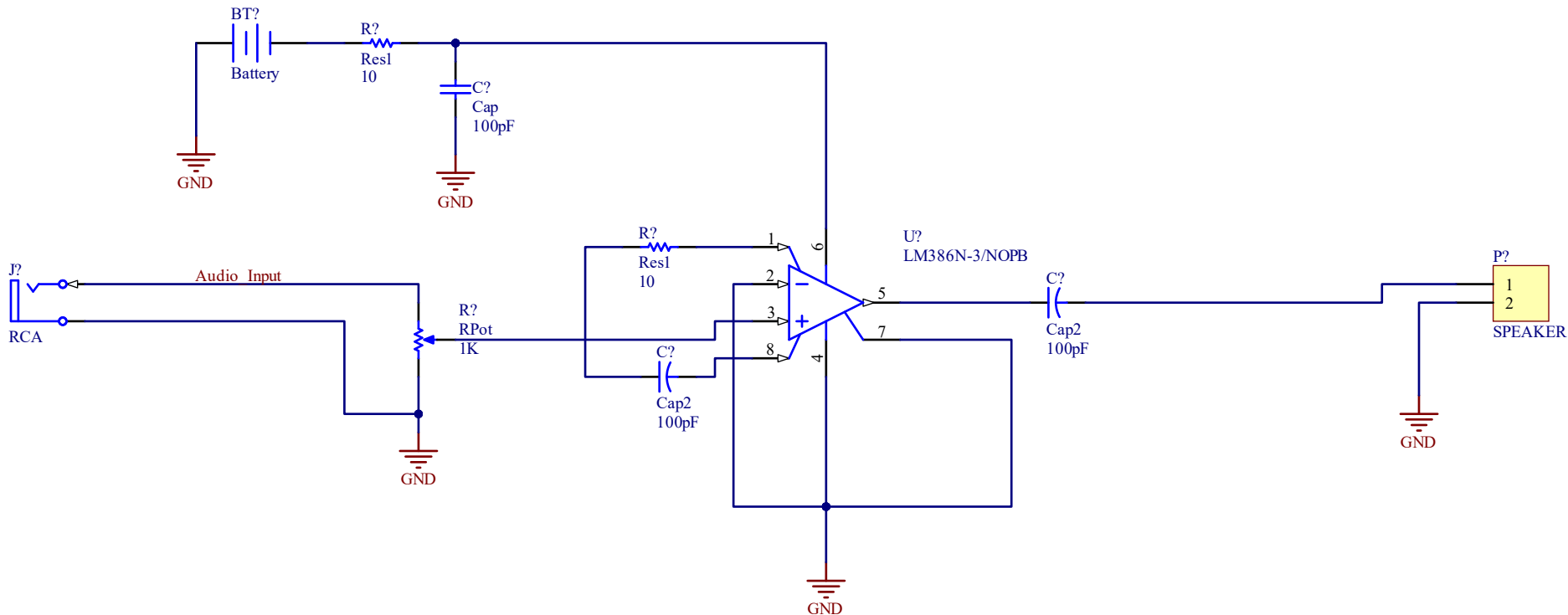
My first design

□ Mi first design: An audio amplifier.



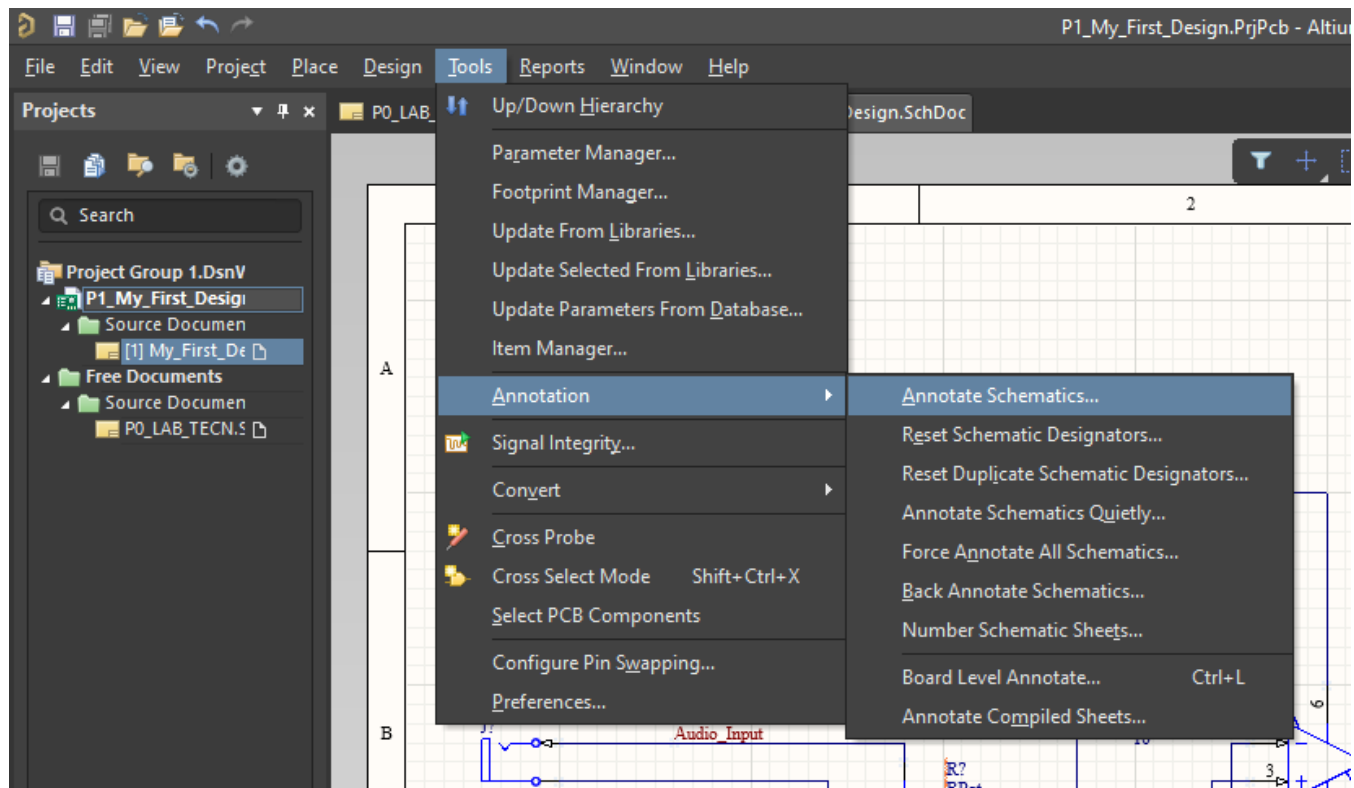
My first design

- With this example power options, wire and net label must be used.

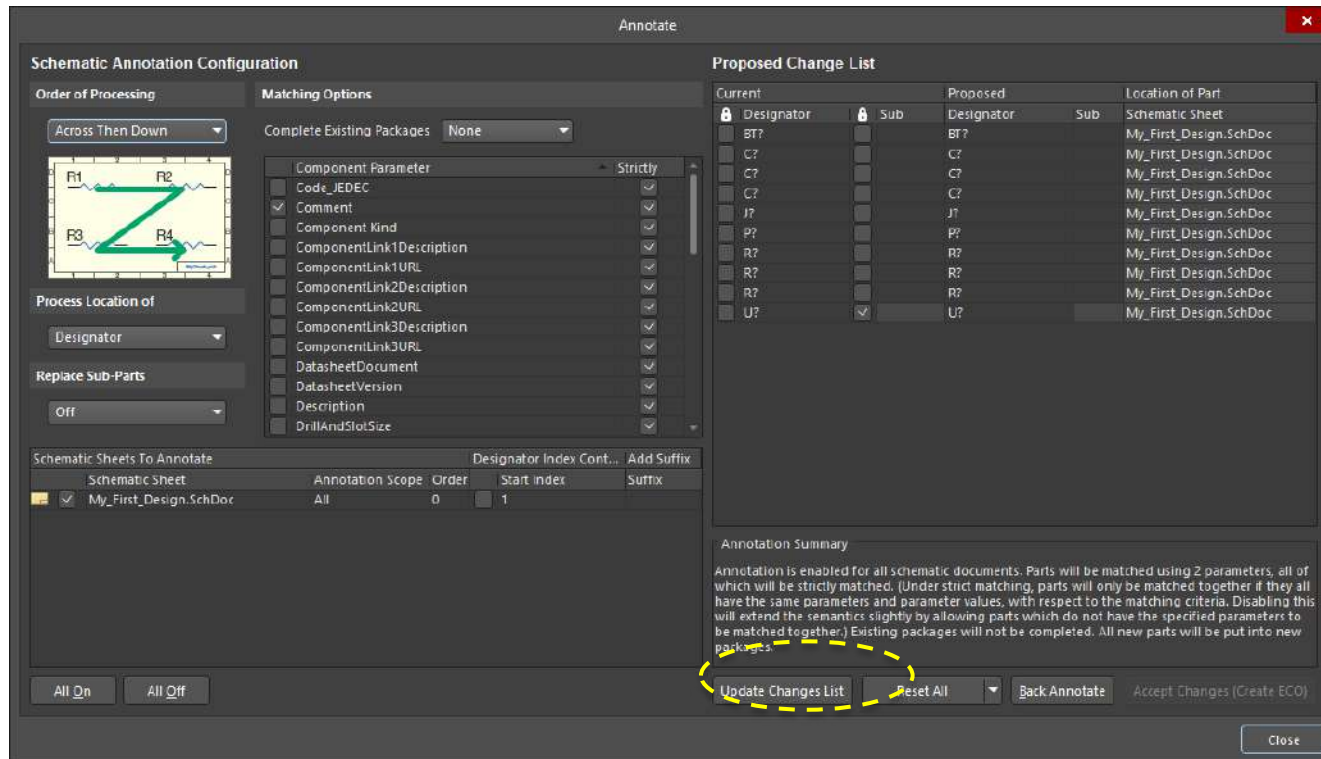


My first design

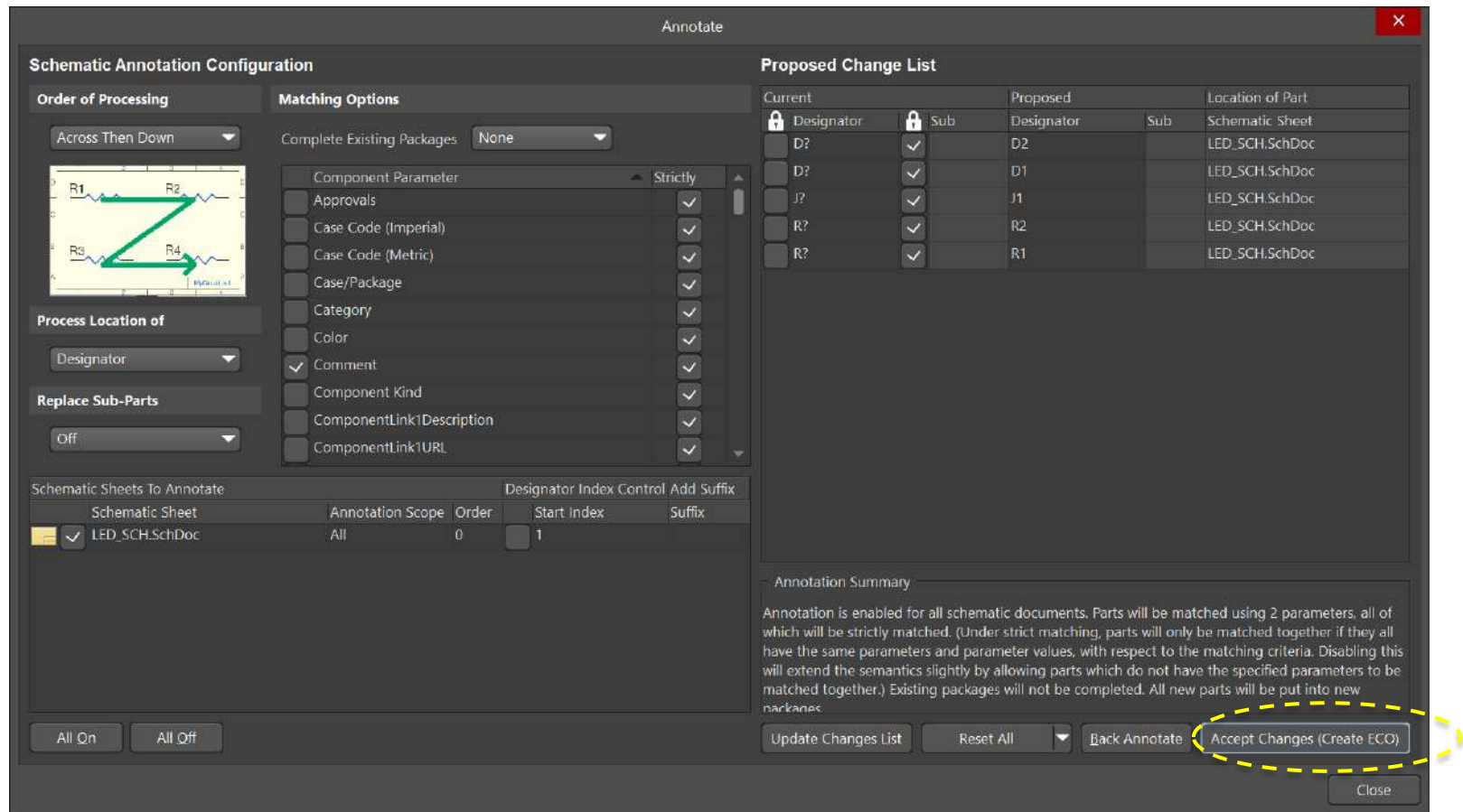
□ Annotation (Tools Menu)



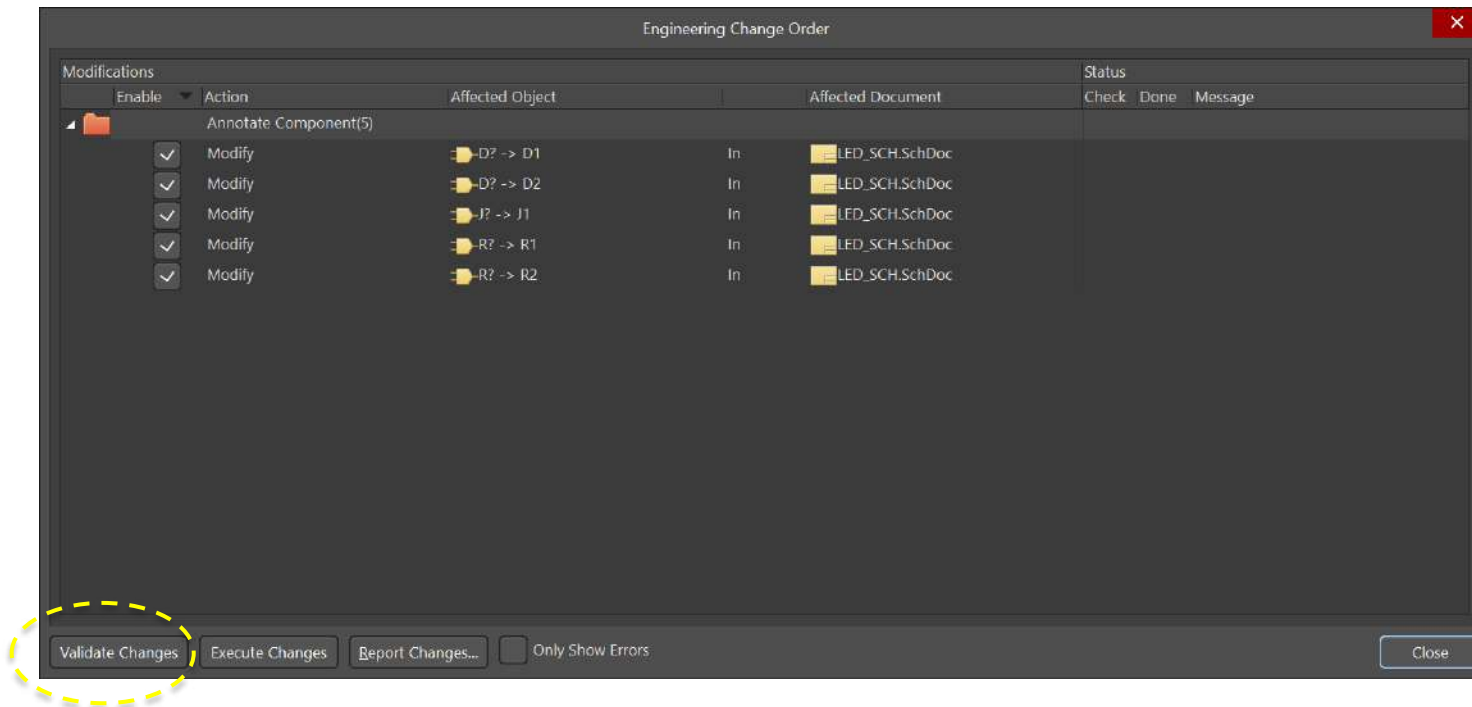
My first design



My first design



My first design



Engineering Change Order

Modifications

Enable	Action	Affected Object		Affected Document	Status
					CheckDoneMessage
Annotate Component(5)					
☒	Modify	D? -> D1	In	LED_SCH.SchDoc	☒
☒	Modify	D? -> D2	In	LED_SCH.SchDoc	☒
☒	Modify	J? -> J1	In	LED_SCH.SchDoc	☒
☒	Modify	R? -> R1	In	LED_SCH.SchDoc	☒
☒	Modify	R? -> R2	In	LED_SCH.SchDoc	☒

Validate Changes

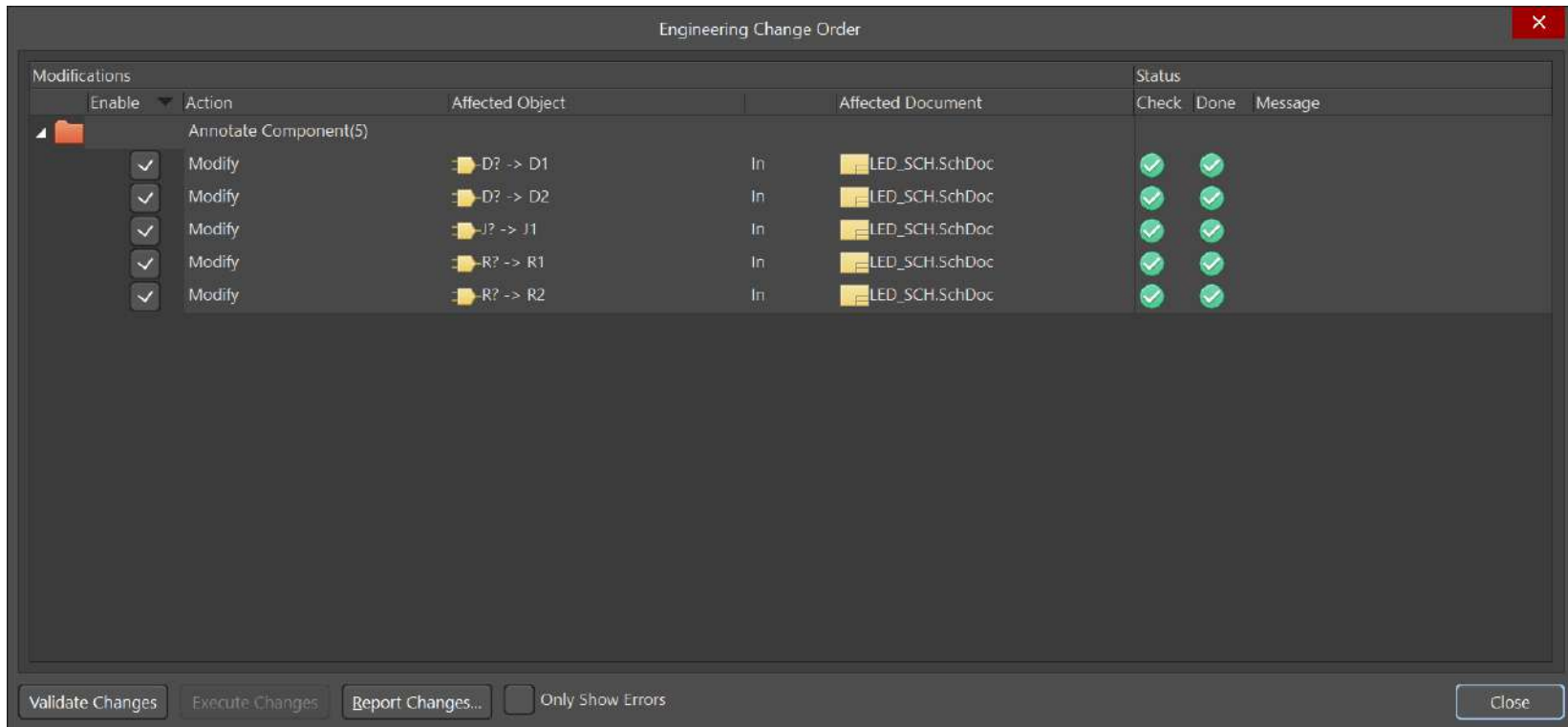
Execute Changes

Report Changes...

☐ Only Show Errors

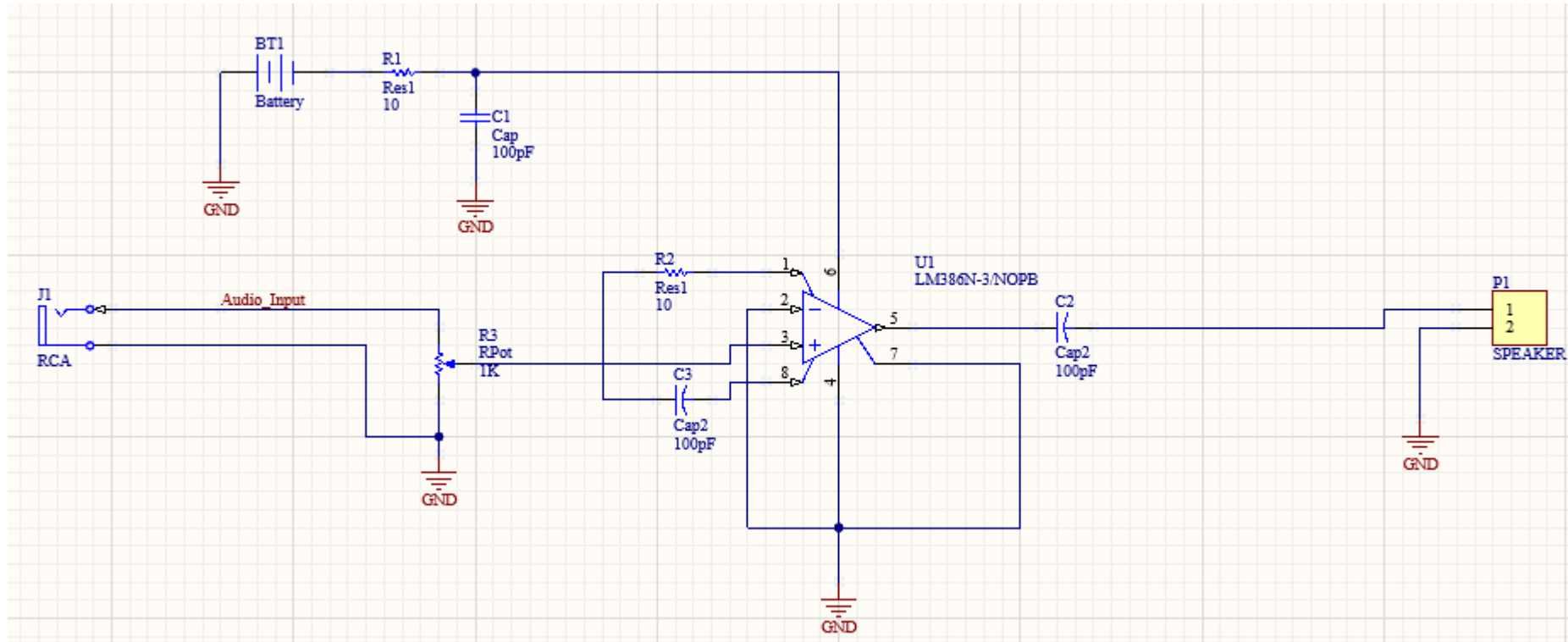
Close

My first design



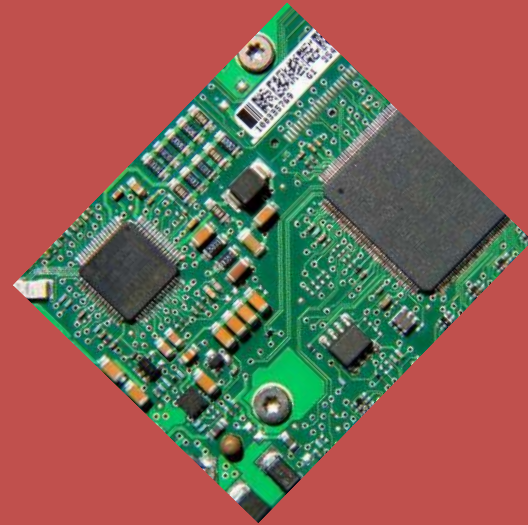
My first design

□ Final design view

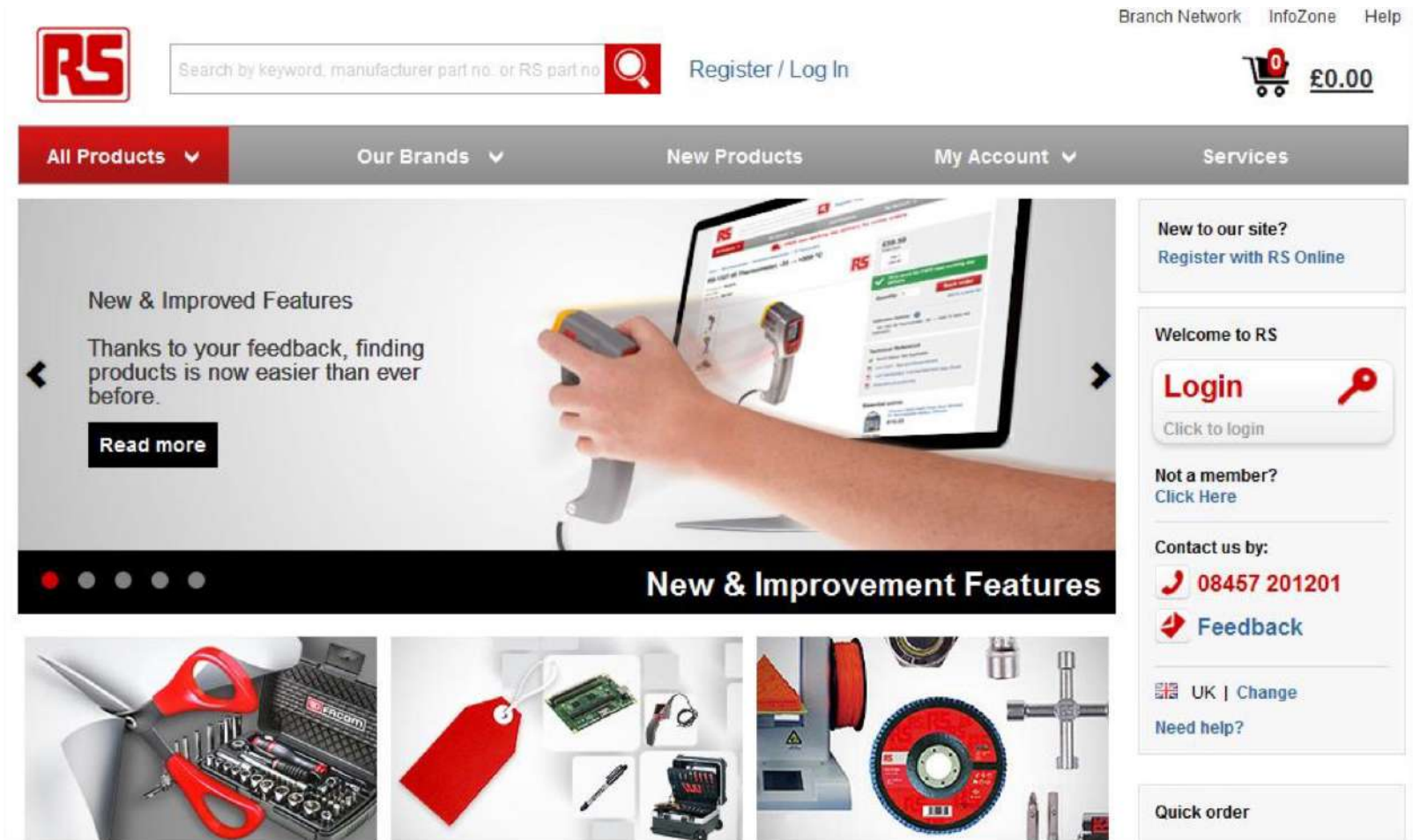


Introduction to Printed Circuit Board (PCB) design with ALTIUM

2.- Electronic Suppliers



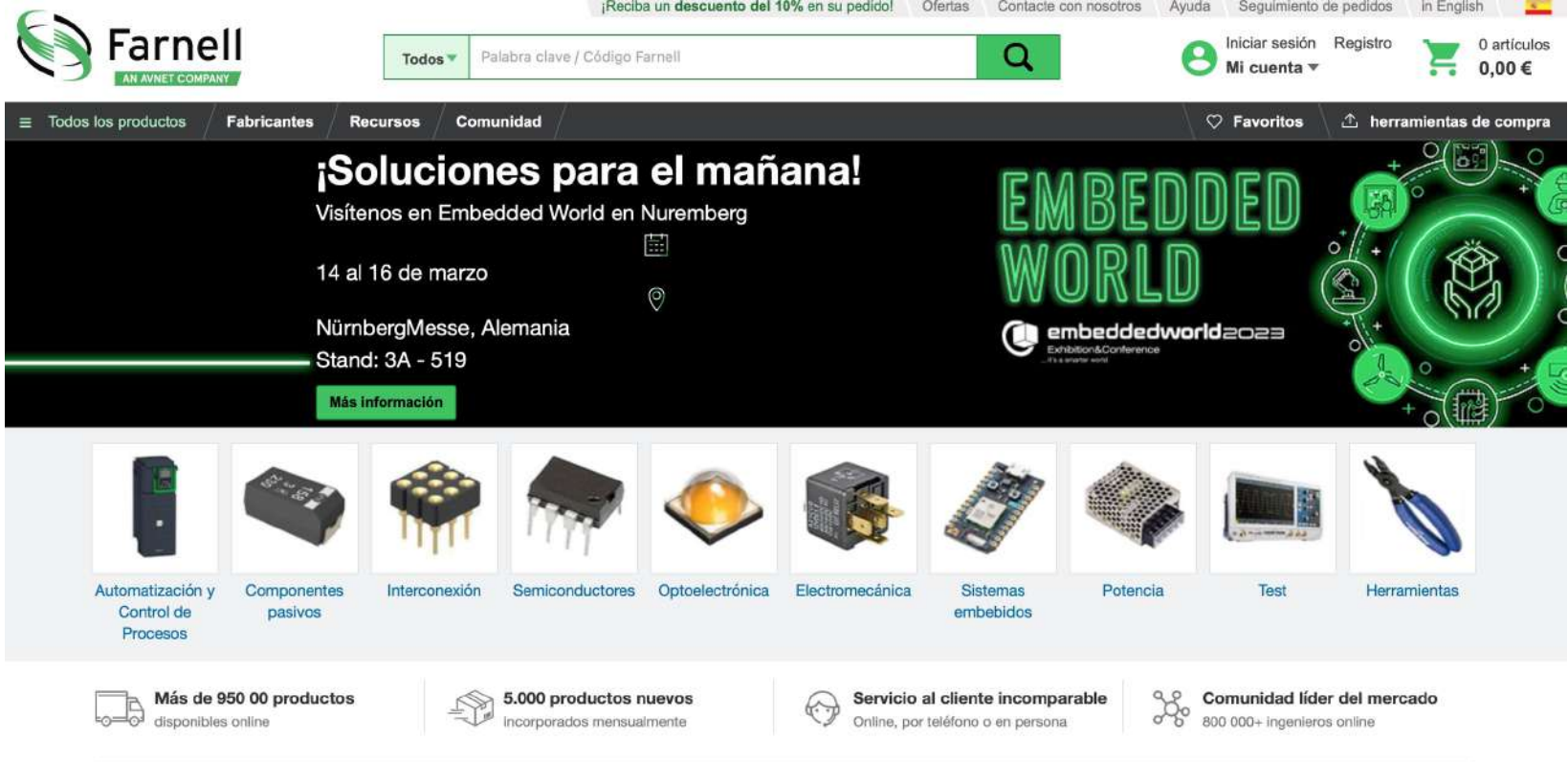
Electronic Suppliers



The screenshot shows the RS Online website homepage. At the top, there is a red navigation bar with the RS logo on the left and links for 'Branch Network', 'InfoZone', and 'Help' on the right. Below the navigation bar is a search bar with the placeholder text 'Search by keyword, manufacturer part no. or RS part no.' and a magnifying glass icon. To the right of the search bar is a 'Register / Log In' link. Further right is a shopping cart icon with a red '0' and the text '£0.00'. Below the search bar is a grey navigation bar with links for 'All Products', 'Our Brands', 'New Products', 'My Account', and 'Services'. The main content area features a large banner with the text 'New & Improved Features' and 'Thanks to your feedback, finding products is now easier than ever before.' A hand is shown using a handheld scanner to scan a product on a laptop screen. Below the banner is a row of four small images: a pair of red-handled scissors, a red tag, a green circuit board, and a red circular component. To the right of the banner is a sidebar with several sections: 'New to our site?' with a 'Register with RS Online' link; 'Welcome to RS' with a 'Login' button and a 'Click to login' link; 'Not a member?' with a 'Click Here' link; 'Contact us by:' with a phone icon and the number '08457 201201' and a 'Feedback' button; and 'Quick order' with a 'Need help?' link. The bottom of the page features a large URL: <https://es.rs-online.com/web/>.

<https://es.rs-online.com/web/>

Electronic Suppliers



The screenshot shows the Farnell website homepage. At the top, there's a navigation bar with the Farnell logo (AN AVNET COMPANY), a search bar with a dropdown menu showing 'Todos' and a search icon, and links for 'Iniciar sesión', 'Registro', and 'Mi cuenta'. A promotional banner at the top right says '¡Reciba un descuento del 10% en su pedido!'. Below the navigation bar, there's a main banner for '¡Soluciones para el mañana!' (Solutions for tomorrow!) featuring the 'EMBEDDED WORLD' exhibition and conference in Nuremberg, Germany, from March 14 to 16. The banner includes a 'Más información' (More information) button. Below the main banner, there's a grid of product categories: Automatización y Control de Procesos, Componentes pasivos, Interconexión, Semiconductores, Optoelectrónica, Electromecánica, Sistemas embebidos, Potencia, Test, and Herramientas. At the bottom, there's a footer with four sections: 'Más de 950 00 productos disponibles online', '5.000 productos nuevos incorporados mensualmente', 'Servicio al cliente incomparable' (Online, por teléfono o en persona), and 'Comunidad líder del mercado' (800 000+ ingenieros online).

Farnell
AN AVNET COMPANY

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Iniciar sesión Registro 0 artículos
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Más de 950 00 productos disponibles online 5.000 productos nuevos incorporados mensualmente Servicio al cliente incomparable Online, por teléfono o en persona Comunidad líder del mercado 800 000+ ingenieros online

<https://es.farnell.com/>

Electronic Suppliers

The screenshot shows the Digi-Key Electronics website. The header features the Digi-Key logo, a search bar with the text "Todos los productos" and "Ingresa una palabra clave o un número de pieza", a search icon, a Spanish flag, and links for "Iniciar sesión o REGISTRARSE". A shopping cart icon shows "0 producto(s)". Below the header, there are navigation links for "Productos", "Fabricantes", and "Recursos", and a banner for "ENTREGA GRATIS para pedidos que superen los 50 EUR.*". The main content area includes a sidebar with a "PRODUCTOS VER TODO" section listing various electronic components, a large central banner titled "La cadena de suministro transformada" with a red "Obtener más información" button, and three right-hand columns for "Artículos", "Nuevos productos", and "Calculadoras de conversión". A vertical sidebar on the far right contains links for "Comentarios" and "¿Necesita ayuda?".

PRODUCTOS VER TODO

- Automatización y control
- Cajas, gabinetes, bastidores
- Arneses de cables
- Cables, alambres
- Conectores, interconectores
- Dispositivos electromecánicos
- Piezas metálicas, piezas de sujeción, accesorios
- Circuitos integrados (CI)
- Soluciones de redes
- Pasivos
- Protección de circuitos, alimentación
- Semiconductores
- Productos de prueba
- Herramienta

La cadena de suministro transformada

¿Qué están haciendo los proveedores de manera diferente para garantizar que los productos más demandados estén disponibles con facilidad para los ingenieros y diseñadores que dependen de ellos?

Obtener más información

Artículos

Visite nuestra biblioteca con las últimas noticias de tecnología, información de productos y actualizaciones de proveedores.

Nuevos productos

Consulte las últimas piezas y proveedores que se agregaron a nuestro extenso inventario

Calculadoras de conversión

Las calculadoras de conversión en línea incluyen código de colores de resistencias, decimal a fracción y más

Comentarios

¿Necesita ayuda?

<https://www.digikey.es>

Electronic Suppliers

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☐ En existencias ☐ RoHS

Productos  Fabricantes Servicios y herramientas Recursos técnicos Ayuda Cuentas y pedidos   0

Productos más recientes 

- Automatización industrial 
- Cables y alambres 
- Componentes pasivos 
- Conectores 
- Dispositivos de RF e inalámbricos 
- Electromecánicos 
- Energía 
- Gestión térmica 
- Herramientas de ingeniería 
- Herramientas y suministros 
- Iluminación LED 
- Memoria y almacenamiento de datos 
- Optoelectrónica 
- Protección de circuitos 
- Prueba y medición 
- Recintos 
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- Computer Products
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- Electromechanical
- Electronic Design
- Fans & Cooling
- ICs & Semiconductors
- Interconnects
- Optoelectronics & LED/Lighting
- Passive Components
- Power Supplies & Wall Adapters
- Test, Tools & Supplies
- Wire & Cable

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Quantity Discounts & Quote Requests
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PRODUCTS IN-STOCK
READY TO SHIP



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- ☐ Hobby Engineering
- ☐ All Electronics
- ☐ BG Micro
- ☐ Jameco Electronics
- ☐ RadioShack
- ☐ Fry's Electronics
- ☐ Goldmine Electronics
- ☐ Futurlec

Industrial Suppliers

- ❑ On-line catalogs and directories
- ❑ ThomasNET
- ❑ A comprehensive directory of industrial suppliers. Look here to find sources for almost anything!

GLOBALSPEC

An "engineering search engine". It is similar to ThomasNET, and is useful to help find vendors and catalog items.

- ❑ SKF : An excellent interactive engineering catalog for bearing (rodamientos) selection.

Stock Drive Components/Sterling Instruments : A comprehensive on- line catalog of mechanical components. You can download 3D CAD models to use in your CAD assembly.

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MatWeb, Your Source for Materials Information

What is MatWeb? MatWeb's [searchable database of material properties](#) includes data sheets of thermoplastic and thermoset polymers such as ABS, nylon, polycarbonate, polyester, polyethylene and polypropylene; metals such as aluminum, cobalt, copper, lead, magnesium, nickel, steel, superalloys, titanium and zinc alloys; ceramics; plus semiconductors, fibers, and other engineering materials.

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How to Find Property Data in MatWeb

Quantitative Searches:

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Text Search:

- Enter a key word or phrase in the box below
(this search is also available at the top of every page).

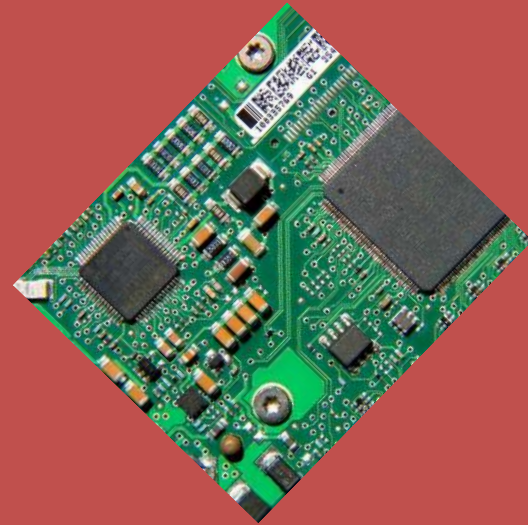
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Introduction to Printed Circuit Board (PCB) design with ALTIUM

2.- Schematic Library



Library concept

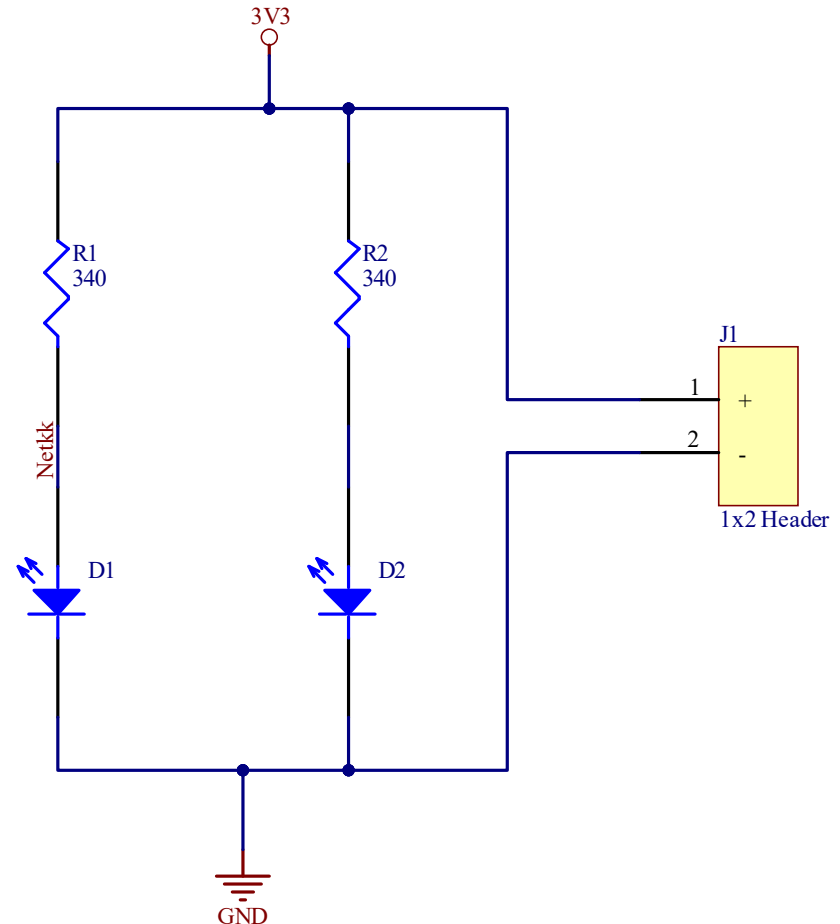
- ❑ A **library** is a repository where components are stored.
- ❑ Several type of library could be handled:
 - ❑ PCB and Schematic
 - ❑ Manufacturer Library
 - ❑ Working Library
 - ❑ Project Library
- ❑ For schematic purposes, we will use commercial components and will design customized elements linked with commercial products (linking with official information and/or models).

Working with libraries

- ❑ It is convenience to choose the components/device to be used in advance for the schematic design.
- ❑ So, we usually link the model, symbol, datasheet, footprint, etc. at the beginning of our design.
- ❑ Methodology:
 - ❑ Creating a SCH Library
 - ❑ Creating SCH symbols
 - ❑ Creating the SCH.

Our basic design

- This is the simple design that we are going to create.
- All the components will be drawn by us without using the official symbol.
- All of them (diode, resistor and header) will be linked with commercial units.



Devices to be used

- The methodology to select the appropriate devices will be as follows:
 - To find the right device (component) through a manufacturer supplier using filter search tool (web).
 - Identify part number or supplier number (web)
 - Finding the chosen device into Manufacturer Search Part (Altium).

Devices to be used

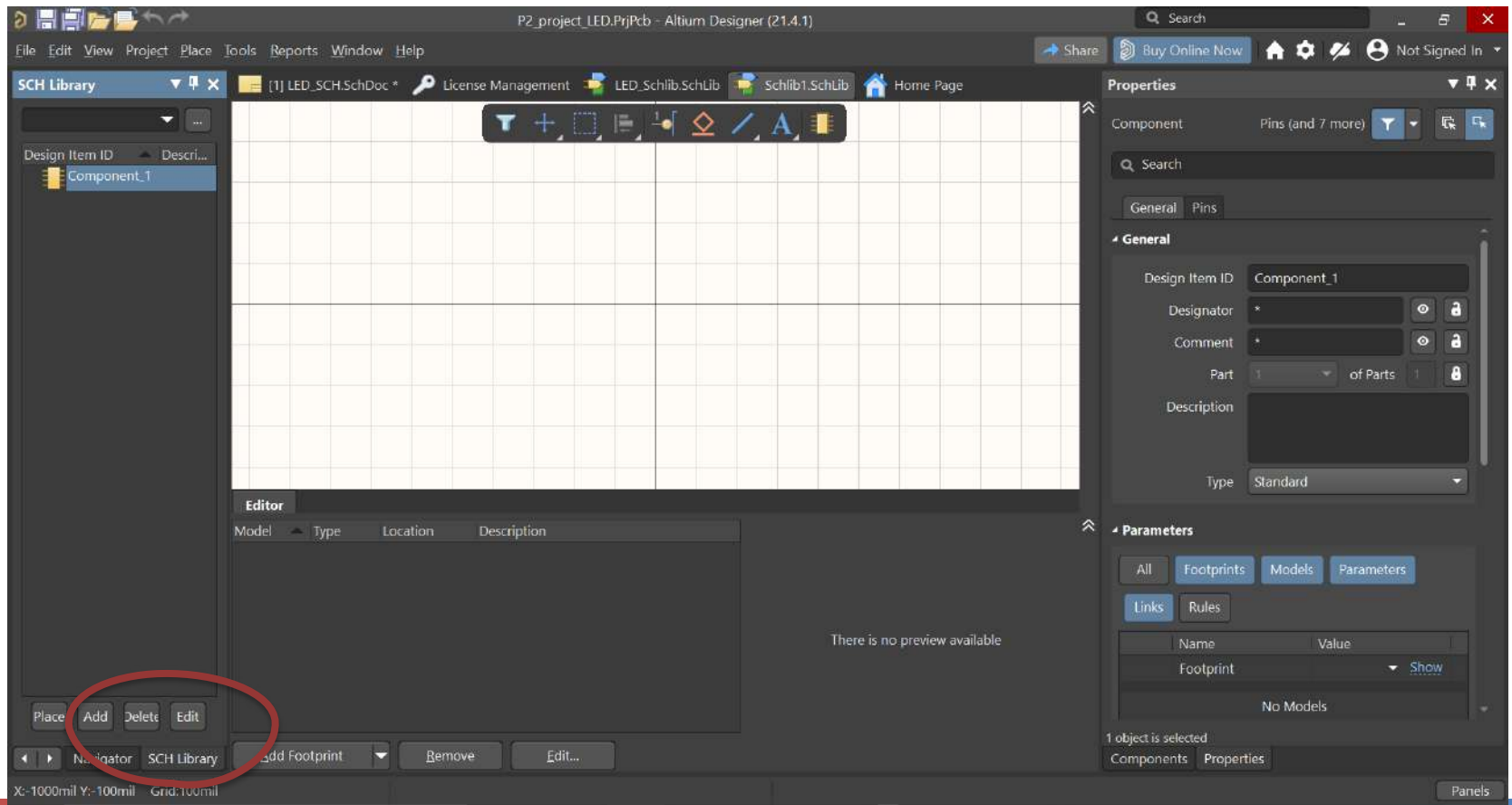
- For this example we will use the next devices:

Name	Manufacturer reference	Manufacturer	Digi-Key Part number
HEADER 2	0022272021	Molex	WM411-ND
RESISTOR	CRCW0805340RF KEA	Vishay Dale	541-340CCT-ND
LED	LNJ337W83RA	Panasonic	LNJ337W83RACT-ND

- Additionally, we will create the schematic symbol for each element.

Starting with the SCH Library

- The first step must be create a SCH Library (File → New → Library → Schematic Library).



Selection of header from a provider

Header 1x2

□ Searching the header component within Digikey

The screenshot shows the Digikey website interface. The browser address bar displays the URL: `digkey.es/es/products/result?s=N4lgTCBcDalBYFMCGATBAnABGAdAVgBYBbIkAXQF8g`. The website header includes the Digi-Key logo, a search bar with the text "header 2.54mm", and navigation links for "Todos los productos", "Iniciar sesión o REGISTRARSE", and "ENTREGA GRATIS para pedidos q". Below the header, a sidebar on the left lists categories under "Ir a:", including "Arneses de cables", "Cables, alambres - administración", "Circuitos integrados (CI)", "Conectores, interconectores", "Kits", "Prototipado, fabricación de productos", and "Sin categoría". The main content area displays "Mostrar 502.036 resultados de 'header 2.54mm'". It features a search bar "Buscar en los resultados" and a "Fabricante" dropdown. Below these are three sections of filters: "Opciones de almacenamiento" (En stock, Normalmente con stock, New Product), "Medios de comunicación" (Hoja de datos, Foto, Modelos EDA/CAD), and "Opciones ambientales" (Cumple con RoHS, No cumple con RoHS). The results section is titled "Resultados principales : header 2.54mm" and lists "Categorías principales" with links to "Conectores rectangulares - Cabezales, clavijas macho" (242.170 Productos), "Conectores rectangulares - Cabezales, receptáculos, enchufes hembra" (119.264 Productos), and "Conectores rectangulares - Placa incorporada, cable directo a placa" (725 Productos). The right side of the results section lists "Conectores, interconectores" three times.

Ir a:

- Arneses de cables
- Cables, alambres - administración
- Circuitos integrados (CI)
- Conectores, interconectores
- Kits
- Prototipado, fabricación de productos
- Sin categoría

Mostrar 502.036 resultados de "header 2.54mm"

Buscar en los resultados

Fabricante

Opciones de almacenamiento

- ☐ En stock
- ☐ Normalmente con stock
- ☐ New Product

Medios de comunicación

- ☐ Hoja de datos
- ☐ Foto
- ☐ Modelos EDA/CAD

Opciones ambientales

- ☐ Cumple con RoHS
- ☐ No cumple con RoHS

Resultados principales : header 2.54mm

Categorías principales

- [Conectores rectangulares - Cabezales, clavijas macho](#) (242.170 Productos)
- [Conectores rectangulares - Cabezales, receptáculos, enchufes hembra](#) (119.264 Productos)
- [Conectores rectangulares - Placa incorporada, cable directo a placa](#) (725 Productos)

Conectores, interconectores

Conectores, interconectores

Conectores, interconectores

Selection of header from a provider

□ Searching the header component within Digikey

Conectores, interconectores | Co: x

digkey.es/es/products/filter/conectores-rectangulares-cabezales-clavijas-macho/314?s=N4lgTCBcDalBYFMCGATBAnABGAdAVgBYBblkAX...

Aplicaciones Bookmarks Nueva pestaña CURSO FPGA AVANZ Evalua_MINECO EU EditorSCH UAH Google SEPIE Otros marcadores Lista de lectura

Resultados: 242.170 1.338 restantes

Buscar en los resultados

Fabricante Serie Embalaje Estado de la pieza Tipo de conector

KYOCERA AVX
METZ CONNECT USA Inc.
Mill-Max Manufacturing Corp.
Molex
Omnetics
Omron Automation and Safety
Omron Electronics Inc-EMC Div
On Shore Technology Inc.
Phoenix Contact
Preci-Dip
Samtec Inc.

1000 Box
1001 Box
2300
2400
2500
2600
3000
301

Bandeja
Bolsa
Caja
Cinta cortada (CT)
Cinta y caja
Cinta y rollo (TR)
Digi-Reel®
Granel
Rollo

Activo
Discontinuo en Digi-Key
No para diseños nuevos
Obsoleto
Última compra

Cabecera
Cabecera, entrada inferior
Cabezal, apilable (2)
Cabezal, cortable
Cabezal, elevado
Cabezal, separador
Clavija de conexión
Puente
Receptáculo

Borrar

Ver precios en: Ingresar la cantidad

Opciones de almacenamiento
☒ En stock
☐ Normalmente con stock
☐ New Product

Medios de comunicación
☐ Hoja de datos
☐ Foto
☒ Modelos EDA/CAE

Opciones ambientales
☐ Cumple con RoHS
☐ No cumple con RoHS

PRODUCTO DEL MERCADO
☐ Excluir

Aplicar todo 1.338 restantes

ENTRADA DE BÚSQUEDA

header 2.54mm

Comentarios

¿Necesita ayuda?

Selection of header from a provider

□ Searching the header component within Digikey

The screenshot shows the Digikey website interface for selecting a header component. The browser address bar shows the URL: [digikey.es/products/filter/conectores-rectangulares-cabezales-clavijas-macho/314?s=N4lgjCBcoLQdIDGUBmBDANgZwKYBoQB7K...](https://www.digikey.es/products/filter/conectores-rectangulares-cabezales-clavijas-macho/314?s=N4lgjCBcoLQdIDGUBmBDANgZwKYBoQB7K...). The page title is "Conectores rectangulares - Cabezales, clavijas macho". The search results show 1,338 results, with 12 remaining after filters are applied.

The filters section includes the following options:

- Cantidad de posiciones:** 2 (selected), 3, 4, 5, 6, 7, 8, 9, 10, 11. [Borrar](#)
- Cantidad de filas:** 1 (selected), 2, 4. [Borrar](#)
- Espaciado de fila - acoplamiento:** 0.100" (2.54mm). [Borrar](#)
- Estilo:** Placa a cable/alambre (selected), Placa a placa o cable, Placa a placa. [Borrar](#)
- Recubrimiento:** Envuelto - 1 pared, Envuelto - 4 paredes, Sin protección. [Borrar](#)
- Tipo de montaje:** Montaje en superficie, Montaje en superficie, ángulo recto, Orificio pasante (selected), Orificio pasante, ángulo recto. [Borrar](#)

The bottom section includes the "Ver precios en:" field with the input "Ingresar la cantidad". Below this are the following options:

- Opciones de almacenamiento:** ☒ En stock, ☐ Normalmente con stock, ☐ New Product.
- Medios de comunicación:** ☐ Hoja de datos, ☐ Foto, ☒ Modelos EDA/CAD.
- Opciones ambientales:** ☐ Cumple con RoHS, ☐ No cumple con RoHS.
- PRODUCTO DEL MERCADO:** ☐ Excluir.

The "Aplicar todo" button is highlighted in red, and the text "12 Restante" is displayed next to it.

Selection of header from a provider

□ Searching the header component within Digikev

Conectores, interconectores | Co x +

digikev.es/es/products/filter/conectores-rectangulares-cabezales-clavijas-macho/314?s=N4lgjCBcoGwJxVAYygMwIYBsDOBTANCAPZ...

Aplicaciones ★ Bookmarks 🌐 Nueva pestaña 🌐 CURSO FPGA AVANZ 🐞 Evalua_MINECO 🇪🇺 EU 📄 EditorSCH 🇺🇦 UAH 📄 Google 📄 SEPIE » 📄 Otros marcadores 📄 Lista de lectura

Comparar	Número de parte del fabricante	Precio	Stock	Proveedor	Fabricante	Cantidad mínima	N.º de pieza de DK	Serie	Paquete	Estado de la pieza	Tipo de conector	Tipo de contactos	Paso:
☐	 1719720002 CONN HEADER VERT 2POS 2.54MM	0,69000 €							Tubo	Activo	Cabecera	Clavija macho	0.100"
☐	 0705430036 CONN HEADER VERT 2POS 2.54MM	0,87000 €							Tubo	Activo	Cabecera	Clavija macho	0.100"
☐	 0901361102 CONN HEADER VERT 2POS 2.54MM	1,03000 €							Bandeja	Activo	Cabecera	Clavija macho	0.100"
☐	 0705410036 CONN HEADER VERT 2POS 2.54MM	1,37000 €							Granel	Activo	Cabecera	Clavija macho	0.100"
☒	 0022272021 CONN HEADER VERT 2POS 2.54MM	0,26000 €							Granel	Activo	Cabecera	Clavija macho	0.100"
☐	 0901361302 CONN HEADER VERT 2POS 2.54MM	1,15000 €							Bandeja	Activo	Cabecera	Clavija macho	0.100"
☐	 0901361202 CONN HEADER VERT 2POS 2.54MM	1,15000 €							Bandeja	Activo	Cabecera	Clavija macho	0.100"



Comentarios

¿Necesita ayuda?

https://www.digikev.es/es/products/detail/molex/0022272021/1130577

Escribe aquí para buscar

60% 11°C ESP 13:14 20/02/2022

Selection of header from a provider

□ Searching the header component within Digikey

0022272021 Molex | Conectores

digikay.es/es/products/detail/molex/0022272021/1130577

Aplicaciones Bookmarks Nueva pestaña CURSO FPGA AVANZ Evalua_MINECO EU EditorSCH UAH Google SEPIE Otros marcadores Lista de lectura

Digi-Key ELECTRONICS

Todos los productos Ingrese una palabra clave o un número de pieza

Iniciar sesión o REGISTRARSE 0 producto(s)

Productos Fabricantes Recursos

ENTREGA GRATIS para pedidos que superen los 50 EUR.*

Índice de productos > Conectores, interconectores > Conectores rectangulares > Cabezales, clavijas macho > Molex 0022272021

Comparta

0022272021

N.º de producto de Digi-Key WM4111-ND

Fabricante Molex

Número de pieza del fabricante 0022272021

Proveedor Molex

Descripción CONN HEADER VERT 2POS 2.54MM

Plazo estándar del fabricante 35 semanas

Descripción detallada Cabezal de conector Orificio pasante 2 posiciones 0.100" (2.54mm)

Referencia del cliente

Hoja de datos

17.189 en stock

Puede enviarse inmediatamente

CANTIDAD

Agregar al carrito

Agregar a la lista

Todos los precios se expresan en EUR

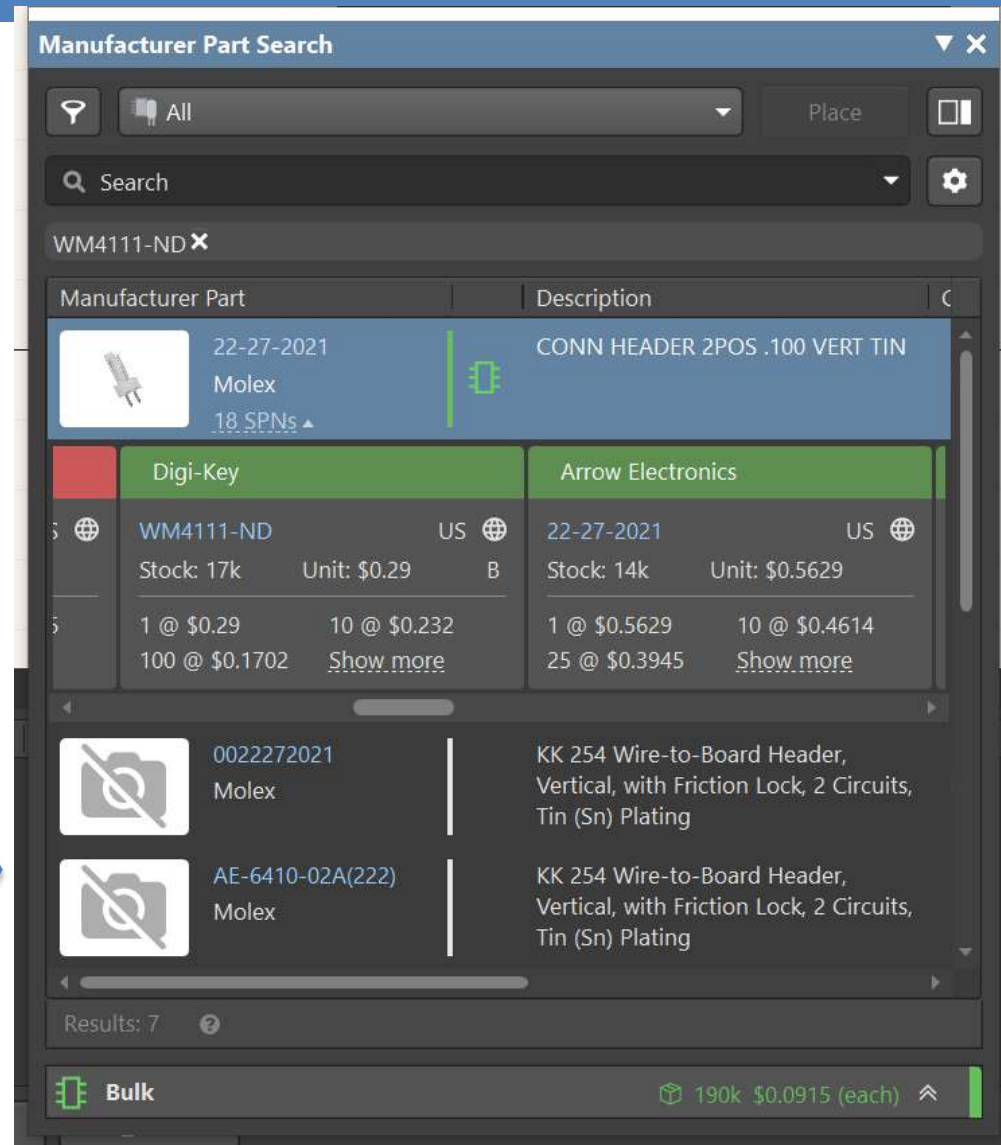
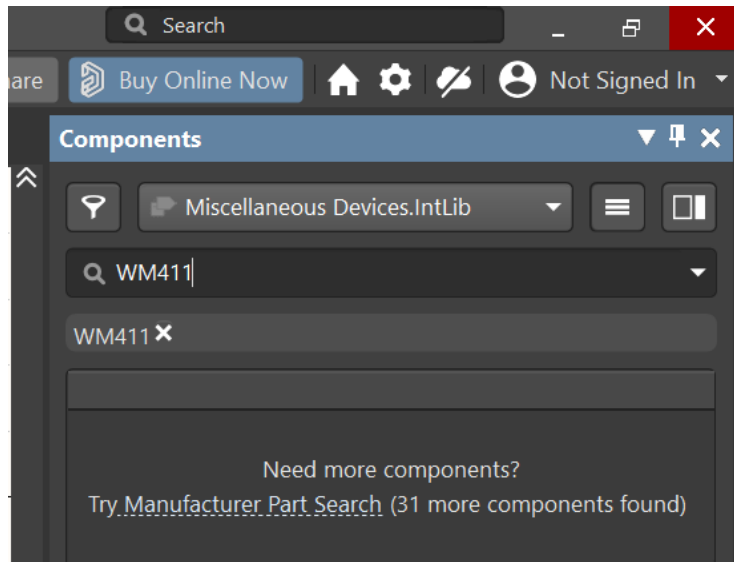
Granel

CANTIDAD	PRECIO POR UNIDAD	PRECIO EXT.
1	0,26000 €	0,26 €
10	0,20800 €	2,08 €
100	0,15230 €	15,23 €

Atributos del producto

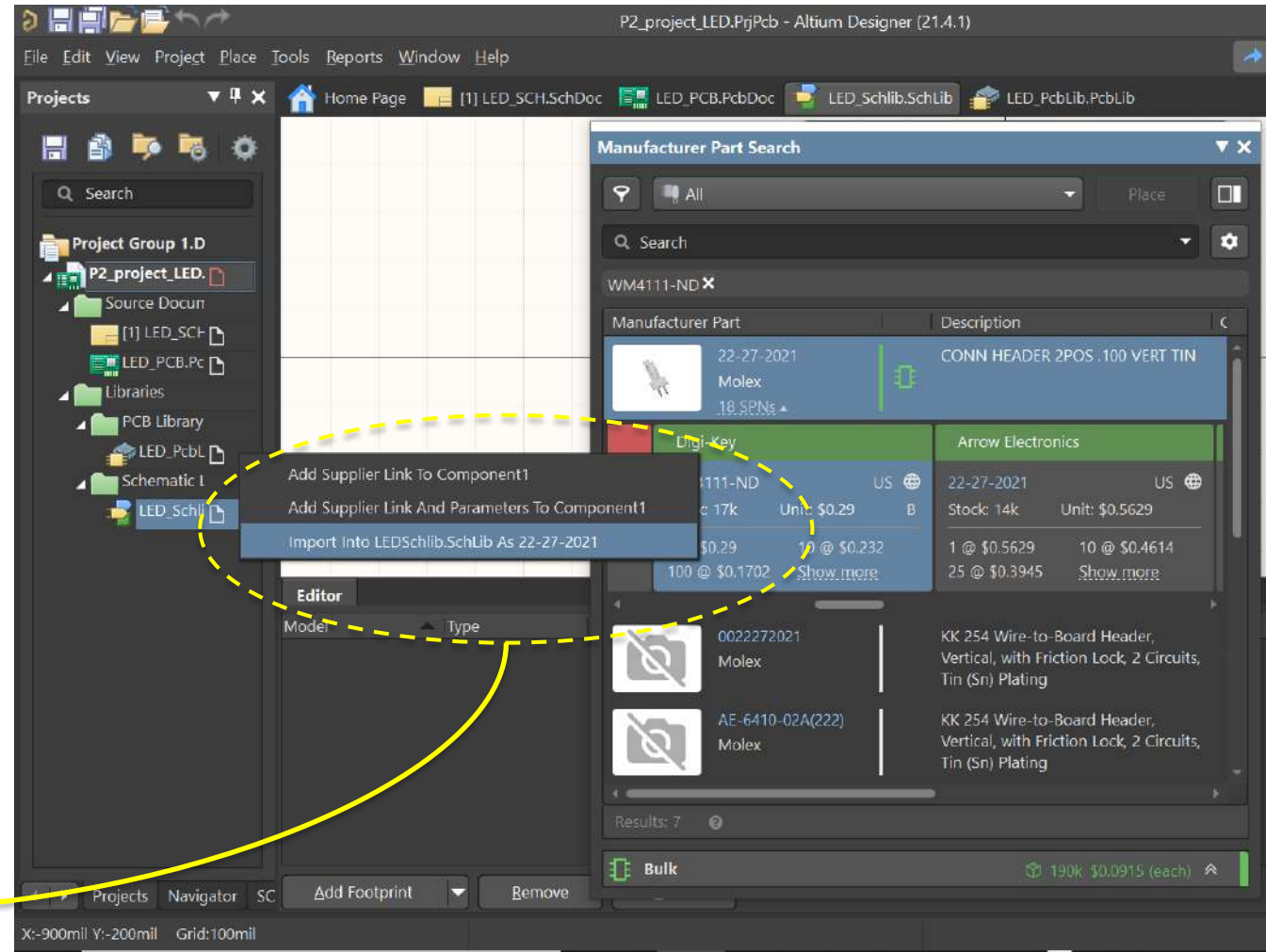
Searching the component

- Then, we will search the model inside of Altium Components (Manufacturer Research)



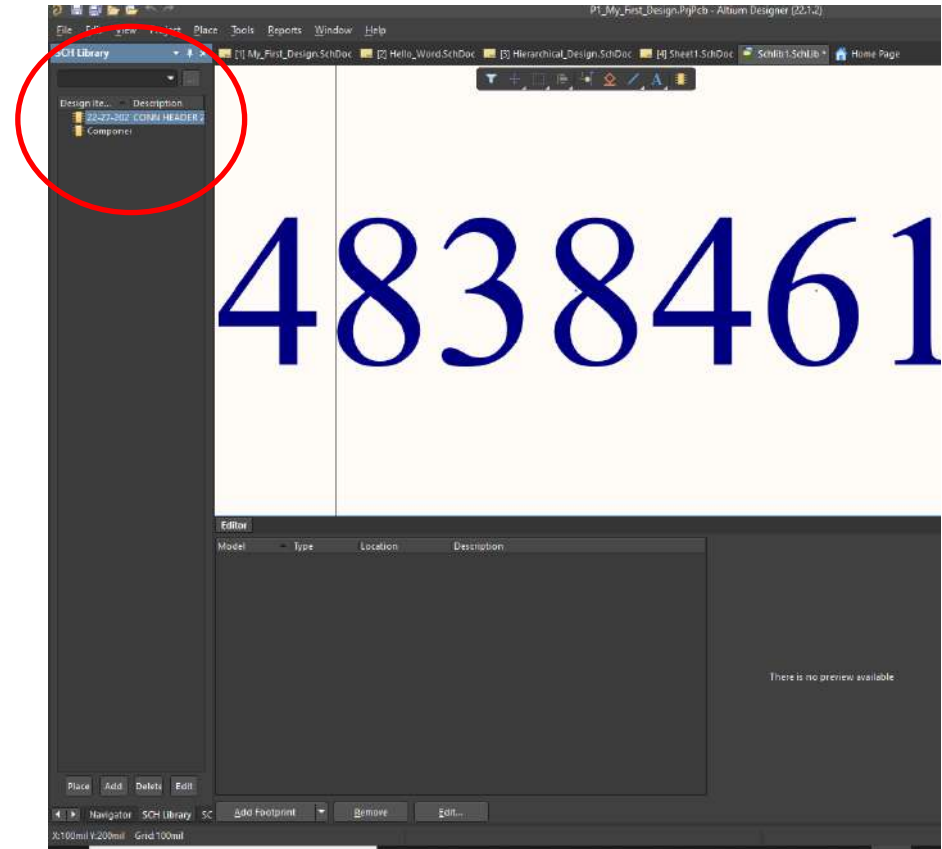
Importing the component properties

□ Now, we should import the component to our Schematic Library (active window)



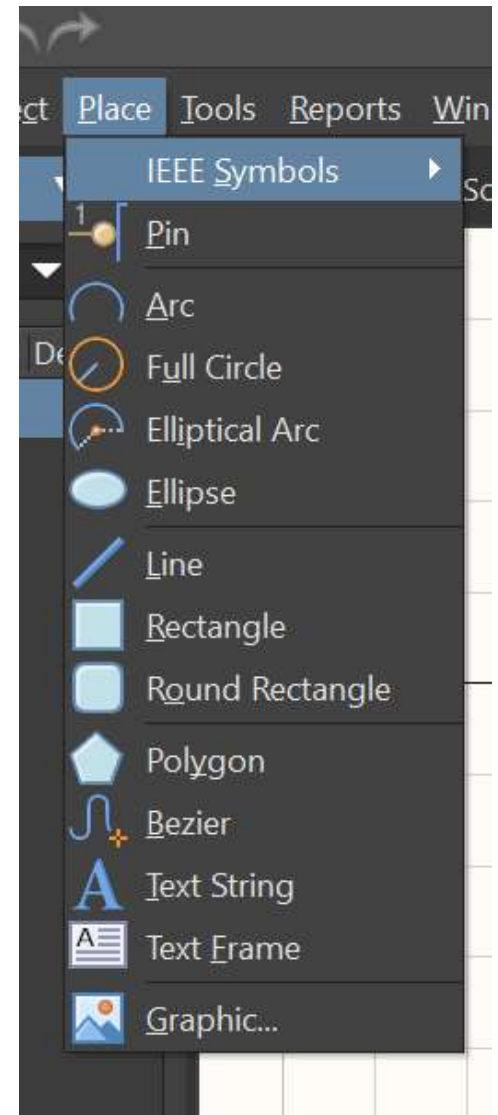
Importing the component properties

- ❑ Component properties are currently included in our SCH Library.
- ❑ However, the symbol should be created



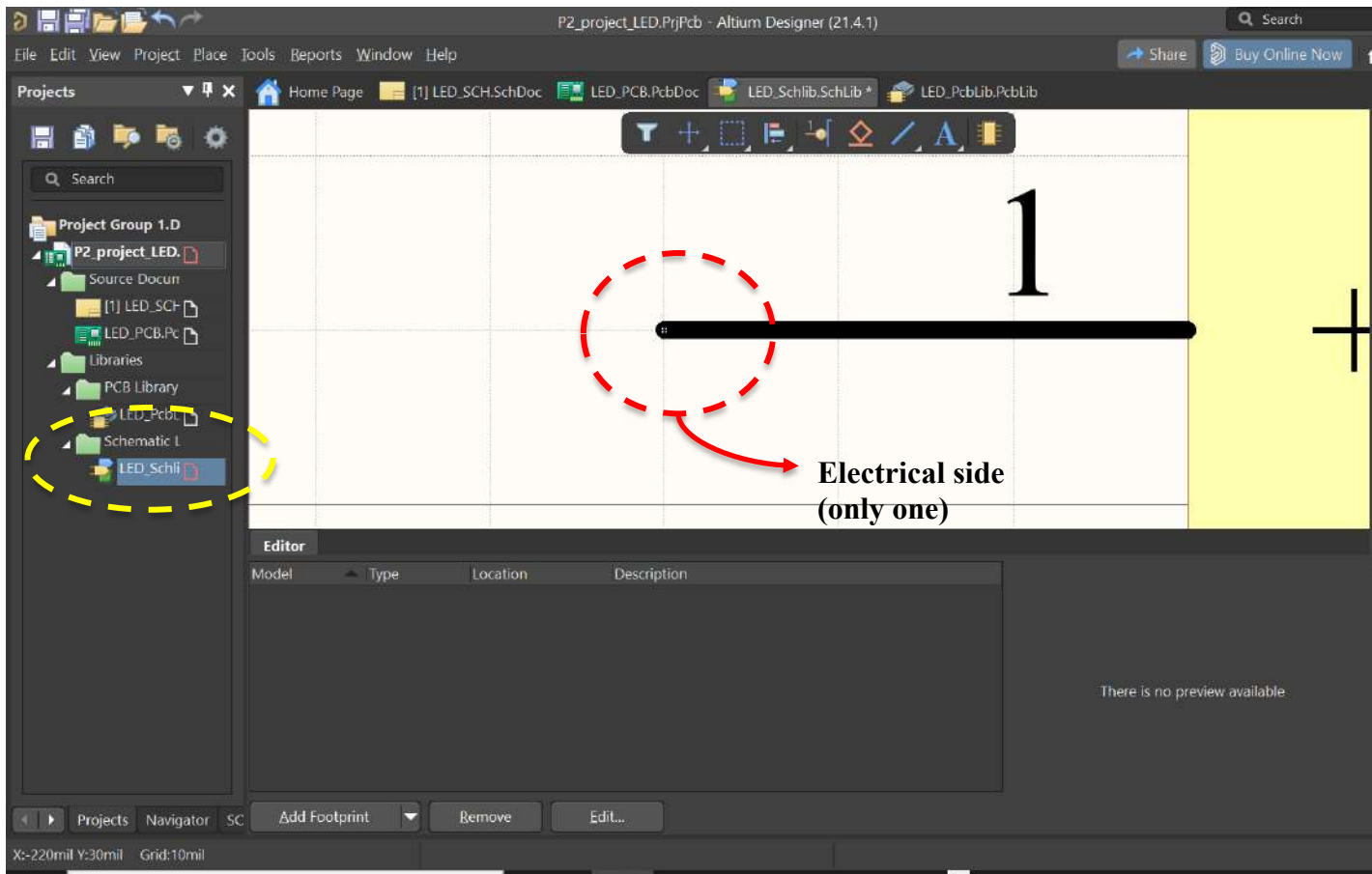
Drawing the component

- ❑ The composition of the component must be fulfilled with Pin and some shape (arc, polygon, etc.).
- ❑ Properties of the pin and shape must be checked.
- ❑ Pay attention to the pin. It must be include a name, number.

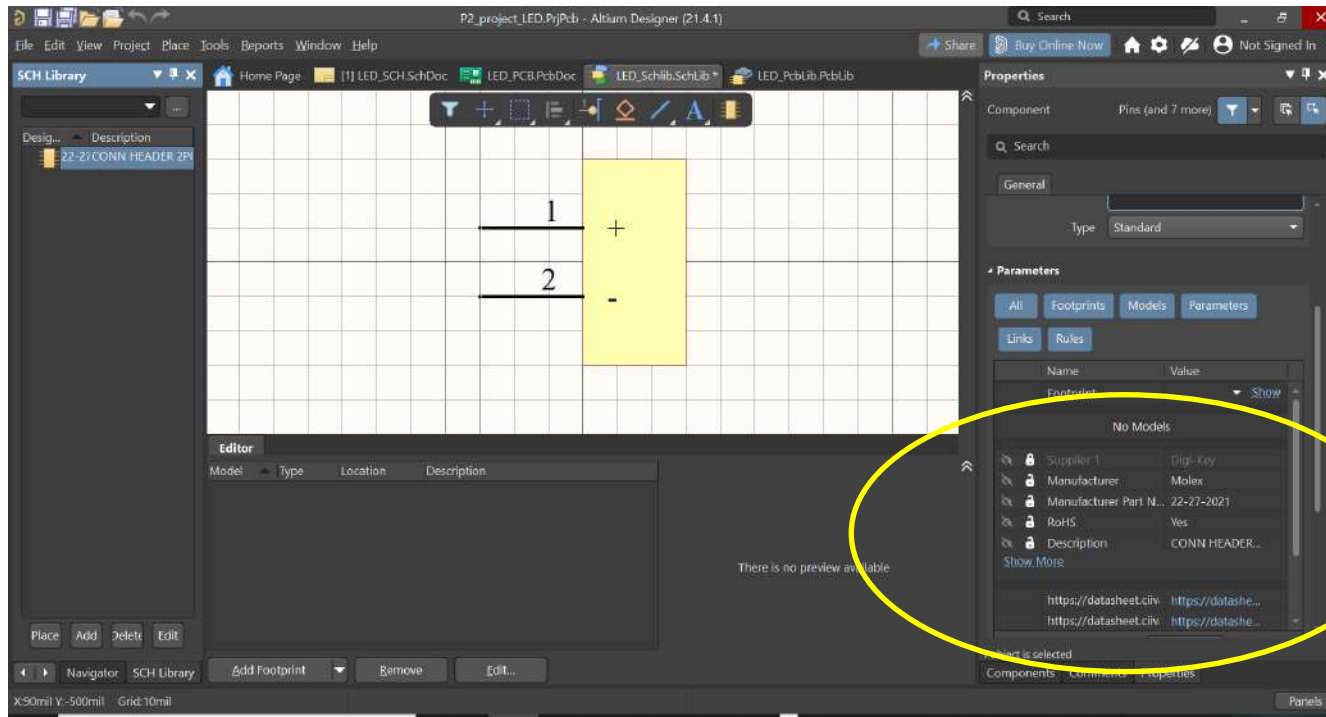


Drawing the component

- We can draw the new component (header) inside of SCH Lib environment



Drawing the component



- Here we can find some imported information.
- It contains relevant data regarding the device features.

Selection of resistor

- We will deploy the same methodology than the diode.
- A SMD version will be selected.

The screenshot shows the Digi-Key website interface for selecting a resistor. The browser tabs include '0022272021 Molex | Conectores' and 'Resistencias | Resistor de chip - montaje en superficie'. The URL is 'digikey.es/es/products/filter/resistor-de-chip-montaje-en-superficie/52?s=N4lgjCBcoLQExV'. The page title is 'Resistor de chip - montaje en superficie'. The search results show 11 results, with 3 remaining. The filters are:

Fabricante	Serie	Embalaje	Tolerancia
Stackpole Electronics Inc	AC	Cinta cortada (CT)	±0.1%
TE Connectivity Passive Product	CPE, Noohm	Cinta y rollo (TR)	±0.5%
Vishay Dale	RC_L	Digi-Reel®	±1%
YAGEO	RMCF		±5%
	RNCF		
	RT		
	TNPW e3		

Below the filters, there is a 'Ver precios en:' section with a text input field 'Ingresar la cantidad'. At the bottom, there are links for 'Opciones de almacenamiento', 'Medios de comunicación', 'Opciones ambientales', and 'PRODUCTO DEL MERCADO'. A red button labeled 'Aplicar todo' and a status '3 Restante' are also visible.

Selection of resistor

- We can choose several options. One criteria to choose can be the availability and quantity

0022272021 Molex | Conectores x AC0805FR-07360RL YAGEO | Res x +

digikay.es/es/products/detail/yageo/AC0805FR-07360RL/5896783

Aplicaciones Bookmarks Nueva pestaña CURSO FPGA AVANZ Evalua_MINECO EU EditorSCH UAH Google SEPIE Otros marcadores Lista de lectura

Índice de productos > Resistencias > Resistor de chip - montaje en superficie > YAGEO AC0805FR-07360RL Comparta

AC0805FR-07360RL

10.000 en stock

Puede enviarse inmediatamente

CANTIDAD

Cantidad

Agregar al carrito

Agregar a la lista

Todos los precios se expresan en EUR

Cinta cortada (CT) & Digi-Reel®

CANTIDAD	PRECIO POR UNIDAD	PRECIO EXT.
1	0,09000 €	0,09 €
10	0,04500 €	0,45 €
100	0,01830 €	1,83 €
1.000	0,00821 €	8,21 €
2.500	0,00712 €	17,80 €

*Todos los pedidos de Digi-Reel añadirán una cuota de carrete de

N.º de producto de Digi-Key 13-AC0805FR-07360RLTR-ND - Cinta y rollo (TR)
13-AC0805FR-07360RLCT-ND - Cinta cortada (CT)
13-AC0805FR-07360RLDKR-ND - Digi-Reel®

Fabricante YAGEO

Número de pieza del fabricante AC0805FR-07360RL

Proveedor YAGEO

Descripción RES SMD 360 OHM 1% 1/8W 0805

Plazo estándar del fabricante 20 semanas

Descripción detallada 360 Ohms ±1% 0.125W, 1/8W Resistencia en microprocesador 0805 (2012 métrico) AEC-Q200 automotriz, resistente a la humedad Película gruesa

Referencia del cliente Referencia del cliente

Hoja de datos Hoja de datos

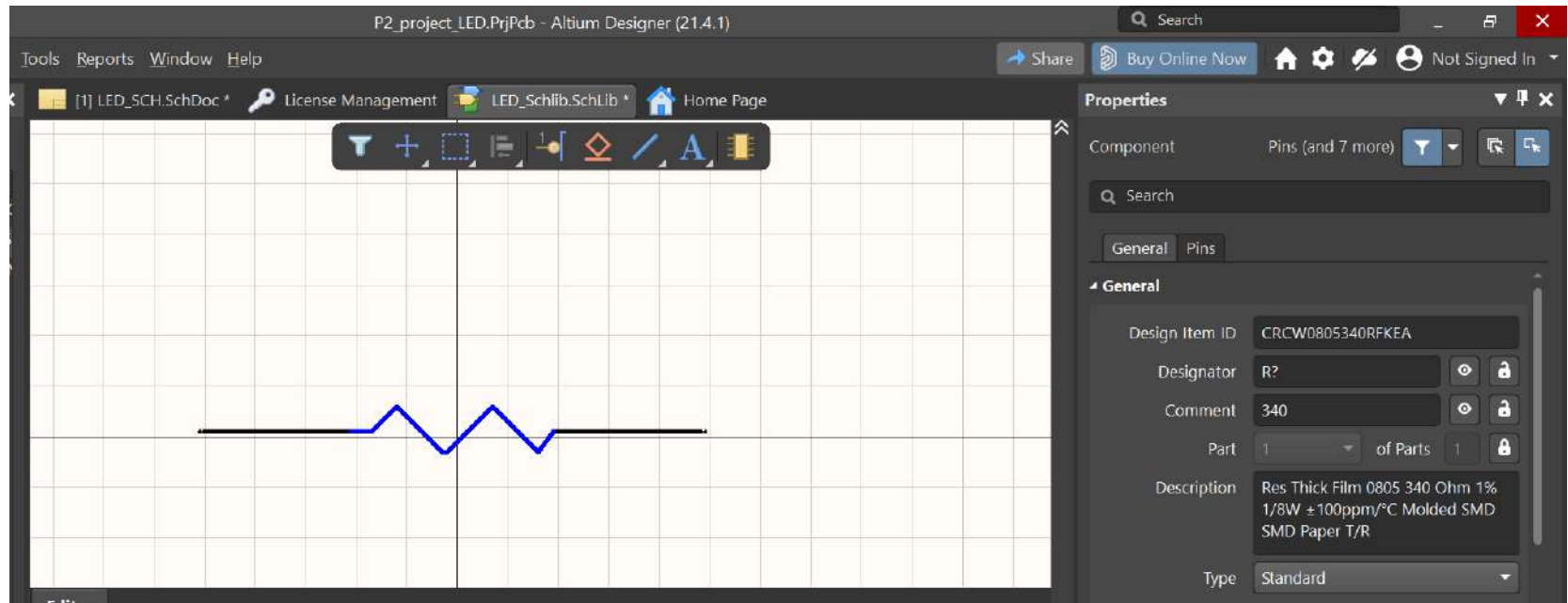
Foto demostrada es una representación solamente. Datos específicos deben ser obtenidos de las hojas de datos del producto.

Atributos del producto

TIPO	DESCRIPCIÓN	SELECCIONAR
Resistencias		☐

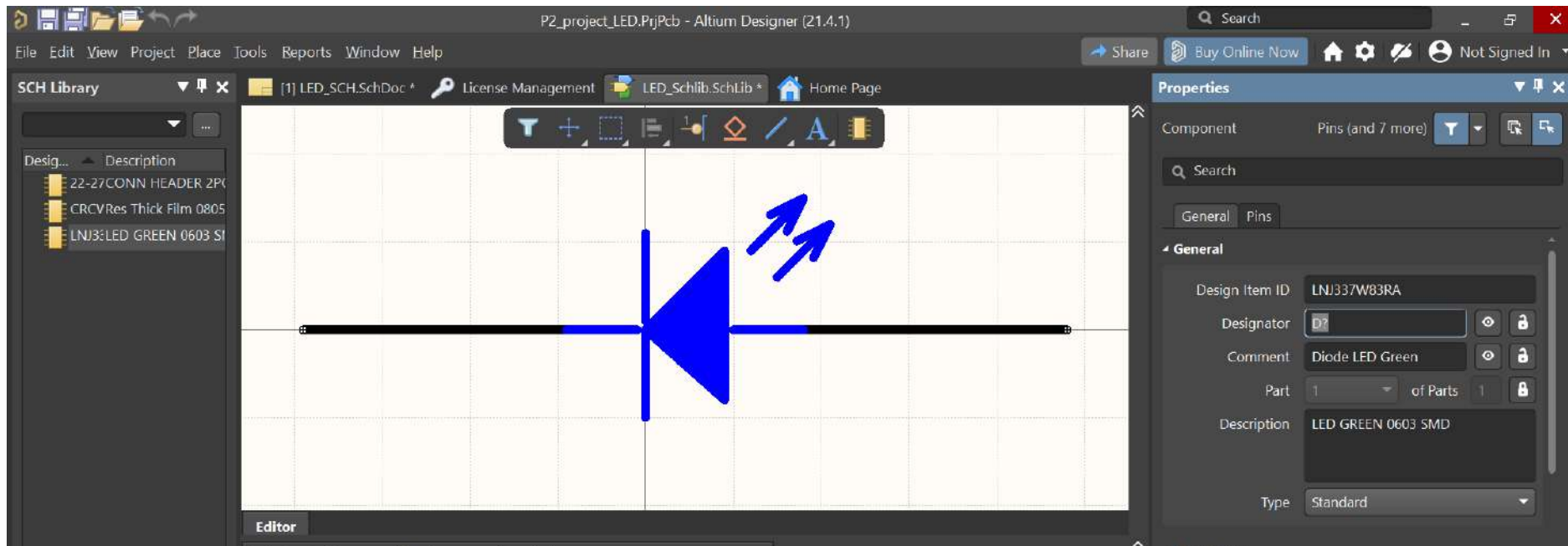
Resistor

- ❑ Remember to import the resistor model to the SCH Library.
- ❑ Then we should draw a resistor



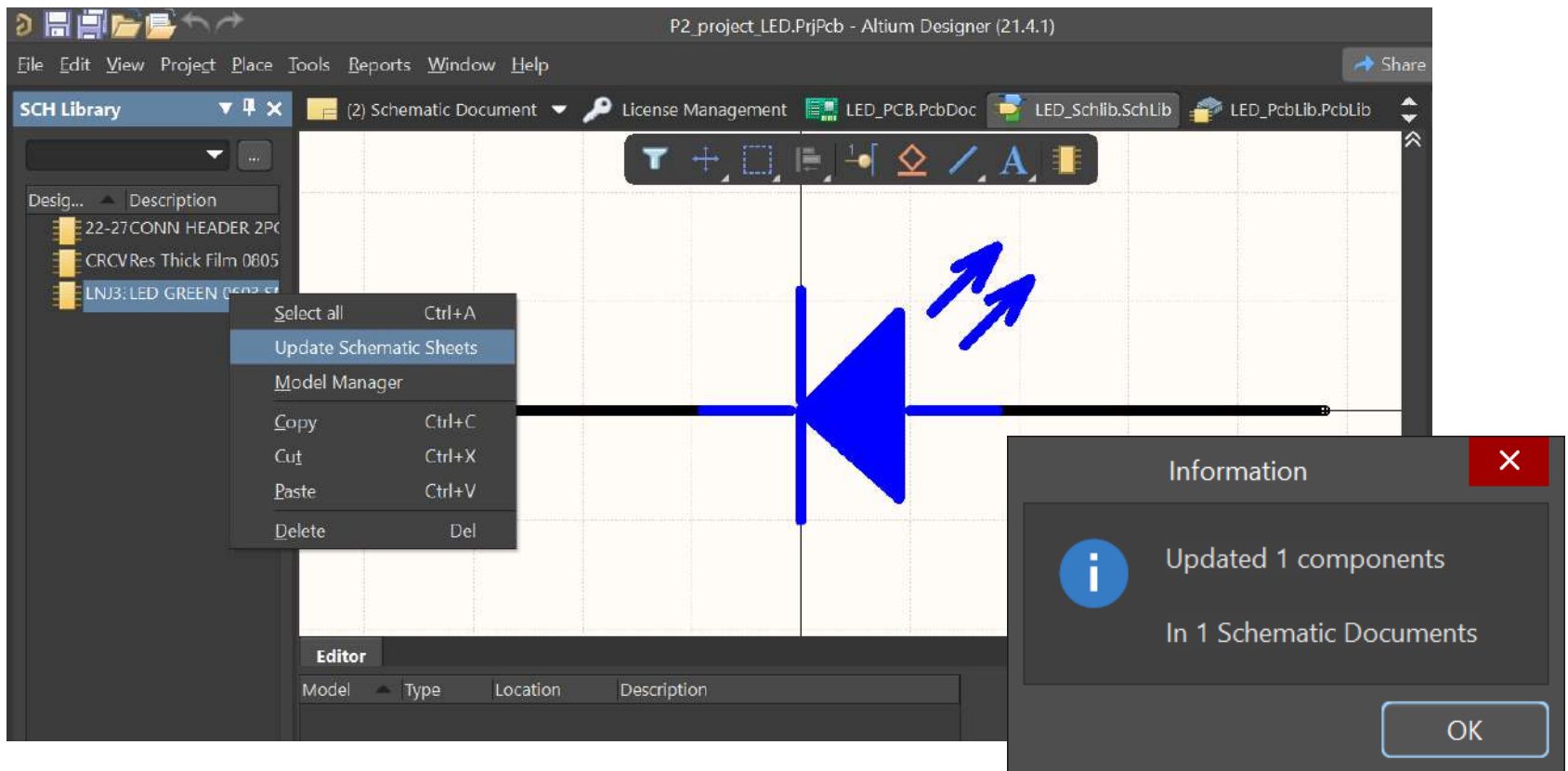
Diode

- We should do the same steps of header with the diode



Updating symbols

- If we wish updating the symbol, an automatic update could be used.

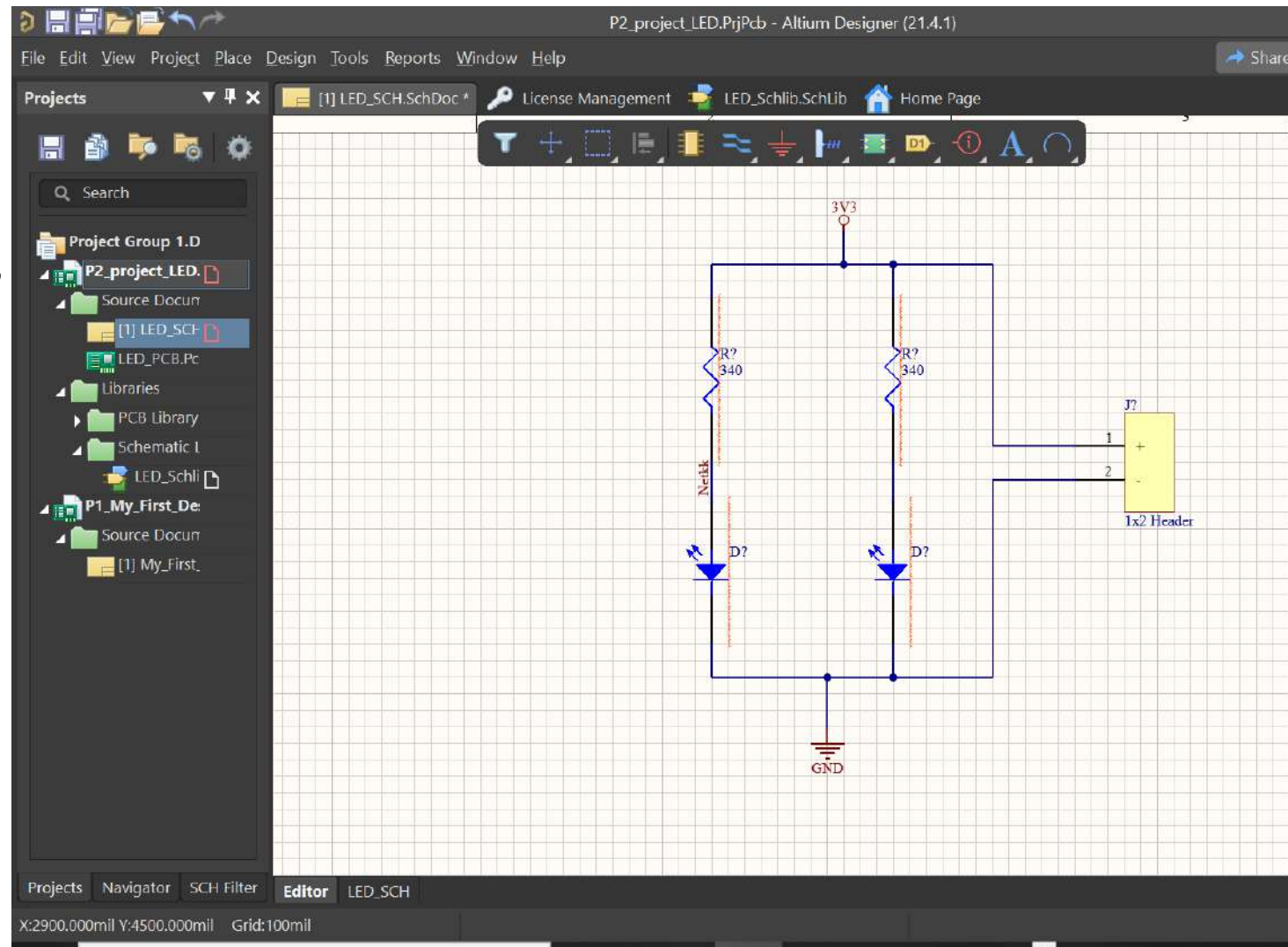


Drawing the schematic

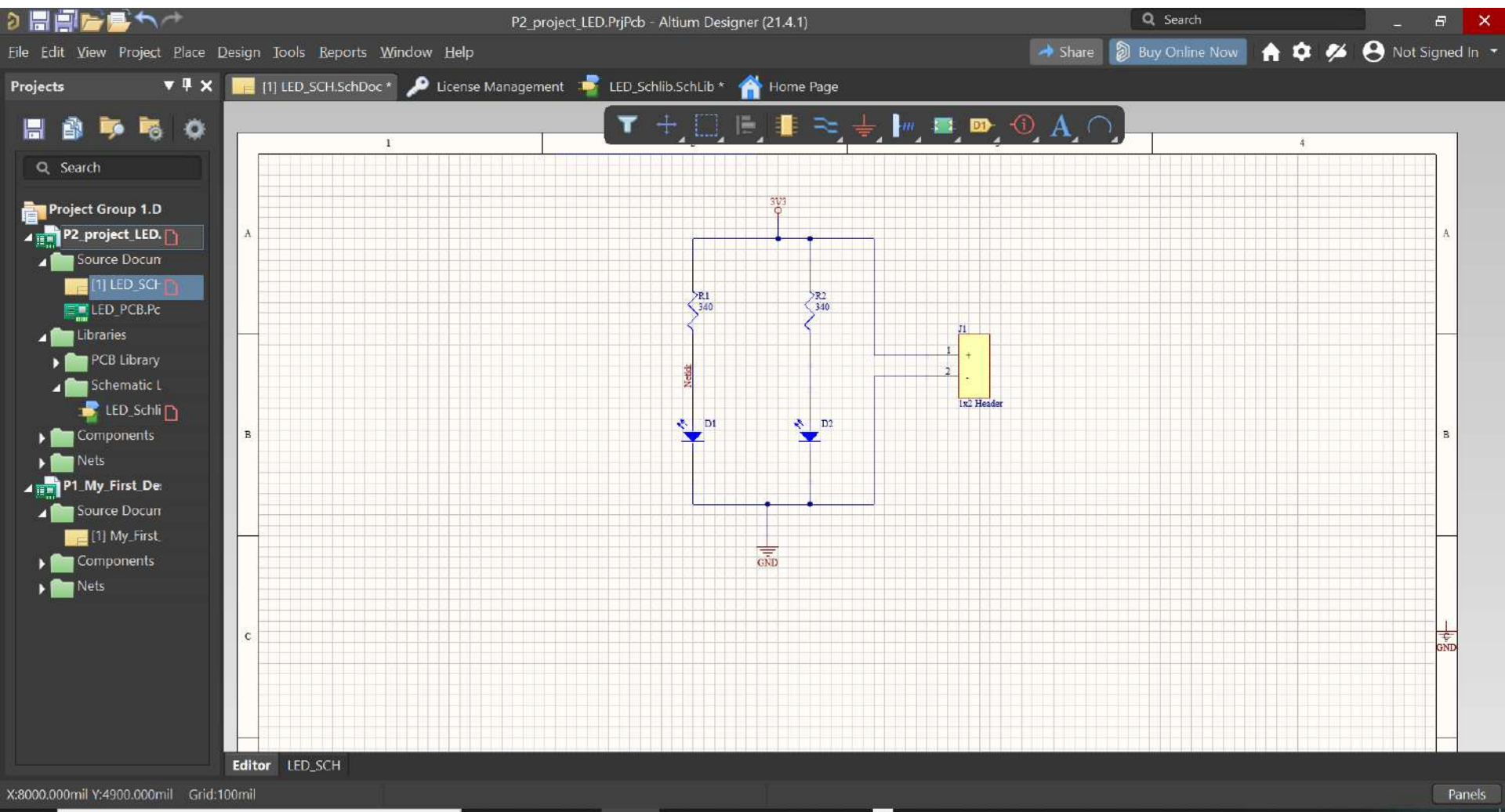
- When we have finished the composition of our schematic library, it is time for drawing of our design.
- Regarding power supply, we will include a 3,3V circle and a GND Power. NOTE: If you will need extra power supplies, it is suggested to copy and paste instead of to create a new one.
- Nets connected to a common point, share the net name.
- We can change the Net Name with Net Label (Place Menu)

Before finishing

- Before annotating the design, errors (red colours) are showed in the sheet.

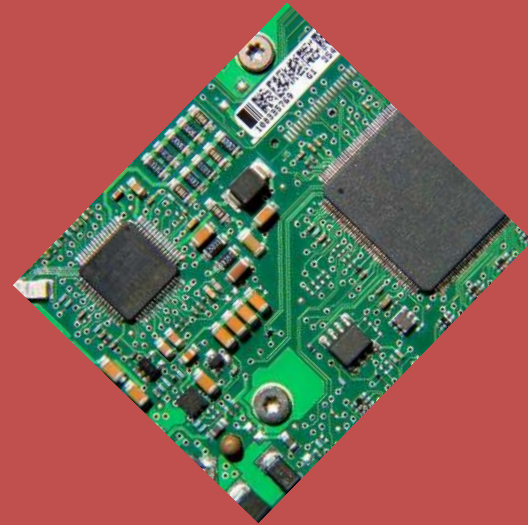


Final schematic view



Introduction to Printed Circuit Board (PCB) design with ALTIUM

3.- Hierarchical Design

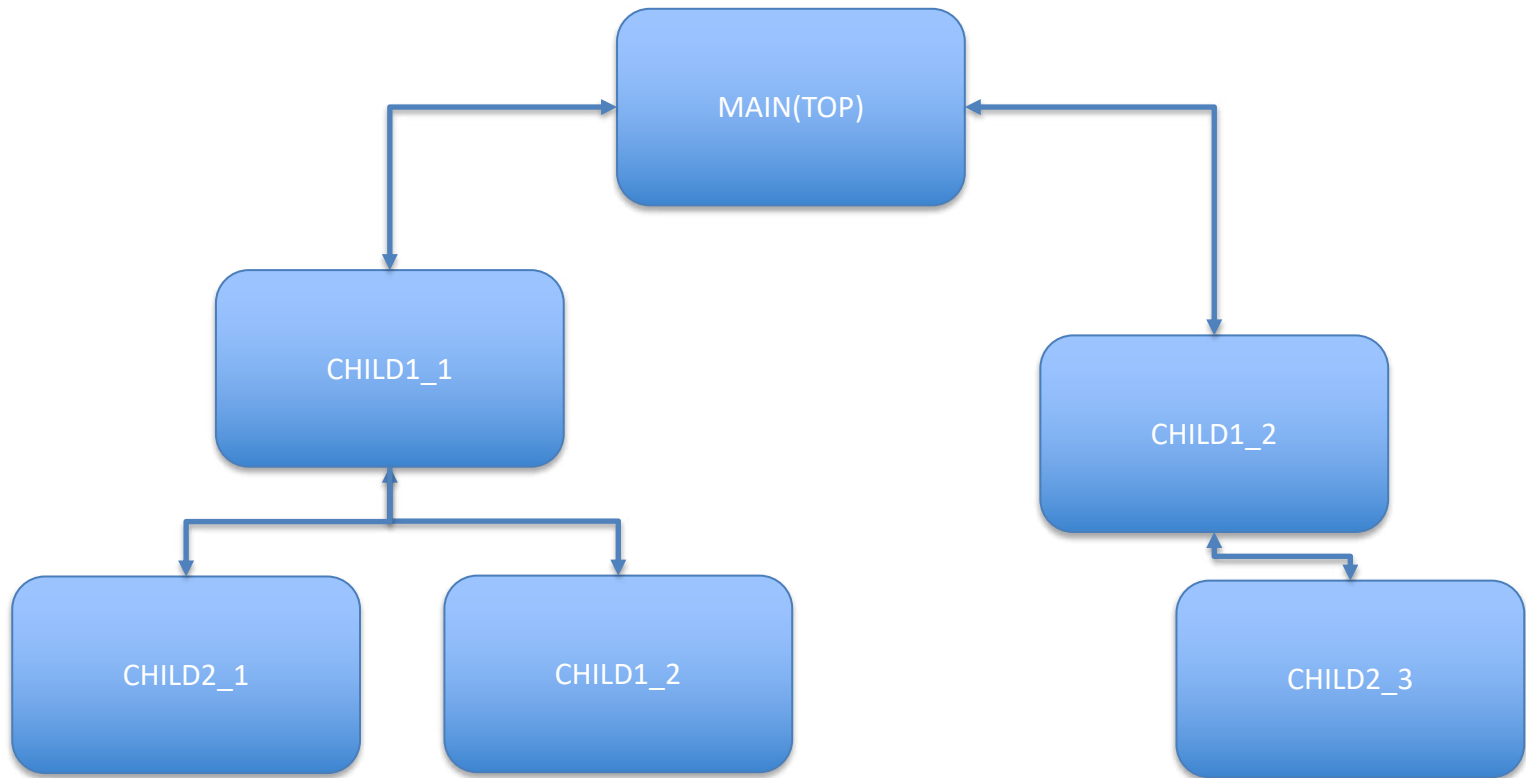


Hierarchical concept

- ❑ When a design is complex and big, it is usual to split in several sheets.
- ❑ Two approaches are possible:
 - ❑ Flat Design (horizontal connectivity)
 - ❑ Hierarchical Design (Top-Down approach).
- ❑ The second one is the most useful and used because has more flexibility and co-working approach is validated.

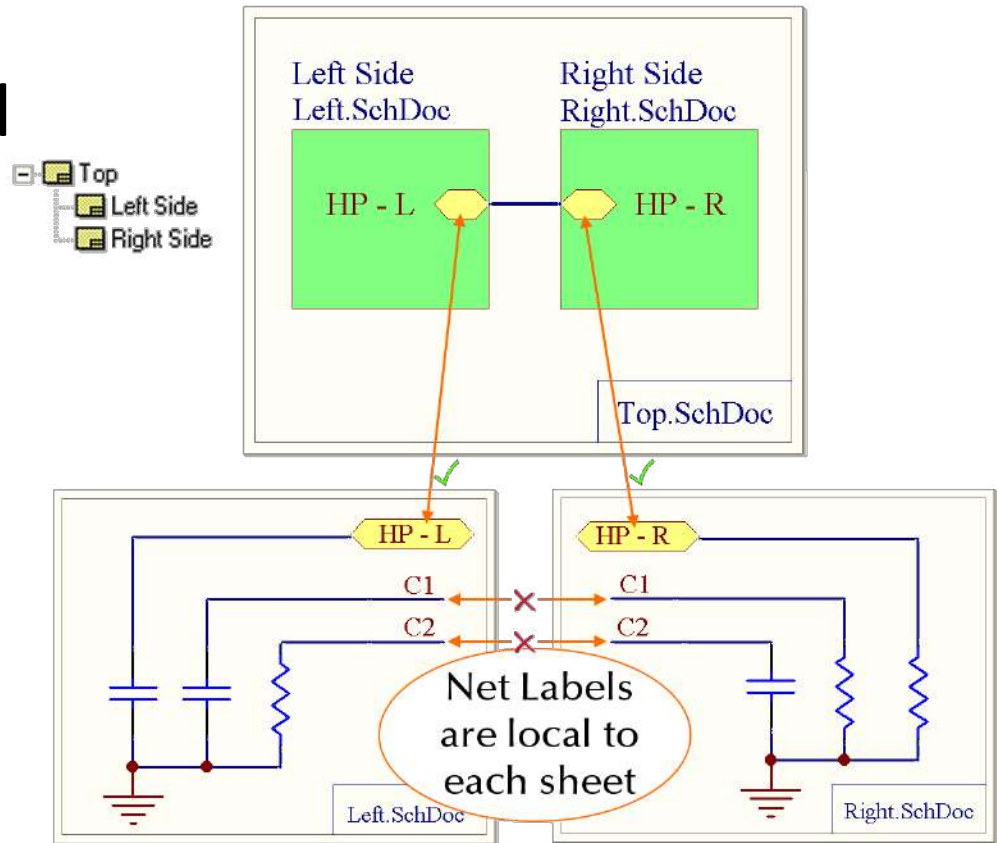
Hierarchical concept

- A top-down topology will have a view like this (maximum 3 levels):



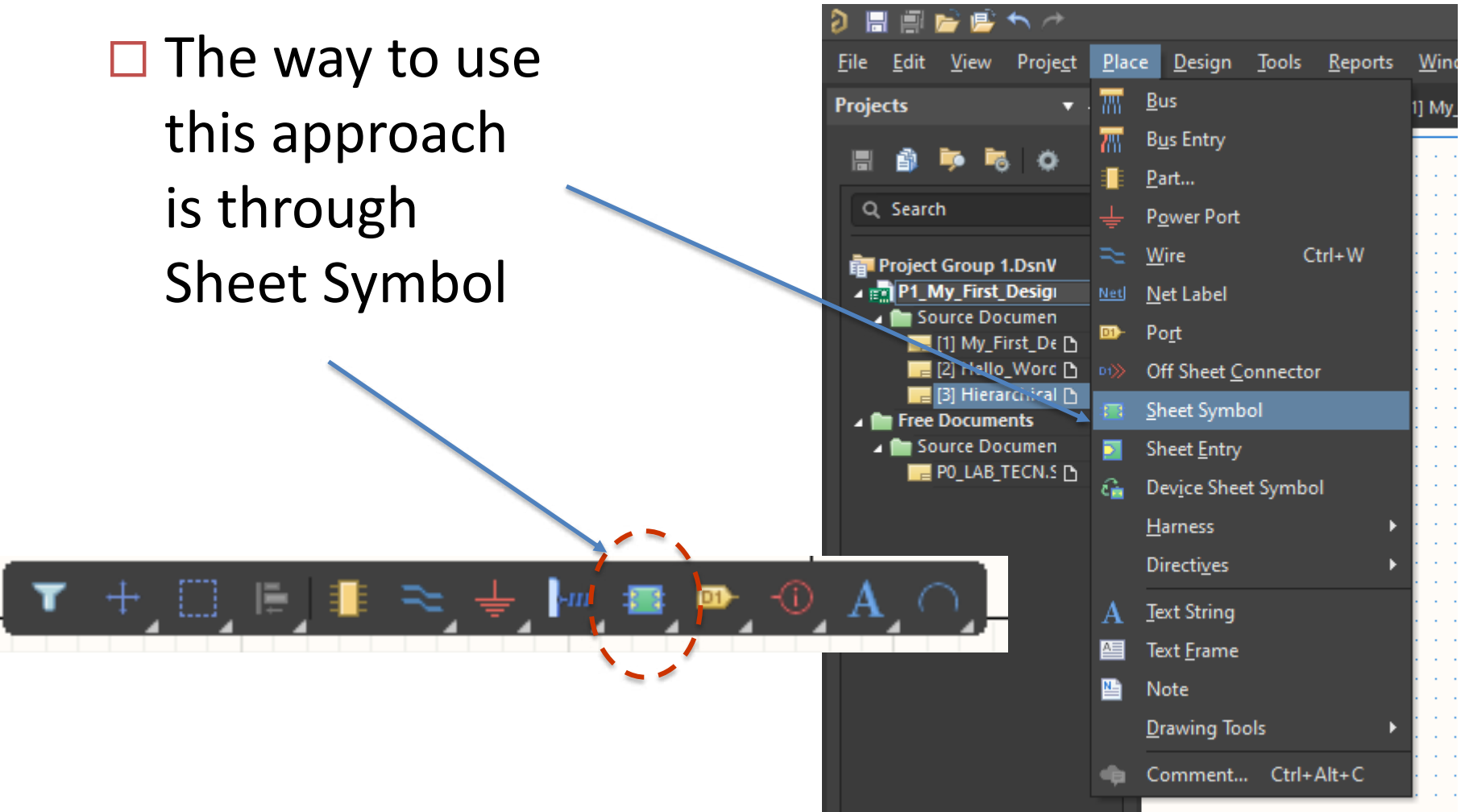
Hierarchical concept

- The connexions between sheets will be done through Ports.
- Local nets are not connected with other sheets.



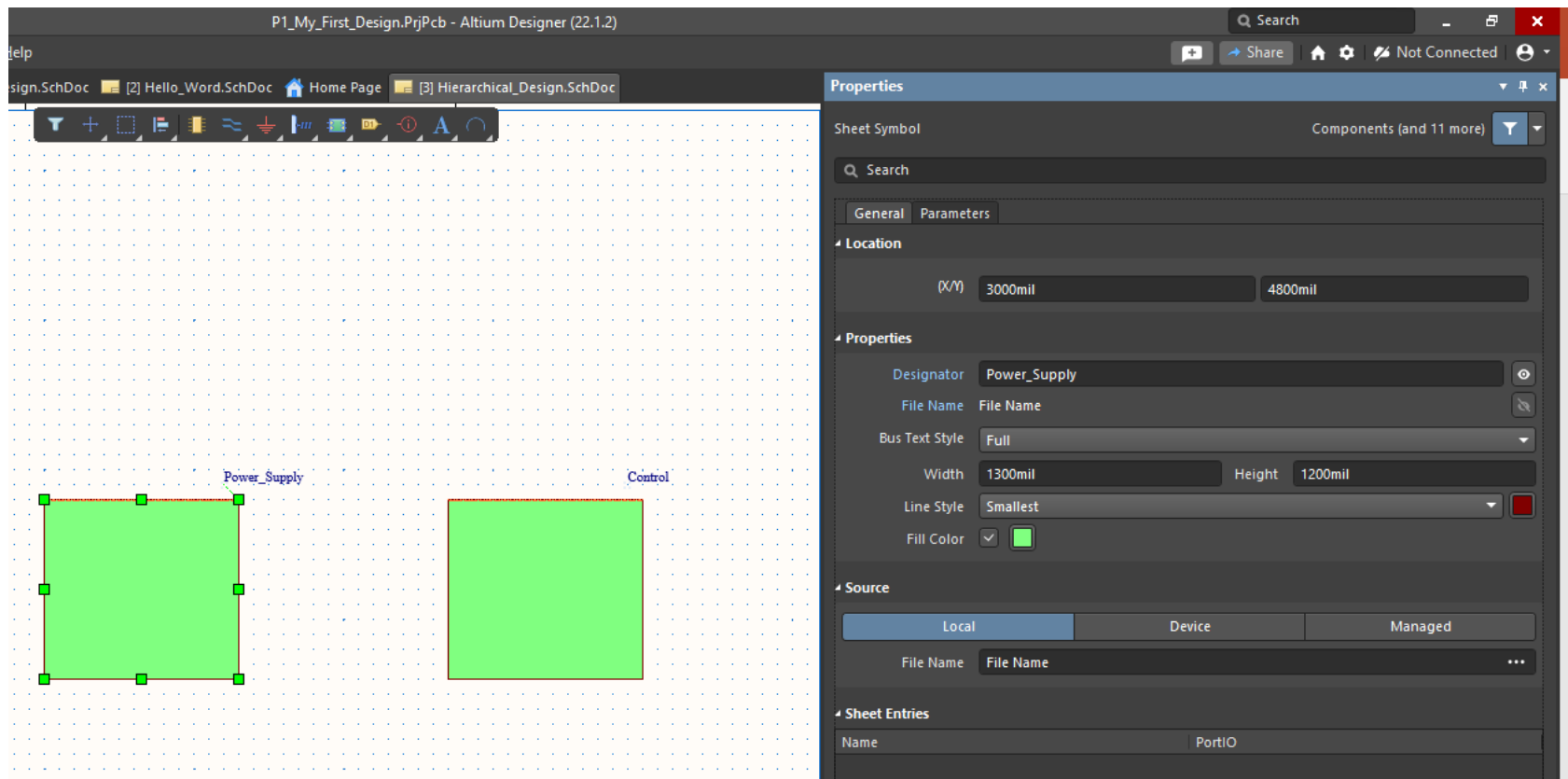
How to create a hierarchical design?

- The way to use this approach is through Sheet Symbol



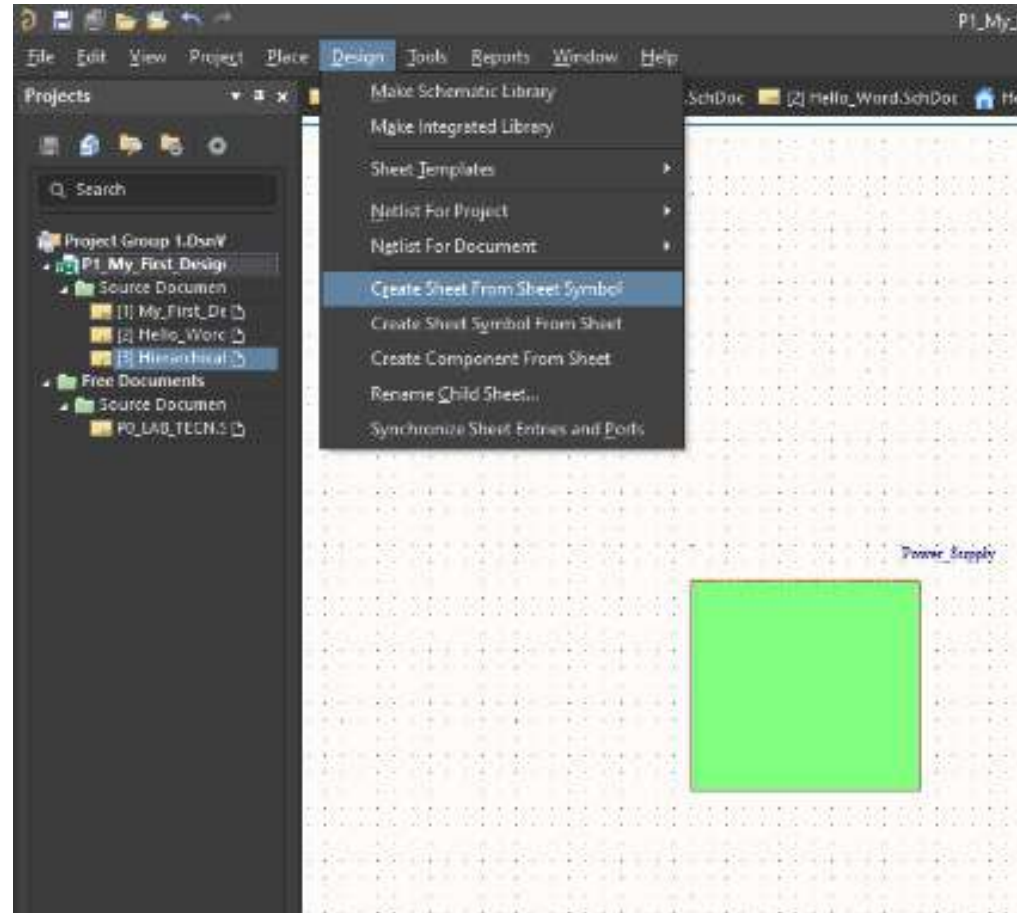
How to create a hierarchical design?

- We will change the sheet name in Properties Panel.



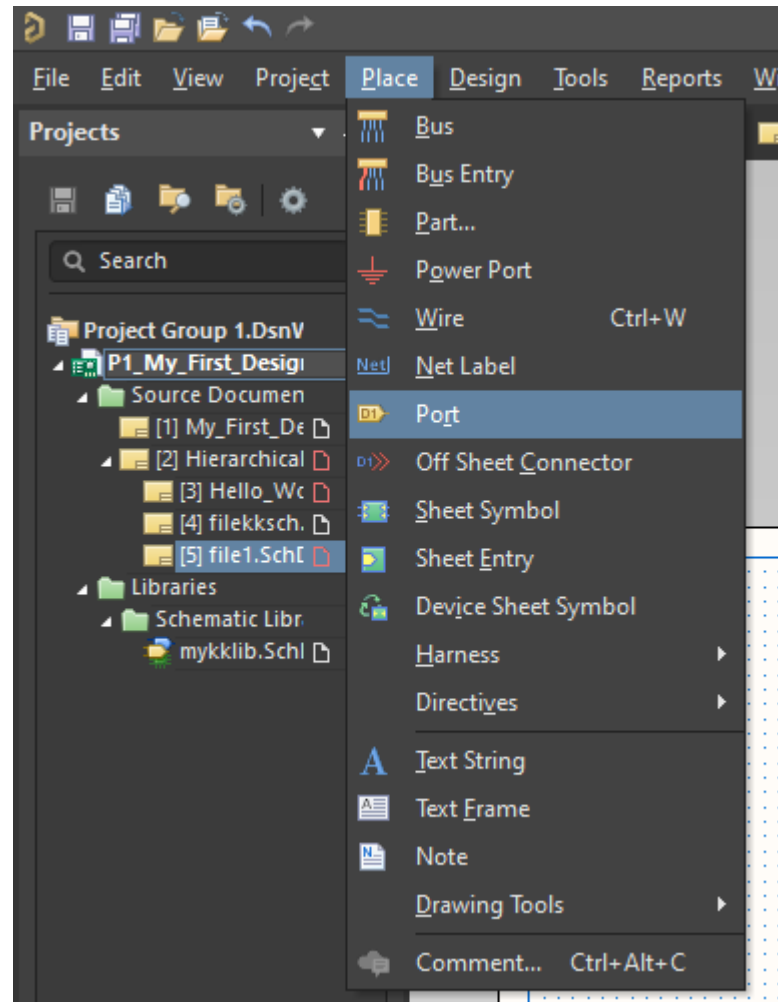
How to create a hierarchical design?

- Then, we need to link the symbols with new sheets



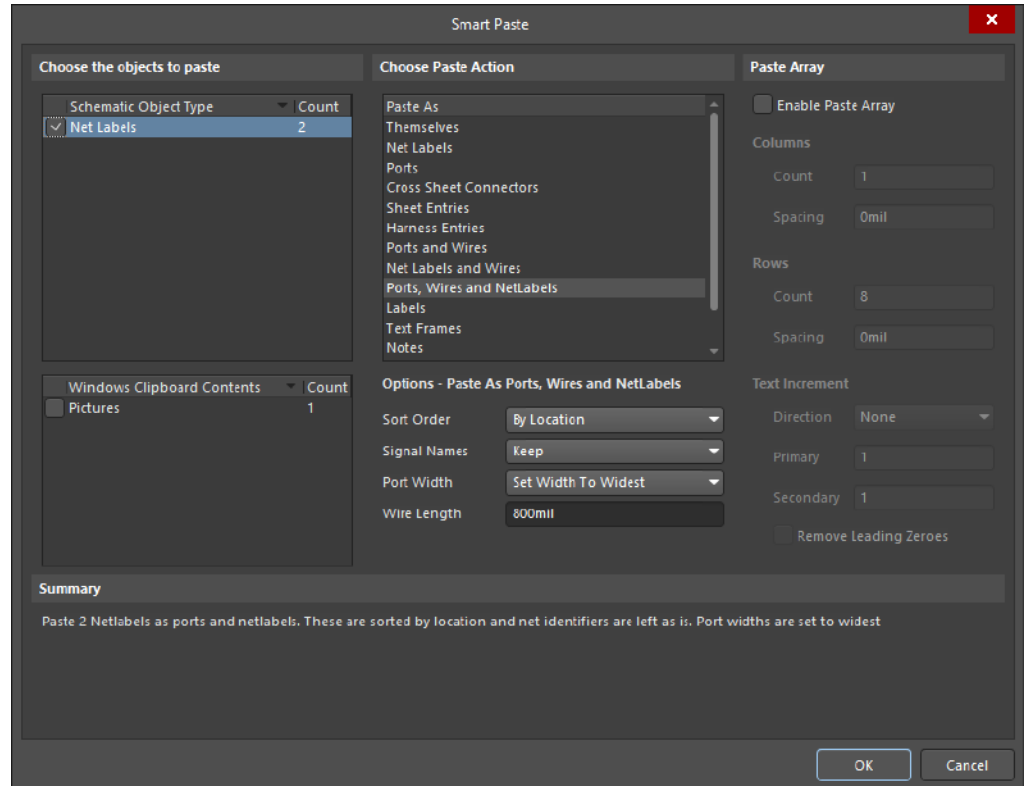
Creating the Ports

- When sheet file is finished, ports must be included.
- One by one can be done using **port** option inside menu Place.



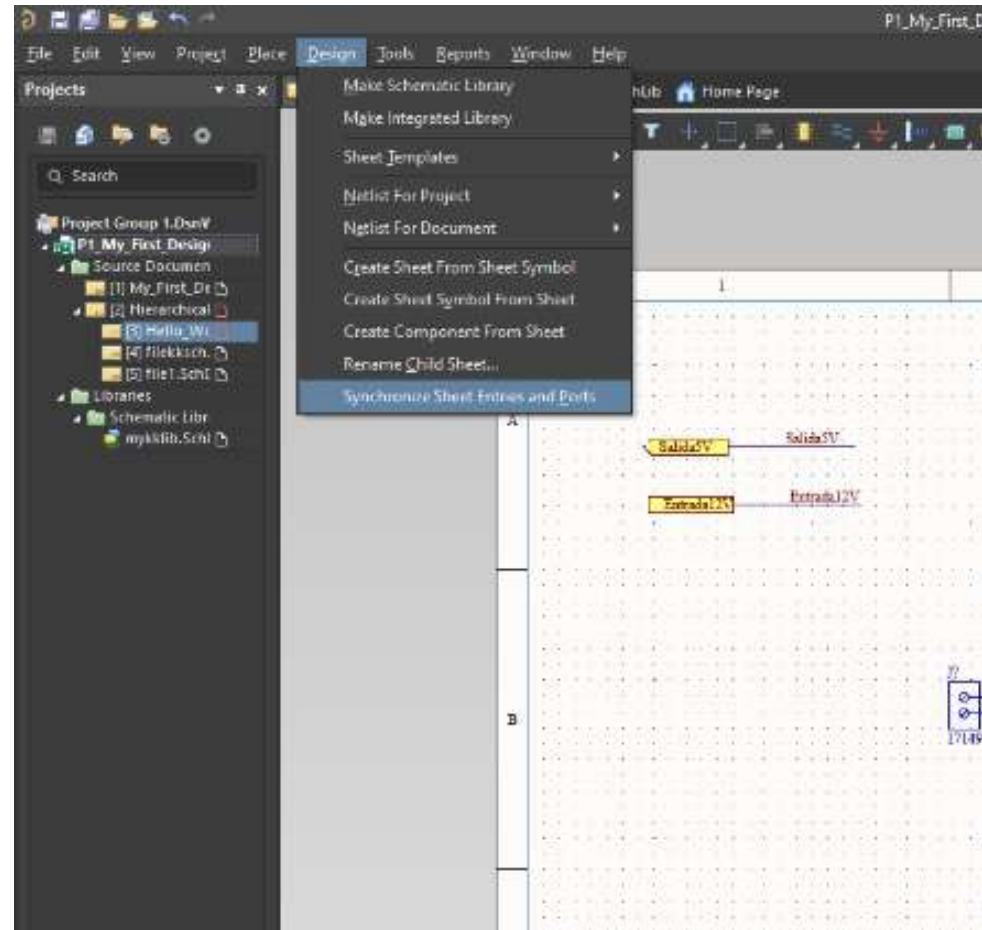
Creating the Ports

- Another way to create several ports in just one click: Smart Paste (Edit Menu),



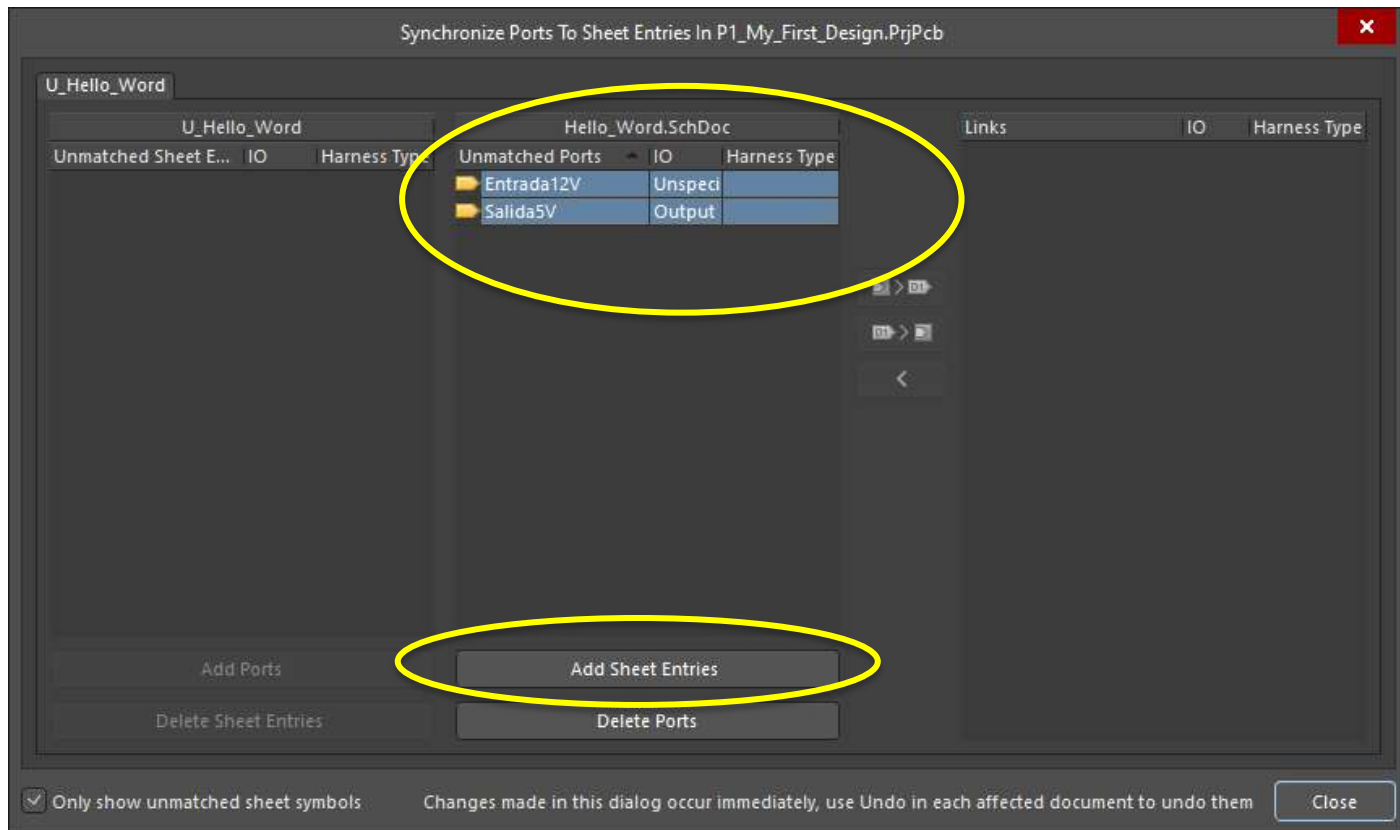
Creating the Ports

- Then, we should include the ports in the Sheet Symbol.
- This can be an automatic process (synchronizing).

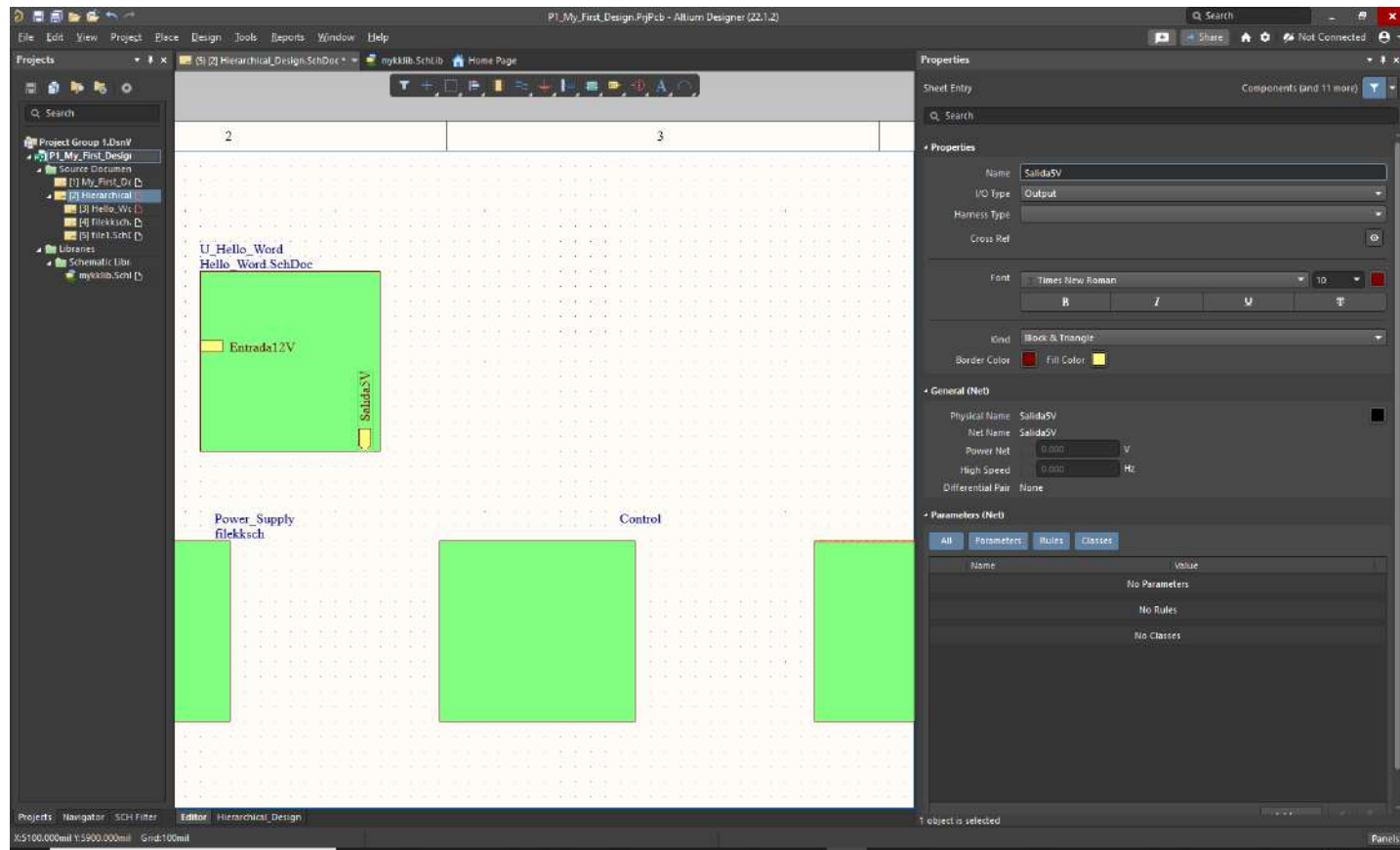


Creating the Ports

- Adding the ports to our symbol sheet

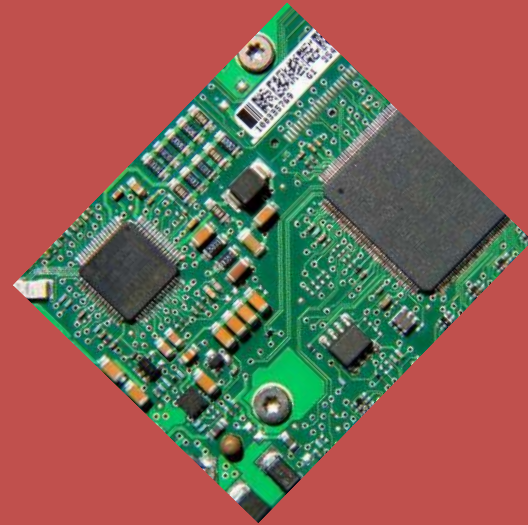


Creating the Ports



Introduction to Printed Circuit Board (PCB) design with ALTIUM

4.- Basic PCB concepts

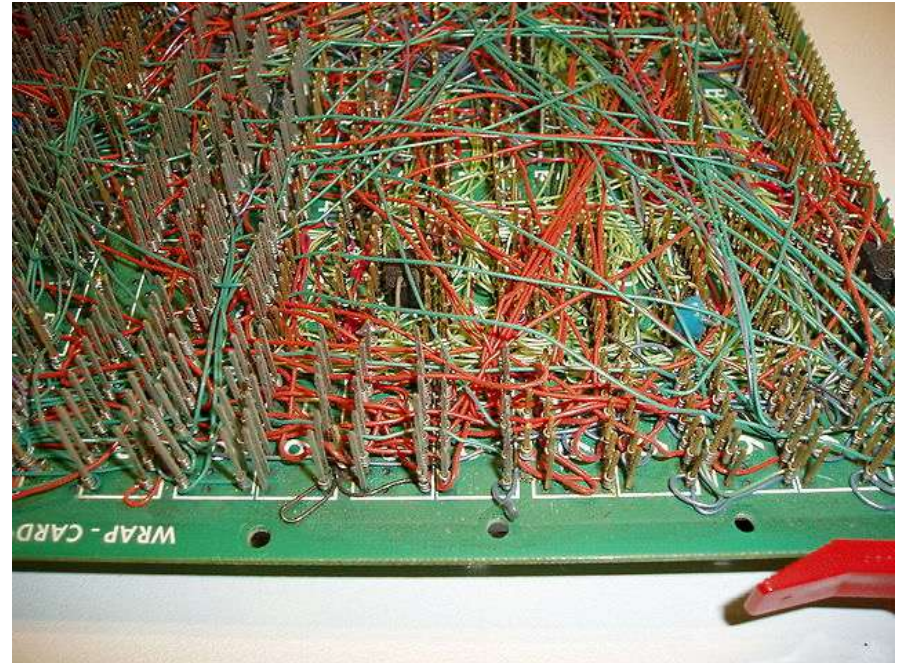


Basic PCB concepts

- What is a printed circuit board (PCB)? A printed circuit board or PCB, is a plate or board used for placing the different elements that conform an electrical circuit that contains the electrical interconnections between them. [1]
- The different “sides” of a PCB is called layer. The simplest PCB is composed by 1 layer. The most common one uses at least 2 layers

PCB vs other options

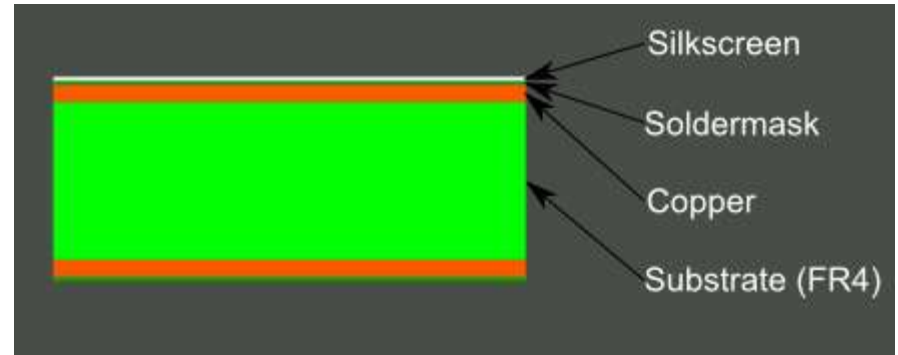
- Why a PCB? If we didn't have it, we should make circuits as the figure → electrical, connection and thermal problems. [2]



PCB basic concepts

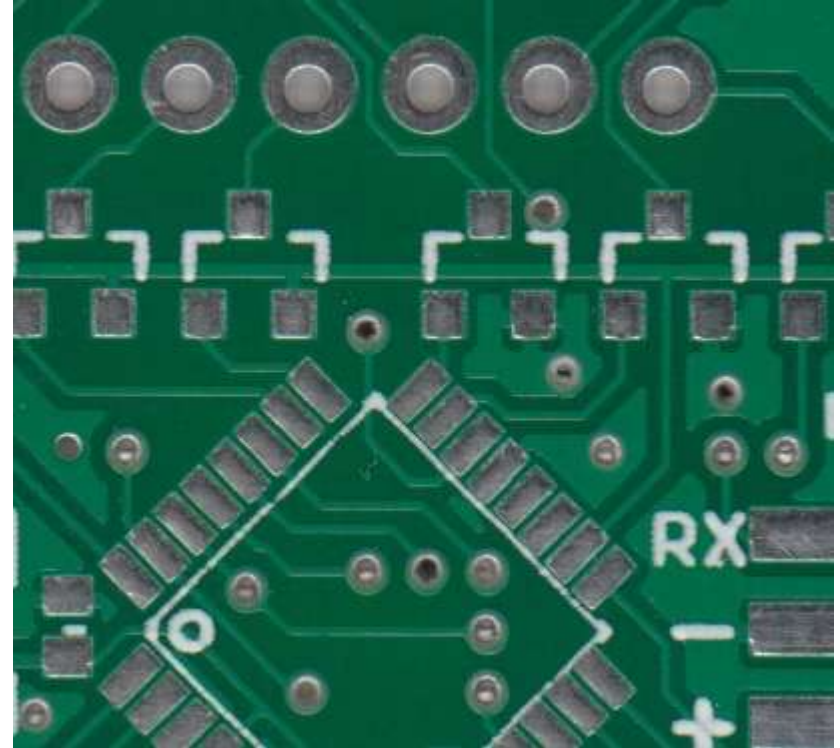
Layers of a PCB

- ❑ The figure shows a 2 layer PCB. [2]
- ❑ Bottom layer and top layer cannot be composed at the same level.
- ❑ FR4 is the substrate is usually fiberglass.
- ❑ Copper is the conductor element.



PCB basic concepts

- ❑ **Soldermask** is the layer on top of the copper.
- ❑ Usually the green colour is used.



PCB basic concepts

Silkscreen

- The white silkscreen layer is applied on top of the soldermask layer. The silkscreen adds letters, numbers, and symbols to the PCB that allow for easier assembly and indicators for humans to better understand the board. [2]



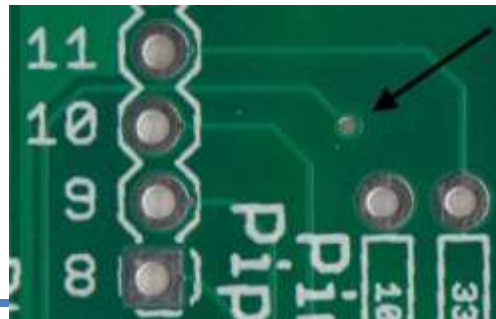
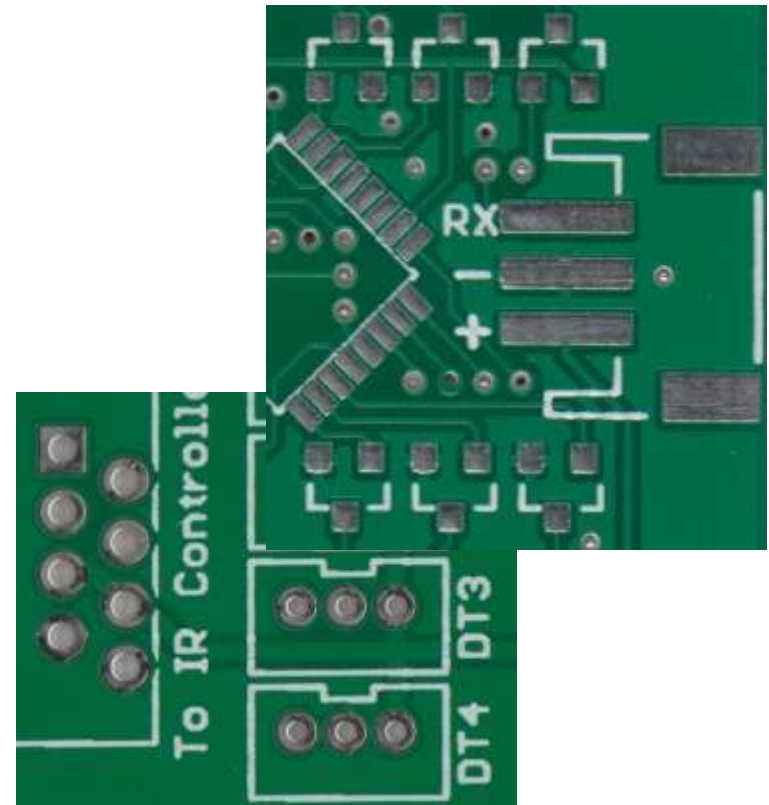
PCB basic concepts

Pad concept.

- A portion of exposed metal on the surface of a board to which a component is soldered. [2]

Via concept.

- A hole in a board used to pass a signal from one layer to another.



A PCB composed by multiple layers

□ Layer Stackup

CAD Layer (conductive and nonconductive)	CAD Layer description
1	Top silkscreen/overlay (nonconductive)
2	Top soldermask (nonconductive)
3	Top paste mask (nonconductive)
4	Layer 1 (conductive)
5	Sustrate (nonconductive)
6	Layer 2 (conductive)
...	...
n-1	Sustrate (nonconductive)
n	Layer n (conductive)
n+1	Bottom paste mask (nonconductive)
n+2	Bottom solder mask (nonconductive)
n+3	Bottom silkscreen/overlay (nonconductive)

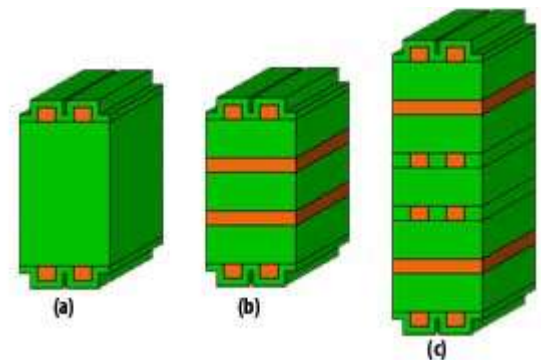


Fig 3. Example of 3 different PCB stackups: 2 layers (a), 4 layers (b) and 6 layers (c)

Types of footprint

Component packages (footprints)

- There are 3 main types of footprints:
- Thru Hole
- Surface Mounting
- BGA (Ball Grid Array)



Types of footprint

- ❑ SMDs are divided in several groups:
- ❑ SOIC (Small Outline Integrated Circuit)
- ❑ SOP (Small Outline Package)

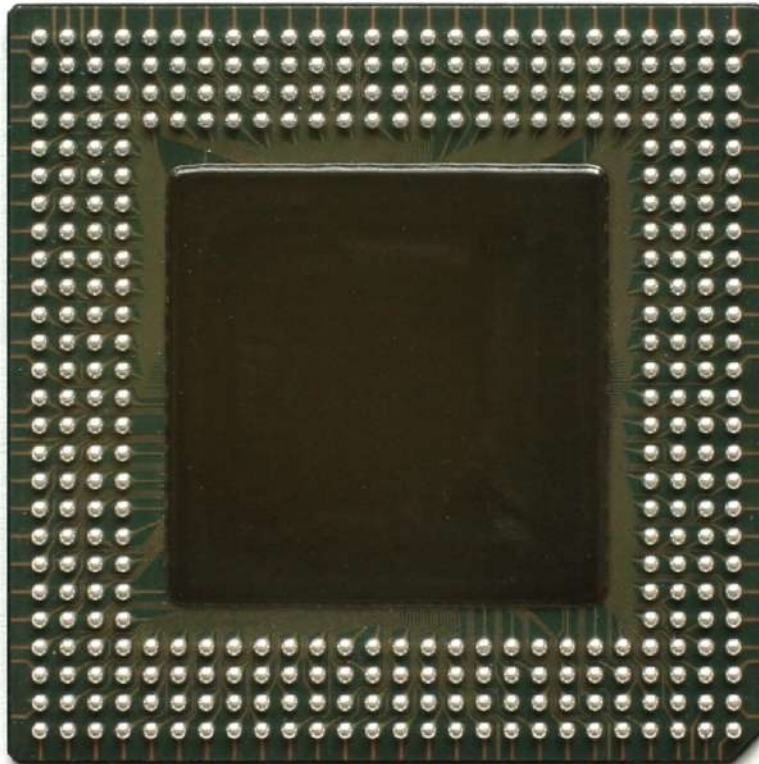


- ❑ QFP (Quad Flat pack)



Types of footprint

□ BGA



Types of footprint

- A summary of footprints (I/II)

IC Chip Packages			
	CBGA - Ceramic Ball Grid Array		PQFP - Plastic Quad Flat Pack
	CCGA - Ceramic Column Grid Array		PSOP - Power Small Outline Package
	CerDIP - Ceramic Dual-in-Line Package		QFN - Quad Flat No Leads Package
	CerPack - Ceramic Package		QSOP - Quarter Size Outline Package
	CLCC - Ceramic Leadless Chip Carrier		SBDIP - Sidebrazed Dual-in-Line Package
	CPGA - Ceramic Pin Grid Array		SC-70 - Small Outline Transistor
	CQFP - Ceramic Quad Flat Pack		SIP - Single-In-Line Package
	D2PAK or DPAK - Double Decawatt Package		SOIC - Small Outline IC Package
	D3PAK - Decawatt Package 3		SOJ - Small Outline J-Lead Package
	DFN - Dual Flat No Leads Package		SOT-23 - Small Outline Transistor
	DPAK - Decawatt Package		SPDIP - Shrink Plastic Dual-in-Line Package
	FBGA - Fine-Pitch Ball Grid Array		SSOP - Shrink Small Outline Package

Types of footprint

□ A summary of footprints (II/II)

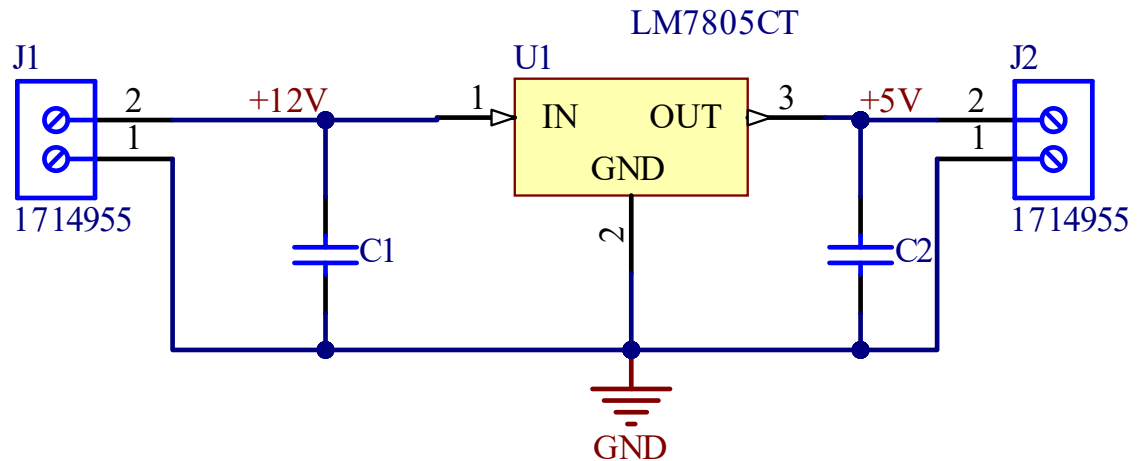
	JLCC - J-Leaded Ceramic Chip Carrier		TDFN - Thin Dual Flat No Leads Package
	LFBGA - Low Profile Fine-Pitch Ball Grid Array		TFBGA - Thin Fine-Pitch Ball Grid Array
	LGA - Land Grid Array		TQFN - Thin Quad Flat No Leads Package
	LQFP - Low-Profile Quad Flat Package		TQFP - Thin Quad Flat Pack
	MLP - Micro Leadframe Package		TSOP - Thin Small Outline Package
	MQFP - Metric Quad Flat Package		TSSOP - Thin Shrink Small Outline Package
	MSOP - Micro Small Outline Package		UTDFN - Ultra Thin Dual Flat No Leads Package
	PBGA - Plastic Ball Grid Array		UTQFN - Ultra Thin Quad Flat No Leads Package
	PDIP - Plastic Dual-in-Line Package		VFBGA - Very Thin Fine-Pitch Ball Grid Array
	PLCC - Plastic Leaded Chip Carrier		VSOP - Very Small Outline Package
	PPGA - Plastic Pin Grid Array		VSOP - Very Small Outline Package
	PQFN - Power Quad Flat No Leads Package		XQFN - Extreme Thin Quad Flat No Leads Package

Several references to know more

- [1] Electrosoft Engineering. “Concepts and terminology used in Printed Circuit Boards (PCB)”
<http://www.pcb.electrosoft-engineering.com/04-articles-custom-system-design-and-pcb/01-printed-circuit-board-concepts/printed-circuit-board-pcb-concepts.html>
- [2] Sparkfun Start Something “PCB Basics”.
<https://learn.sparkfun.com/tutorials/pcb-basics/all>
- [3] MadPCB “Chip Package”
<https://madpcb.com/glossary/chip-package/>

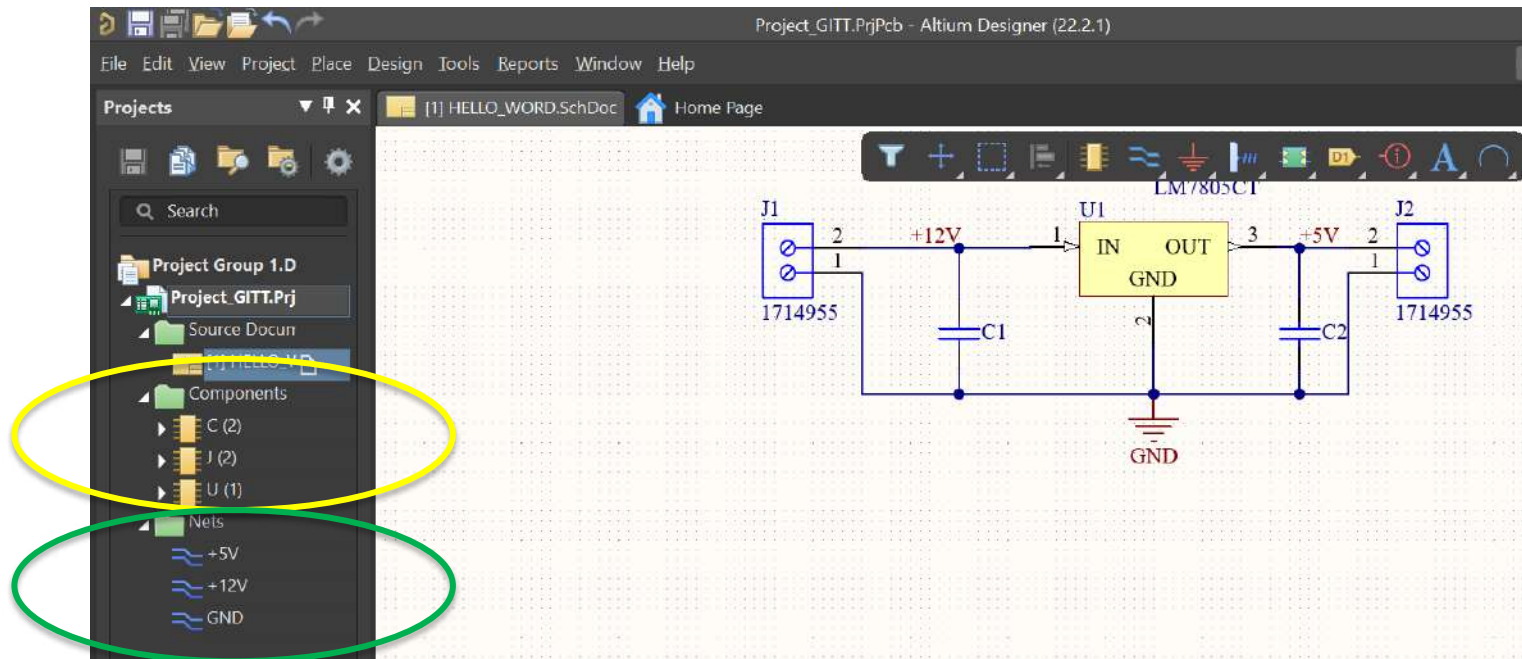
My first Layout

- Based on a simple schematic, the plan is to implement a basic layout design using commercial footprints and devices.



My first Layout

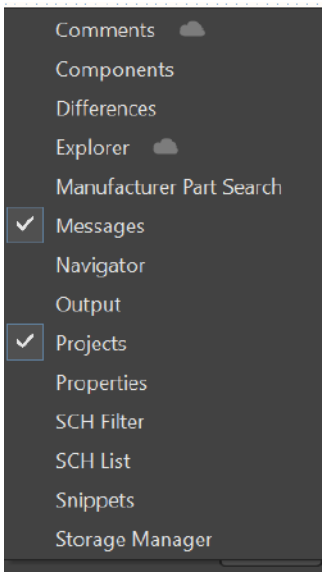
- Before doing anything else we can check the components and nets used in the Project.



PCB OVERVIEW

My first Layout

- Additionally, we will check any type of electrical error (Project → Validate PCB Project).
- Messages can be displayed through panel tab.



Class	Document	Source	Message	Time	Date	No.
[Warning]	HELLO_WORD.SchDoc	Compiler	Net +12V has no driving source (Pin C1-2, Pin J1-2, Pin U1-1)	14:27:04	20/03/2022	1
[Warning]	HELLO_WORD.SchDoc	Compiler	Off grid C1 at 122.46mm,149.92mm	14:27:04	20/03/2022	2
[Warning]	HELLO_WORD.SchDoc	Compiler	Off grid C2 at 157.46mm,149.92mm	14:27:04	20/03/2022	3
[Warning]	HELLO_WORD.SchDoc	Compiler	Off grid J1 at 109.92mm,154.92mm	14:27:04	20/03/2022	4
[Warning]	HELLO_WORD.SchDoc	Compiler	Off grid J2 at 170.08mm,154.92mm	14:27:04	20/03/2022	5
[Warning]	HELLO_WORD.SchDoc	Compiler	Off grid Pin C1-1 at 125mm,147.38mm	14:27:04	20/03/2022	6
[Warning]	HELLO_WORD.SchDoc	Compiler	Off grid Pin C2-1 at 160mm,147.38mm	14:27:04	20/03/2022	7
[Warning]	HELLO_WORD.SchDoc	Compiler	Off grid Pin U1-1 at 132.3mm,160.16mm	14:27:04	20/03/2022	8
[Warning]	HELLO_WORD.SchDoc	Compiler	Off grid Pin U1-3 at 157.7mm,160.16mm	14:27:04	20/03/2022	9
[Warning]	HELLO_WORD.SchDoc	Compiler	Off grid U1 at 137.38mm,162.7mm	14:27:04	20/03/2022	10
[Info]	Project_GITT.PrjPcb	Compiler	Compile successful. no errors found.	14:27:04	20/03/2022	11

Details

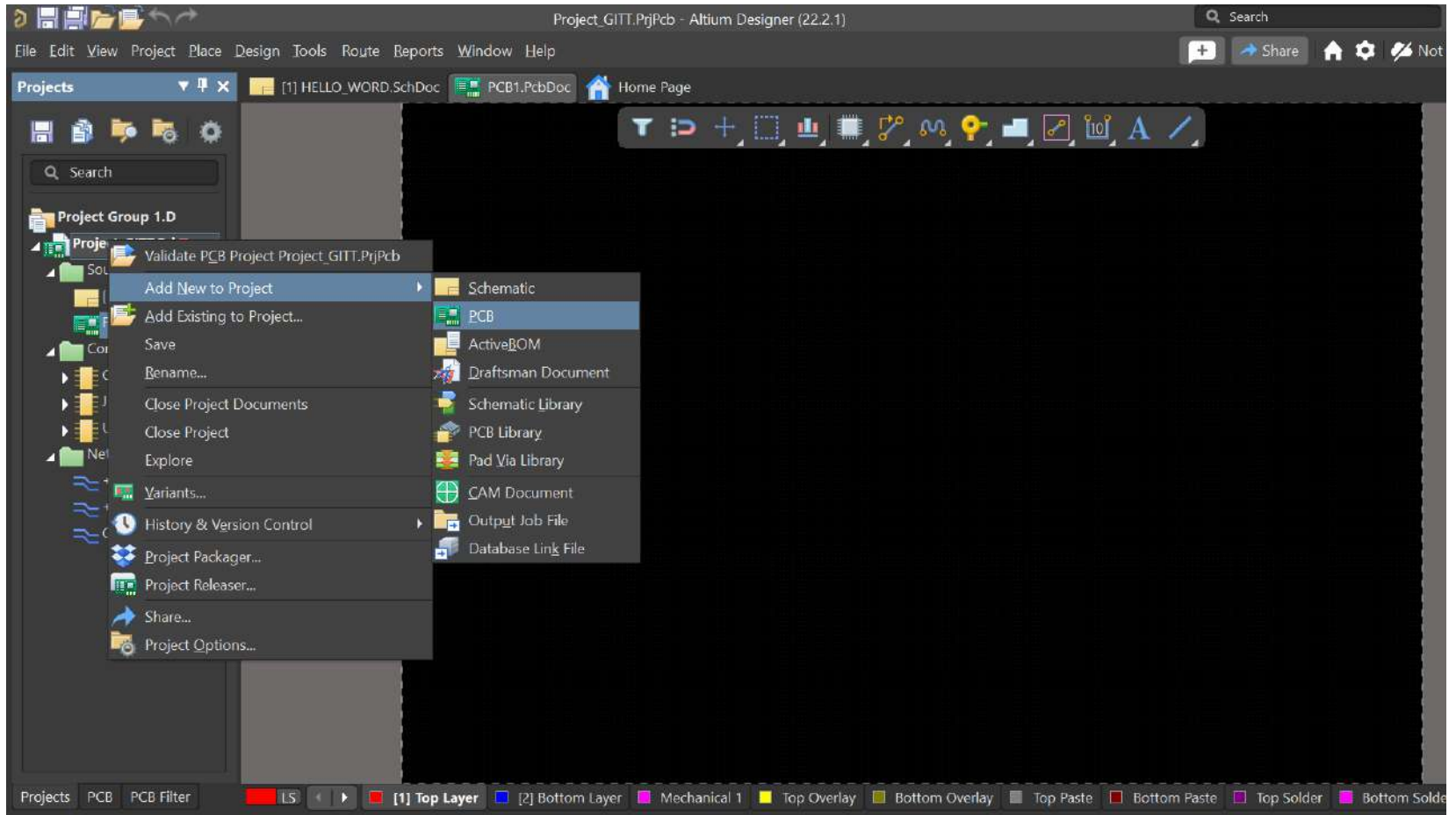
✖ Net +12V has no driving source (Pin C1-2, Pin J1-2, Pin U1-1)

Wire NetC1.2

Pin C1-2

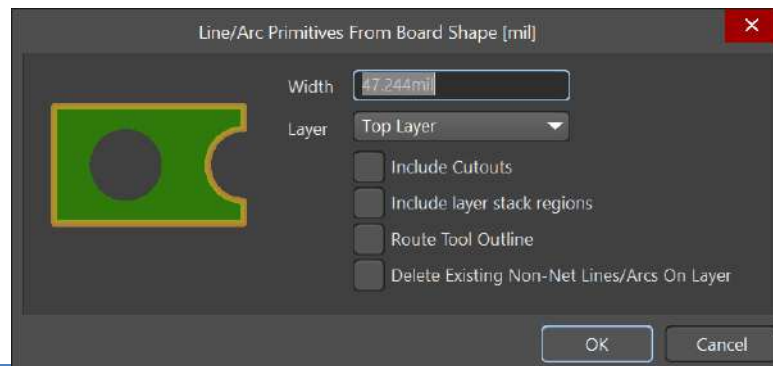
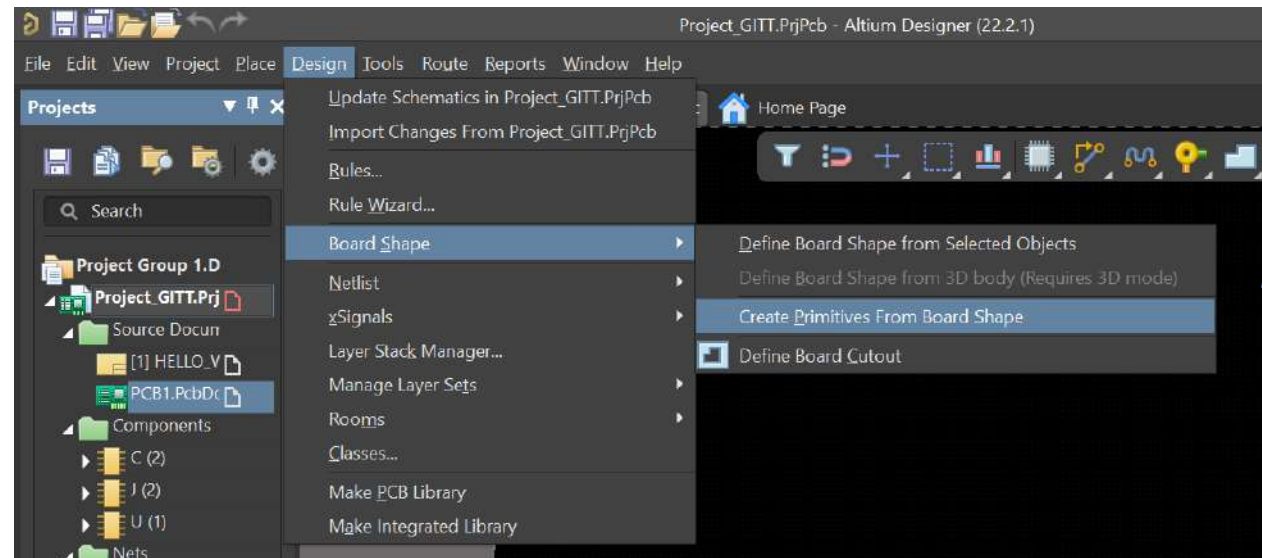
My first Layout

□ Creating a PCB “Sheet”

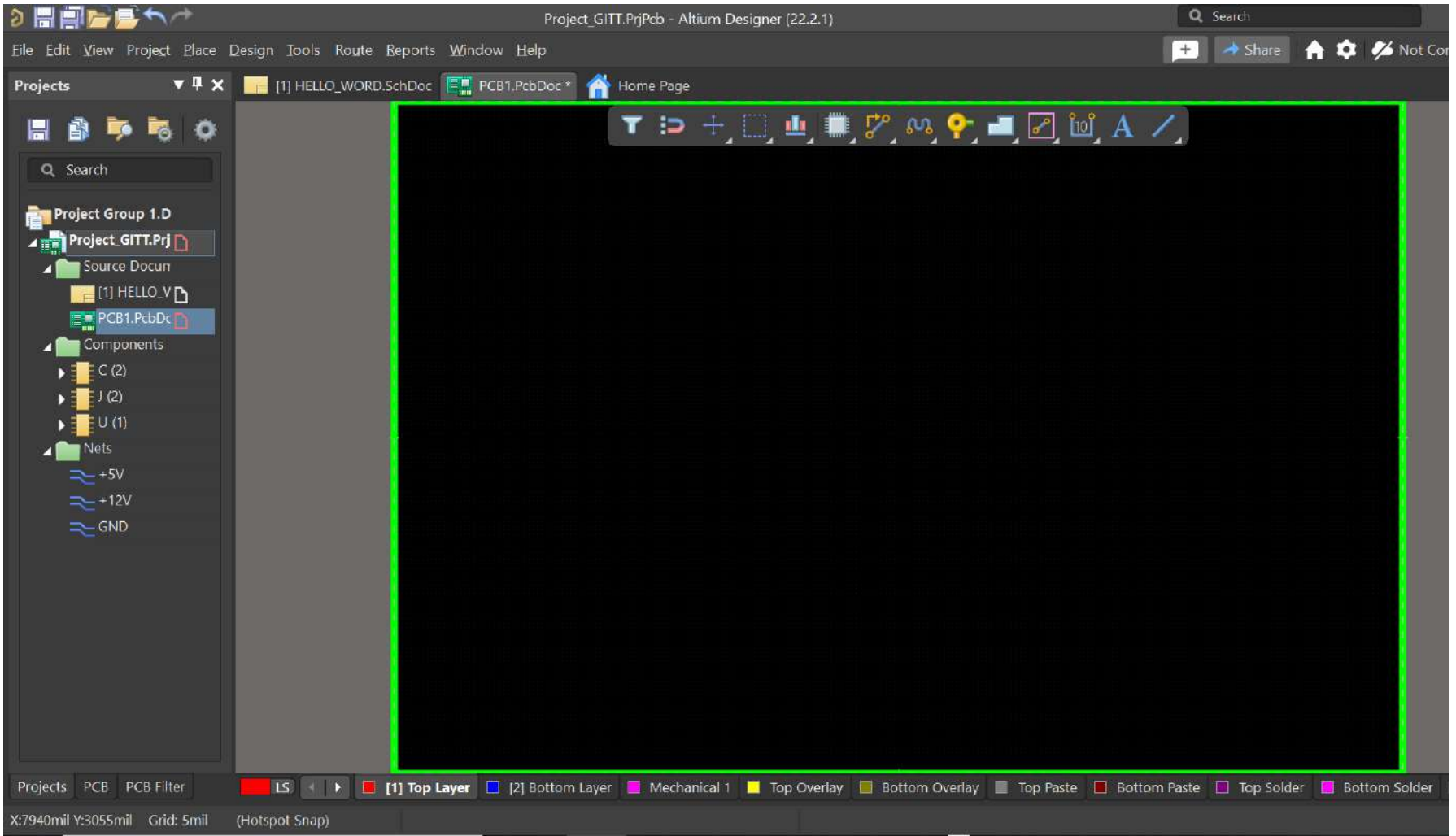


My first Layout

□ Defining the size of the PCB

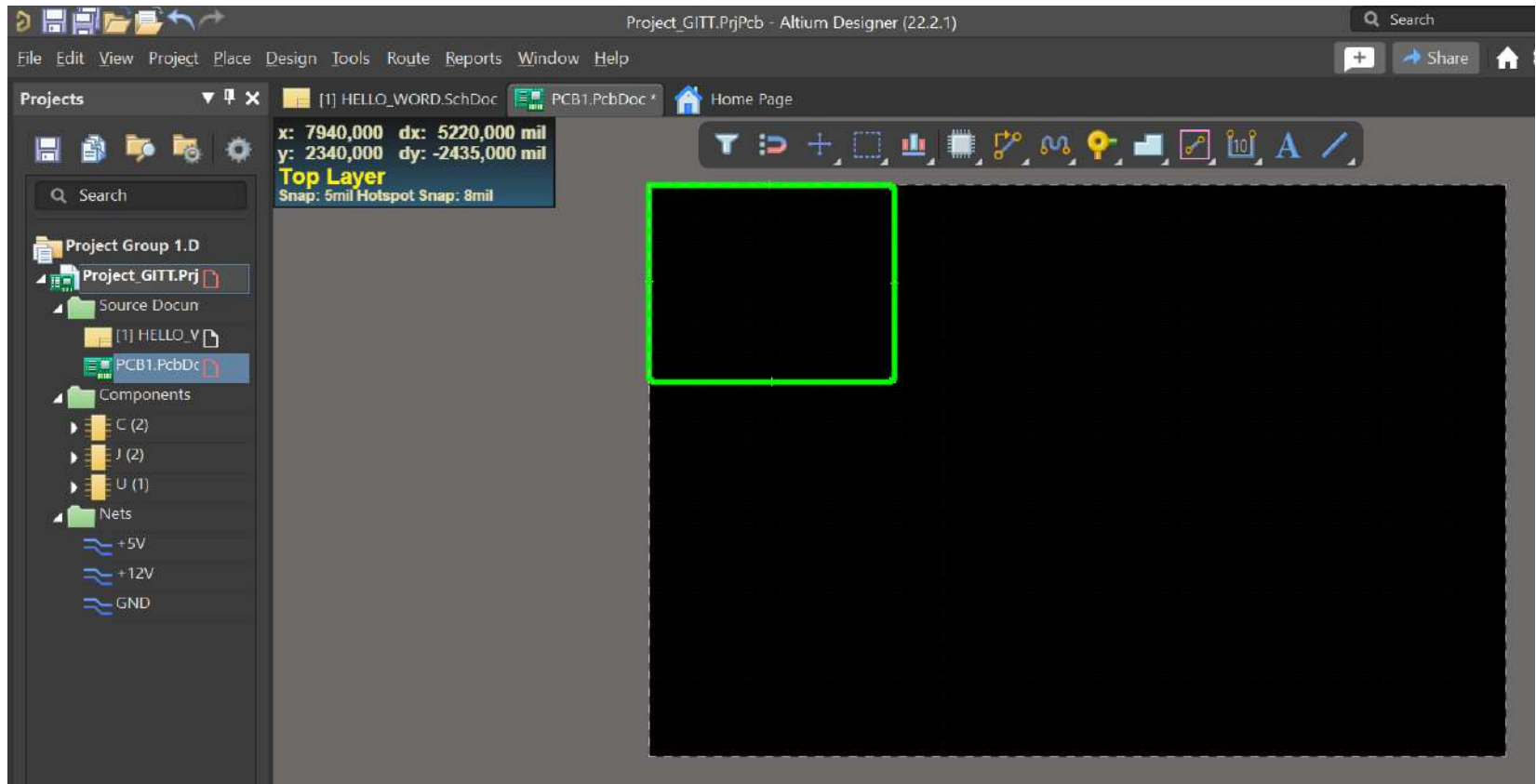


My first Layout



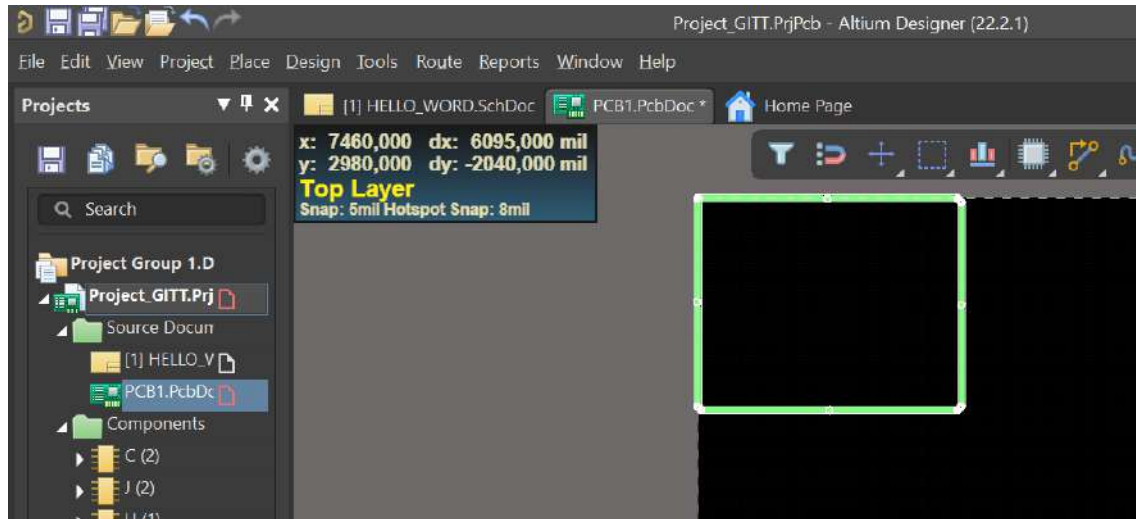
My first Layout

- The default are for our PCB is too big. Then, we should reduce the size moving the green rectangle.

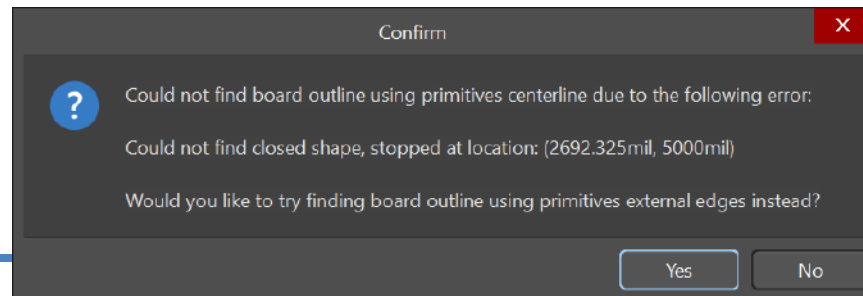


My first Layout

- So, we should click each border to hold on....

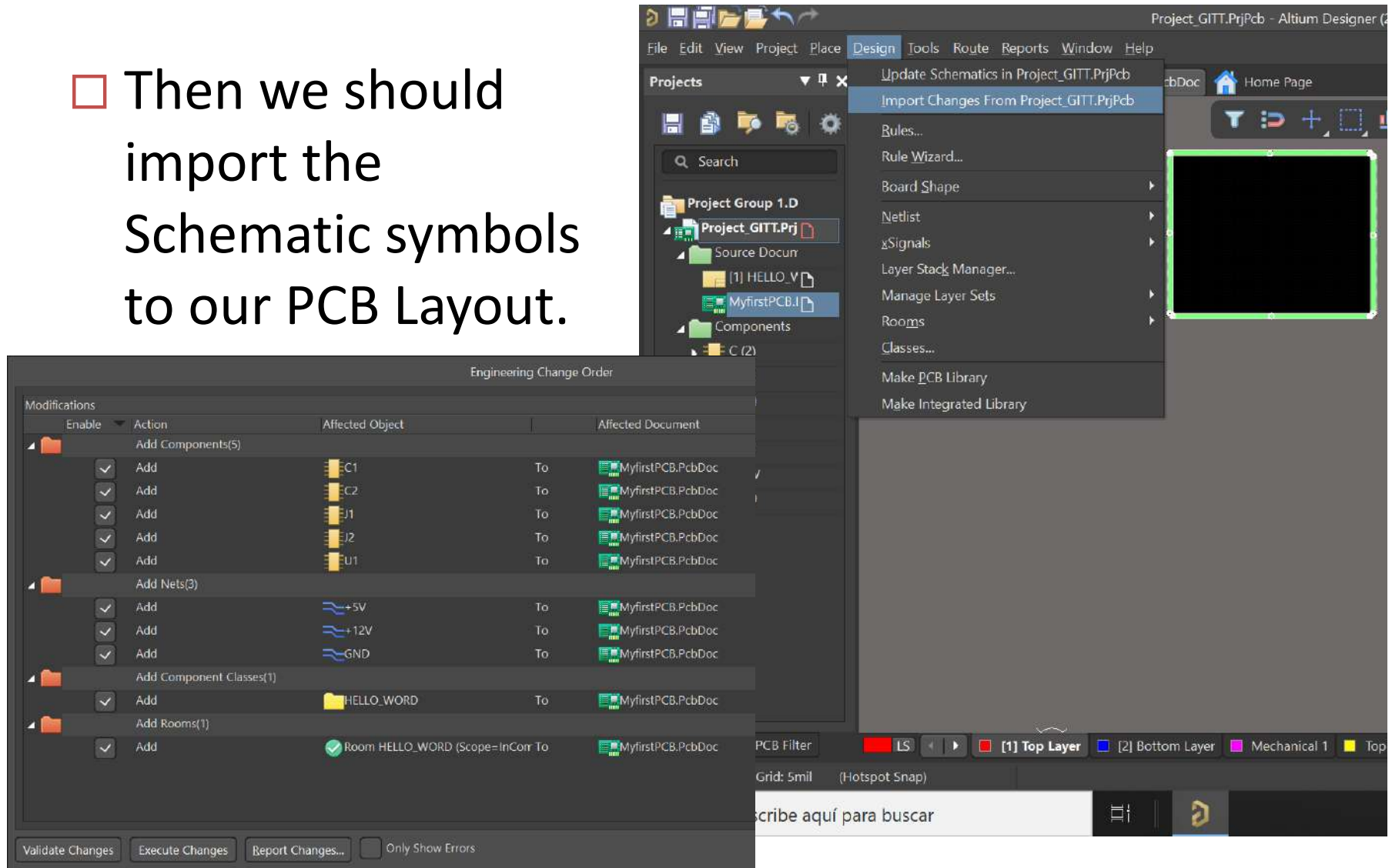


- ...and create a new size for our board. (Design → Board Shape → Define Board Shape from Selected obj.)



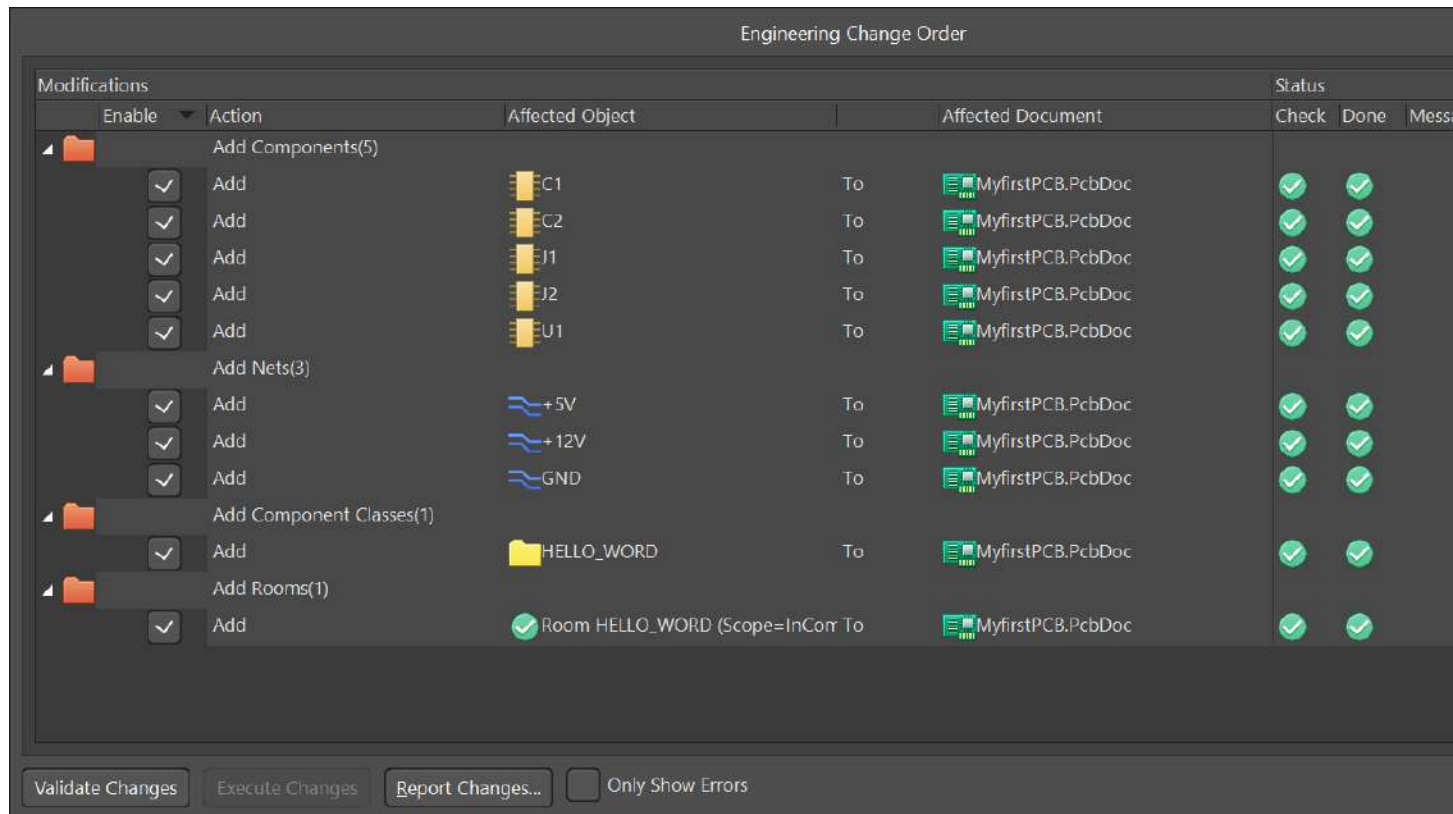
My first Layout

- Then we should import the Schematic symbols to our PCB Layout.



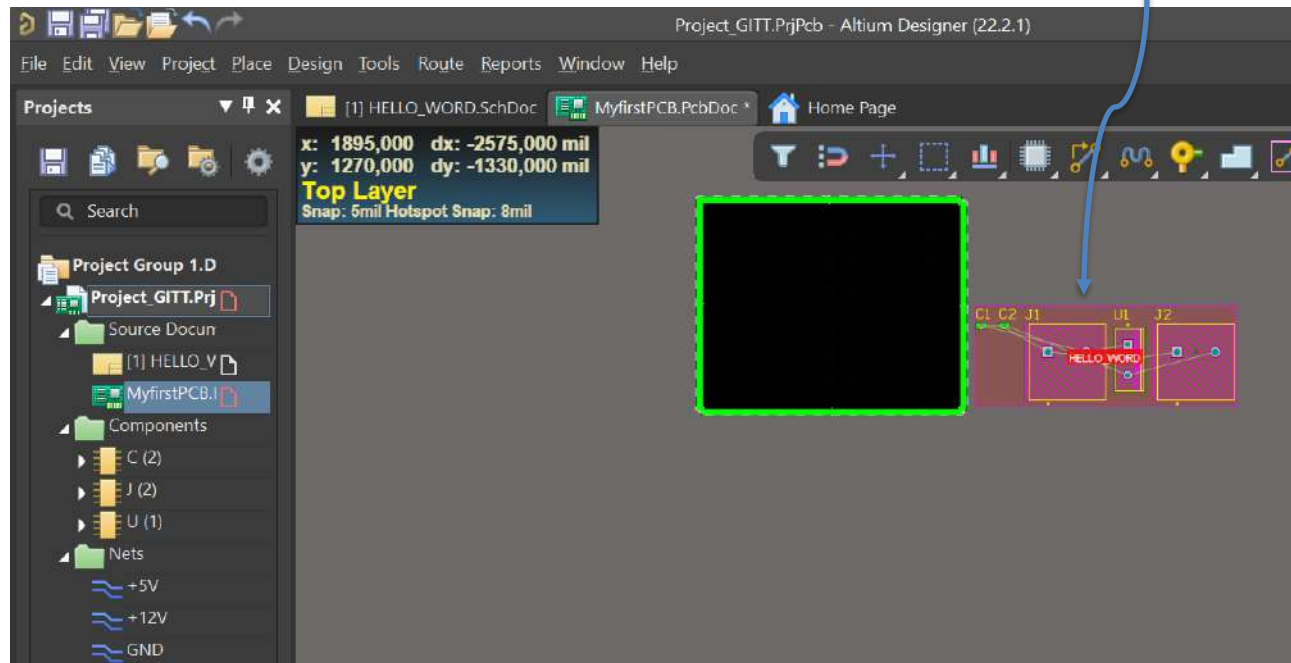
My first Layout

- ❑ Validate changes and our nets, components and romos are imported to layout.



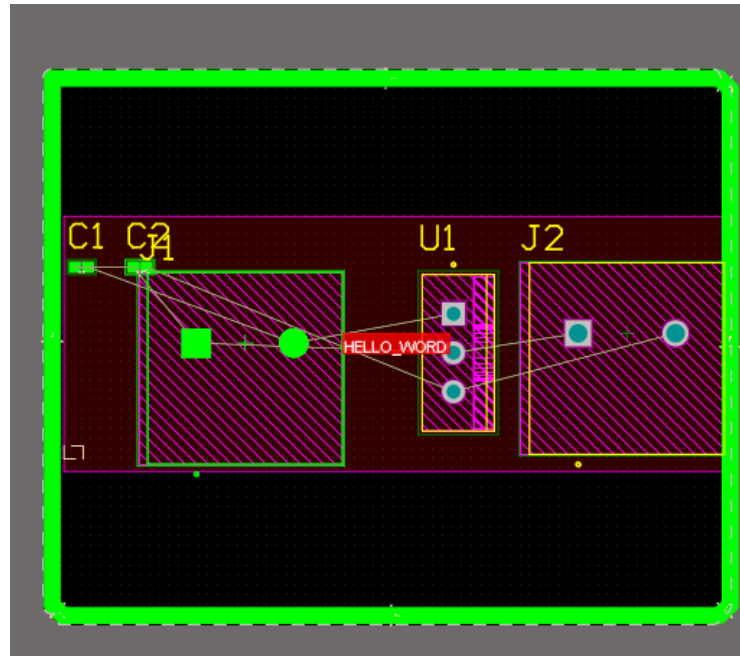
My first Layout

Imported elements from Schematic Tool



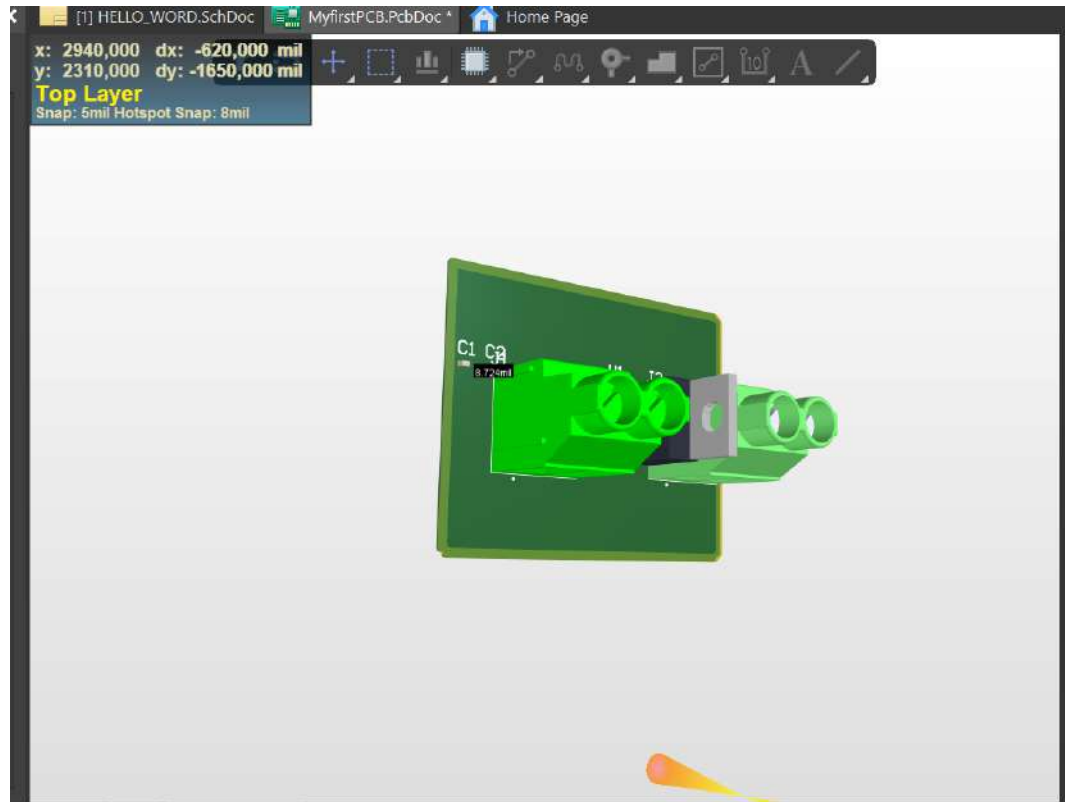
My first Layout

- We should move in the room in the board area.



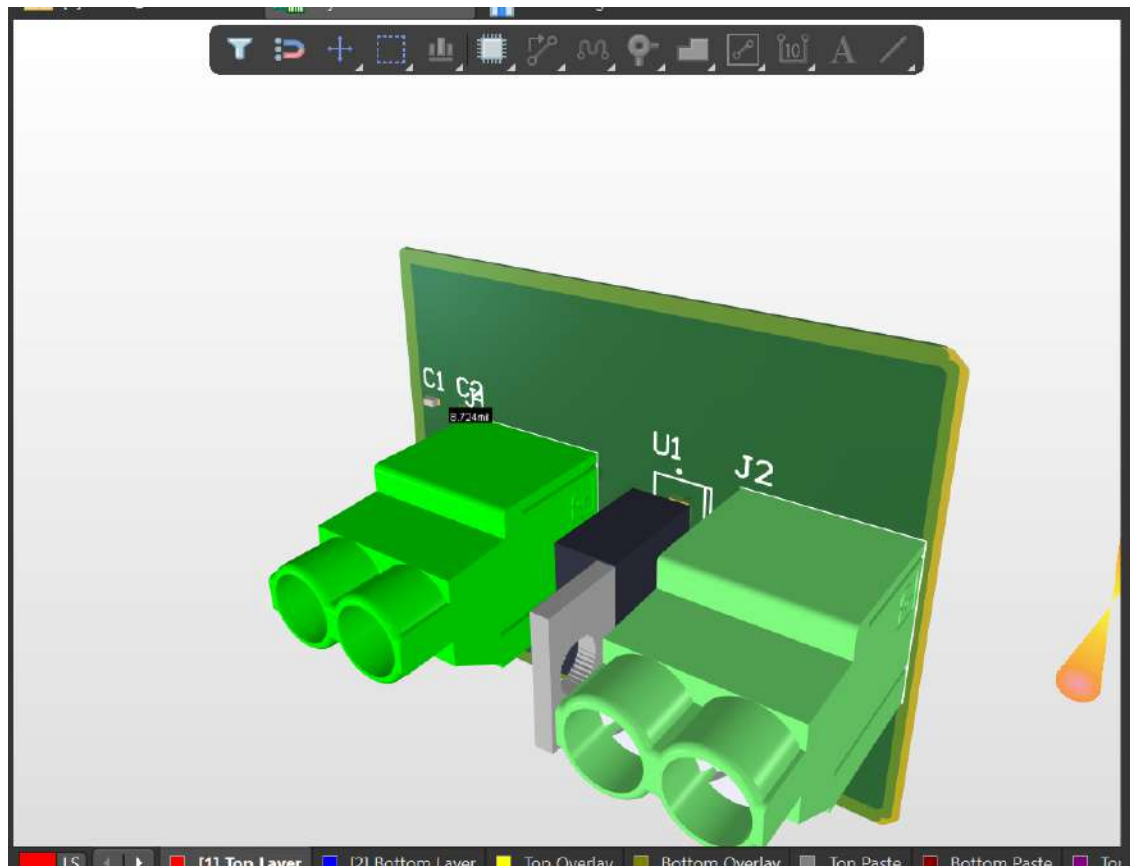
My first Layout

- We can switch between 2D and 3D view with 2 or 3 keys.

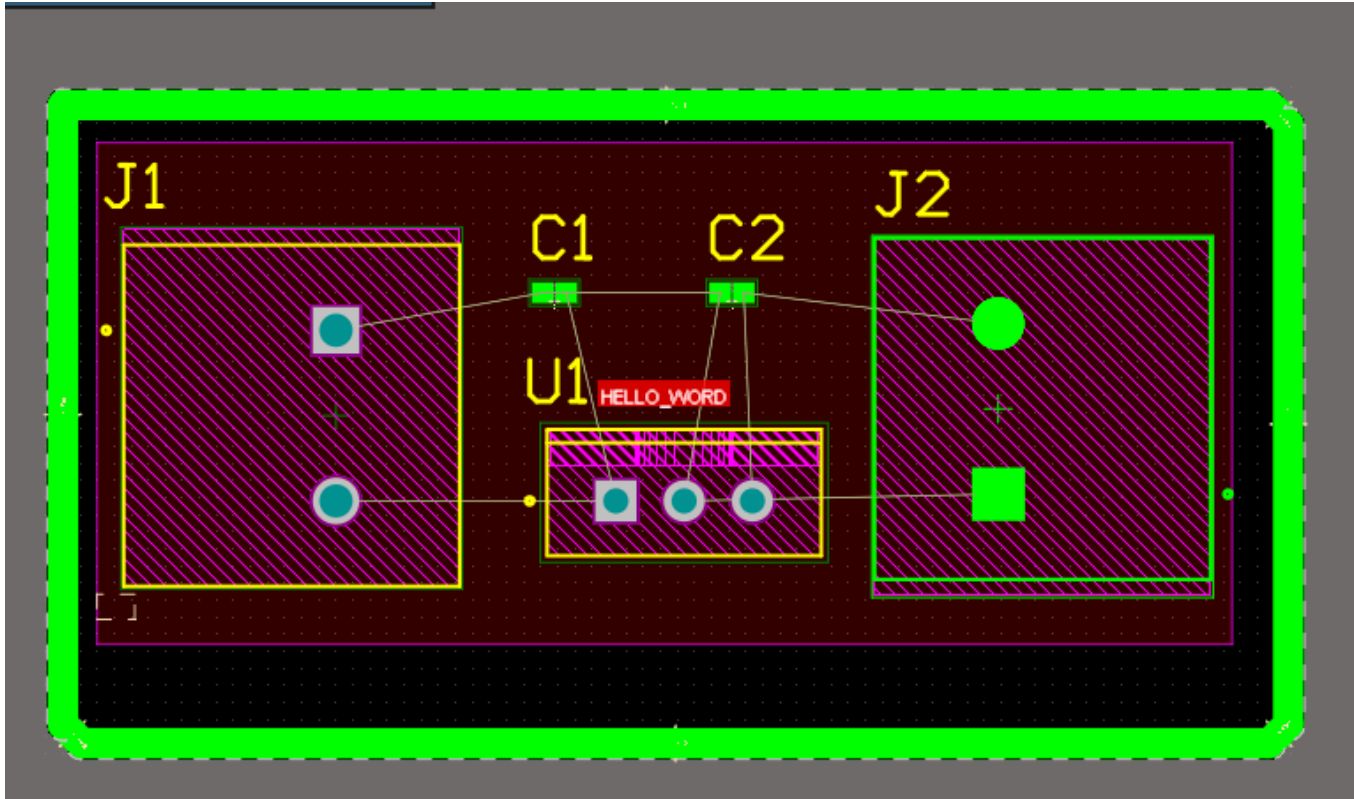


My first Layout

- To move the 3D view in several ways, press SHIFT + Right click

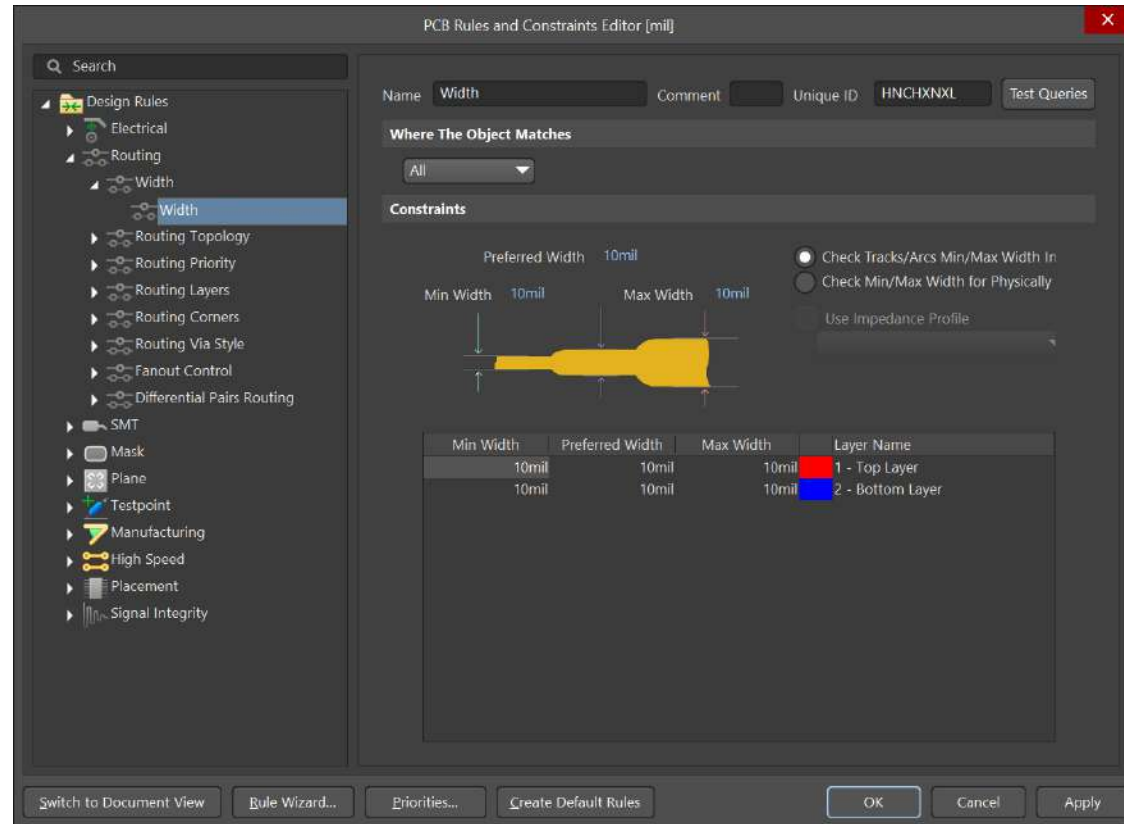


My first Layout



My first Layout

- Now it is time to start the routing process. First of all, we should define the track widths.



My first Layout

- ❑ What is the right width for the tracks? It depends on the maximum current through them
- ❑ Digital signals rarely exceed a maximum current of 500mA (examples: Arduino output pins maximum current of 500mA, Arduino output pins: maximum current 40mA, raspberry Pi: 16mA, DC motor of 9volt DC motor, max. current consumption 500mA)

Current	Inches	Millimetres
10 A	0,15748	4 mm
5 A	0,0787402	2 mm
4 A	0,0590551	1.5 mm
3 A	0,0393701	1 mm
2 A	0,019685	0,5 mm
0.5 A	0,00787402	0,2 mm

My first Layout

Other recommendations:

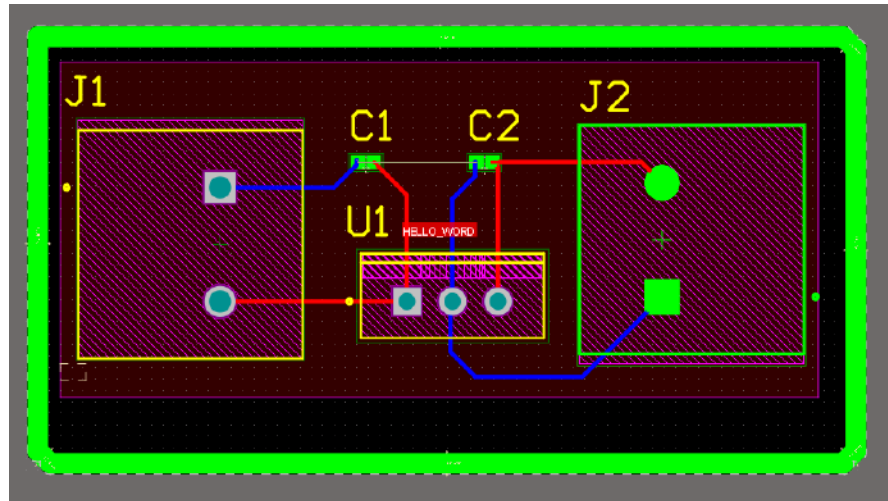
- ☐ Distance between tracks: It depends on the voltage. For digital signals 0,3mm is enough.
- ☐ Routing with 45° avoiding 90°.
- ☐ Choose the right width track according to the type of via.
- ☐ Take care when you route tracks for SMD components. Are you in the right layer?
- ☐ Colour the priority tracks (GND, VCC and others).

Further details:

<https://www.youtube.com/watch?v=AhJEJWTKUVI>

My first Layout

- Then ... we are ready to route (CTRL + W or Route → Interactive Routing).

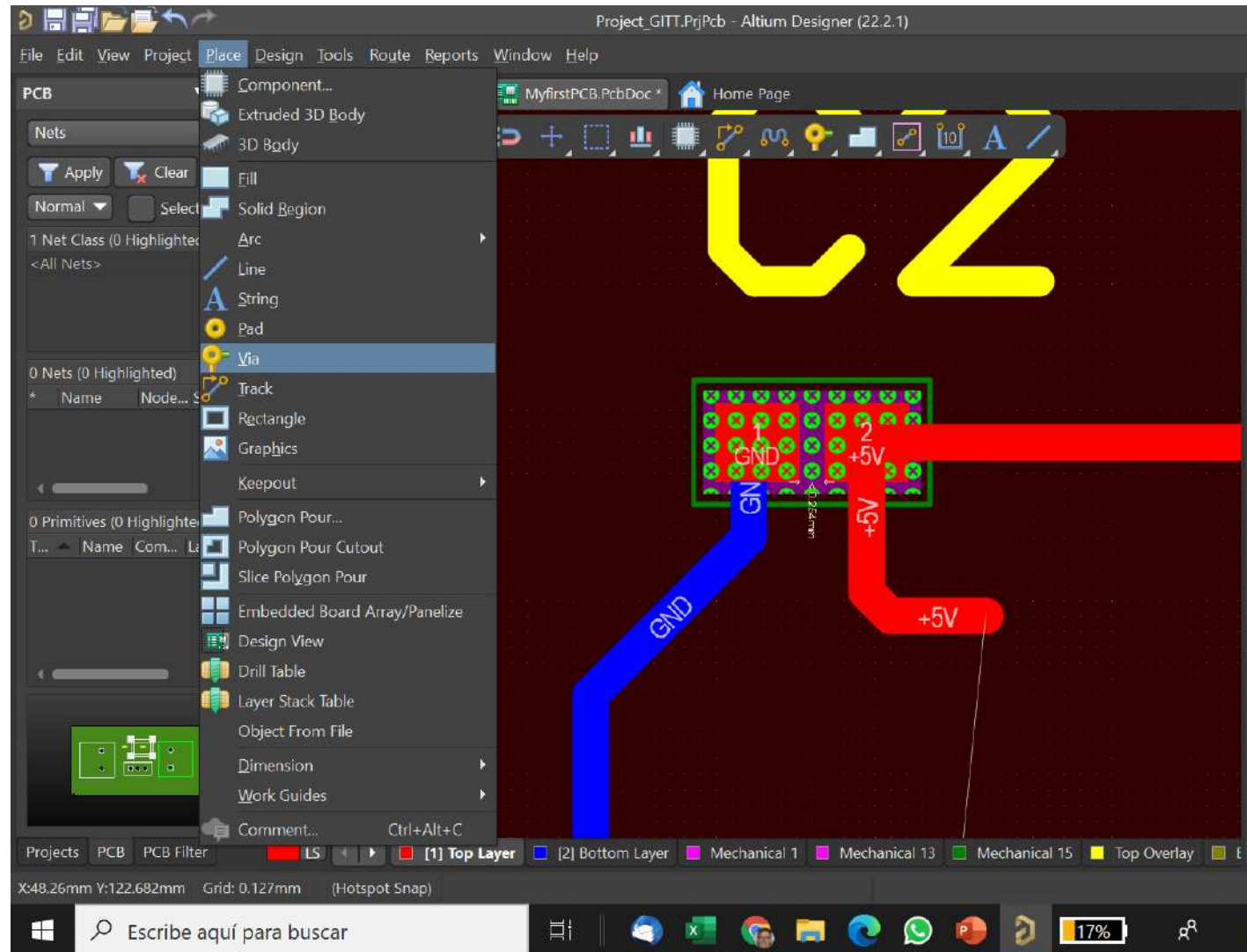


- We can change the layer to be routed.
- Track width can be dynamically changed pressing 2 key.

My first Layout

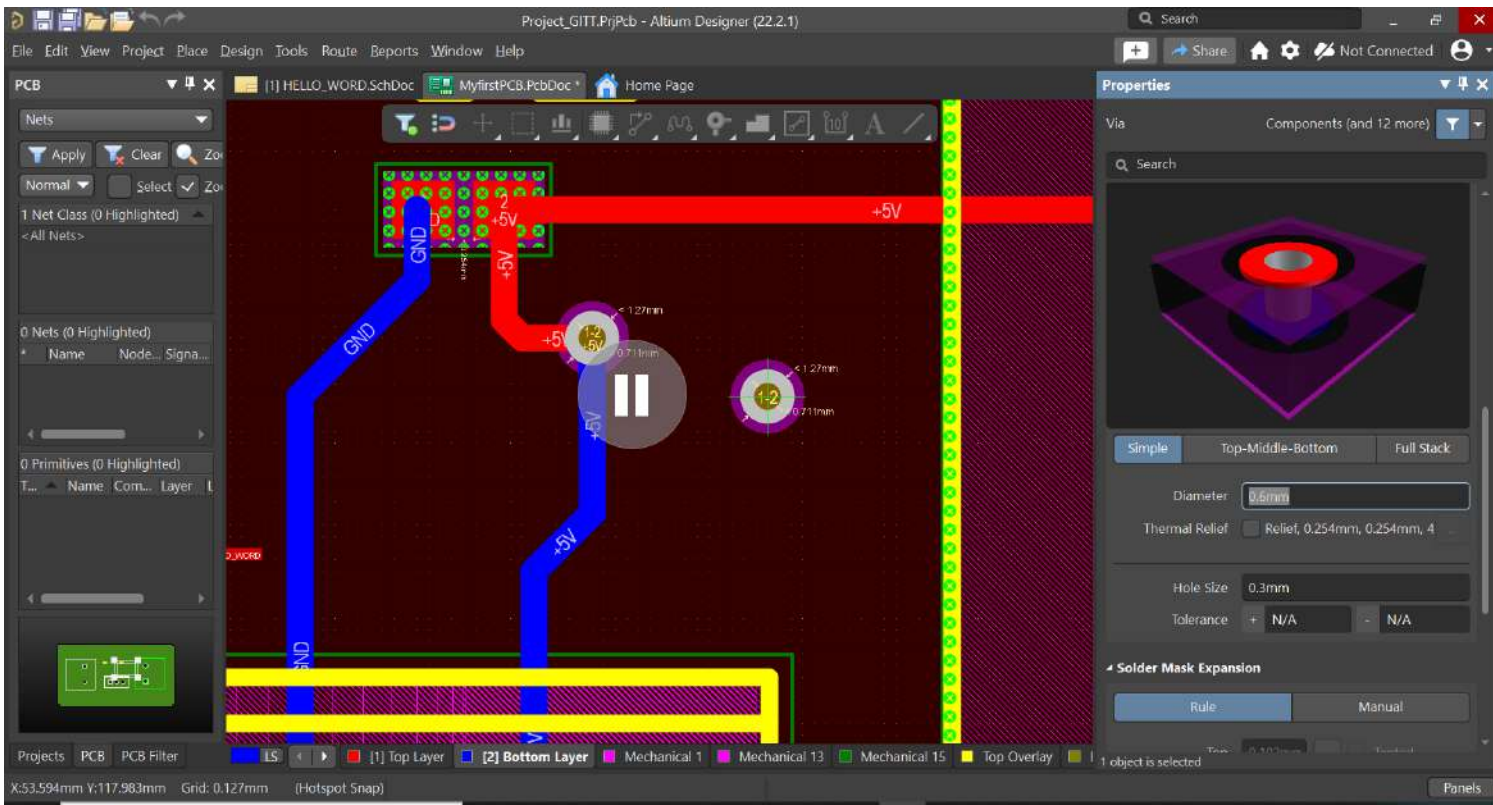
Placing vias

- ❑ Via is a hole to communicate a same track through several layers.



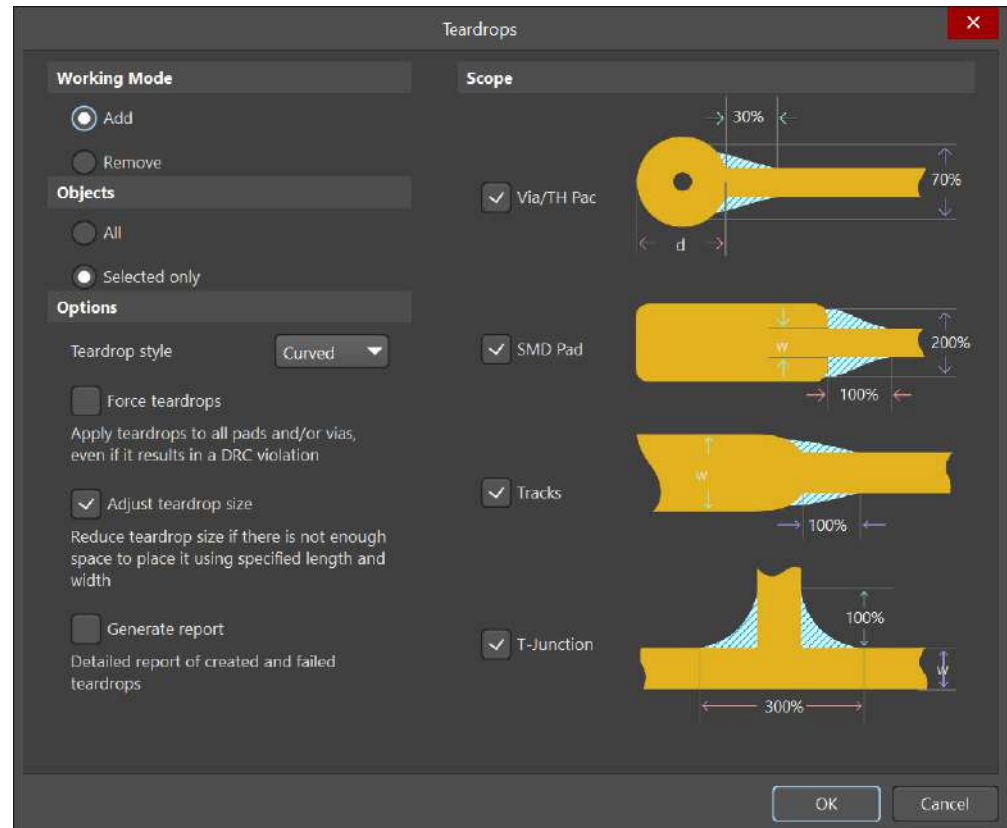
My first Layout

- ❑ The via width and size must be equivalent to the track width.



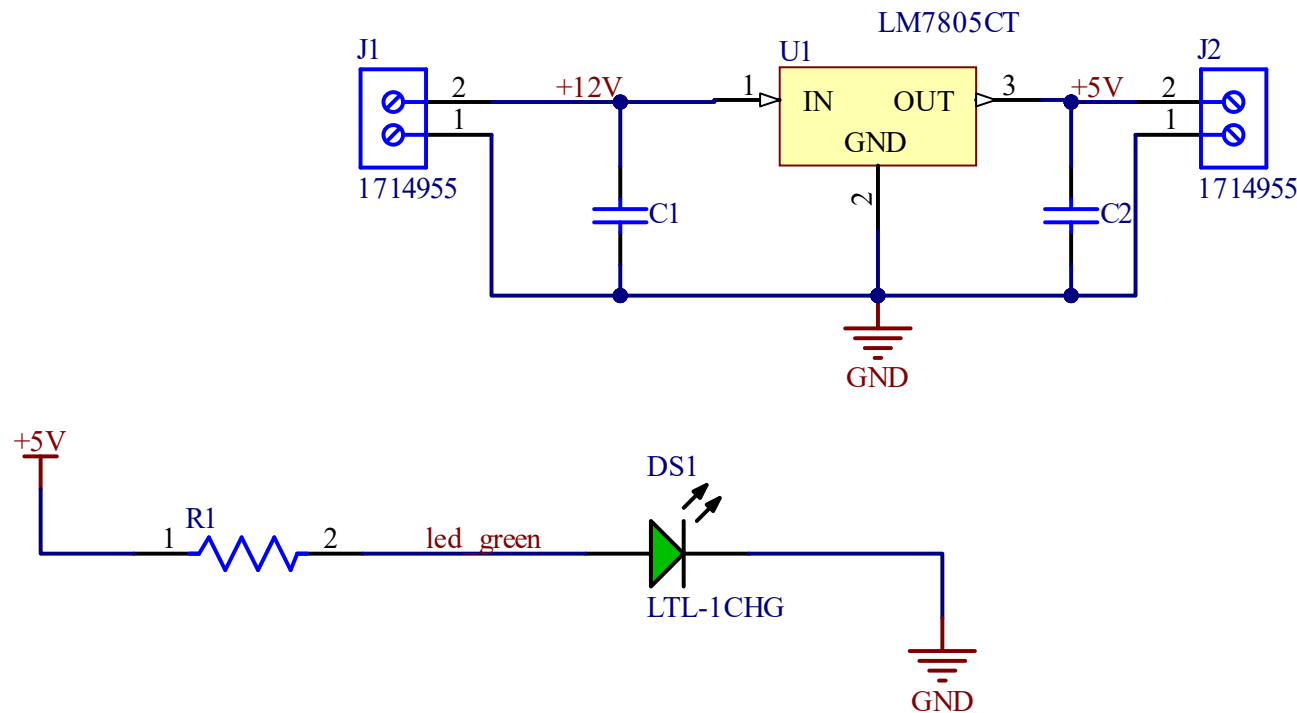
My first Layout

- When routing stage has finished we could do a “track smooth” using Tools → Teardrops. Default options can be validated to improve the track connexions.



My first Layout

- ❑ Eco feature is still active between Schematic and Layout Tool.
- ❑ Now, we are going to add a LED and a resistor.



My first Layout

- ❑ We must update our PCB information with the new added components and their nets.

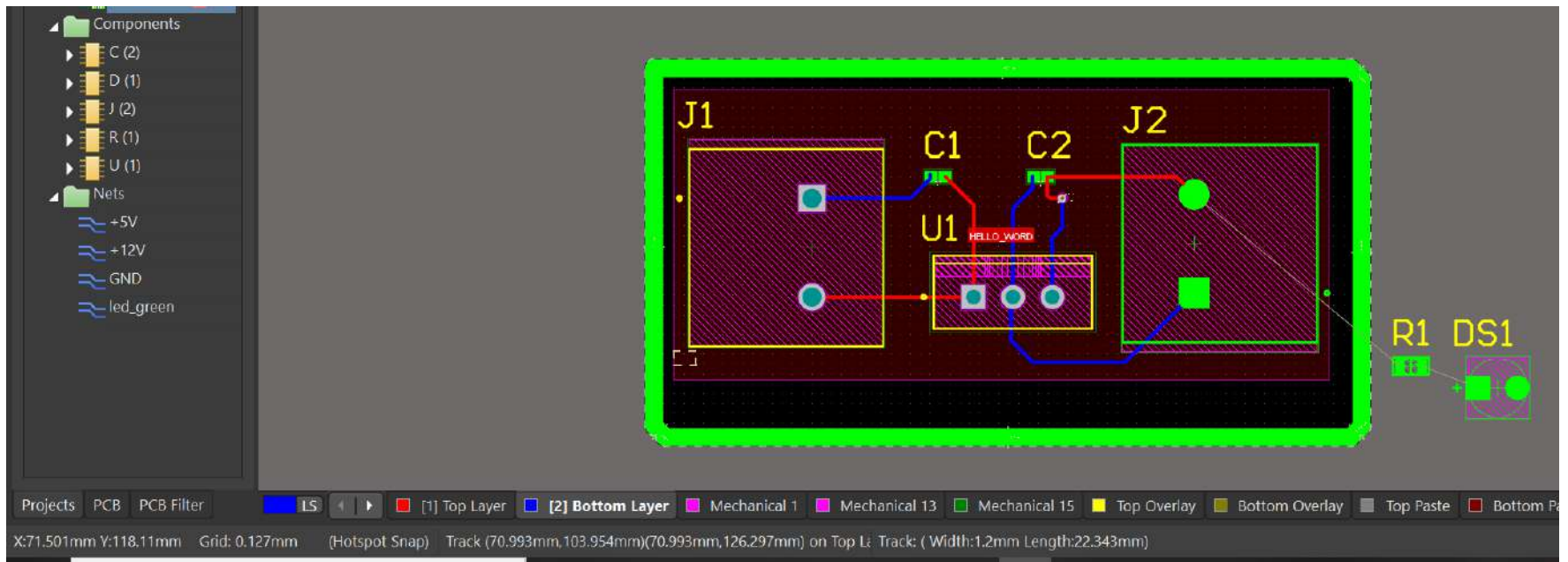
The screenshot displays the Altium Designer interface. The 'Design' menu is open, showing the 'Update PCB Document MyfirstPCB.PcbDoc' option. Below it, the 'Engineering Change Order' dialog is visible, listing modifications to the PCB document. The modifications include adding components (DS1, R1) and pins to nets (DS1-2 to GND, R1-1 to +5V), adding a net (led_green), and adding component class members (DS1 to HELLO_WORD, R1 to HELLO_WORD).

Engineering Change Order

Modifications	Enable	Action	Affected Object		Affected Document
Add Components(2)					
	✓	Add	DS1	To	MyfirstPCB.PcbDoc
	✓	Add	R1	To	MyfirstPCB.PcbDoc
Add Pins To Nets(2)					
	✓	Add	DS1-2 to GND	In	MyfirstPCB.PcbDoc
	✓	Add	R1-1 to +5V	In	MyfirstPCB.PcbDoc
Add Nets(1)					
	✓	Add	led_green	To	MyfirstPCB.PcbDoc
Add Component Class Members(2)					
	✓	Add	DS1 to HELLO_WORD	In	MyfirstPCB.PcbDoc
	✓	Add	R1 to HELLO_WORD	In	MyfirstPCB.PcbDoc

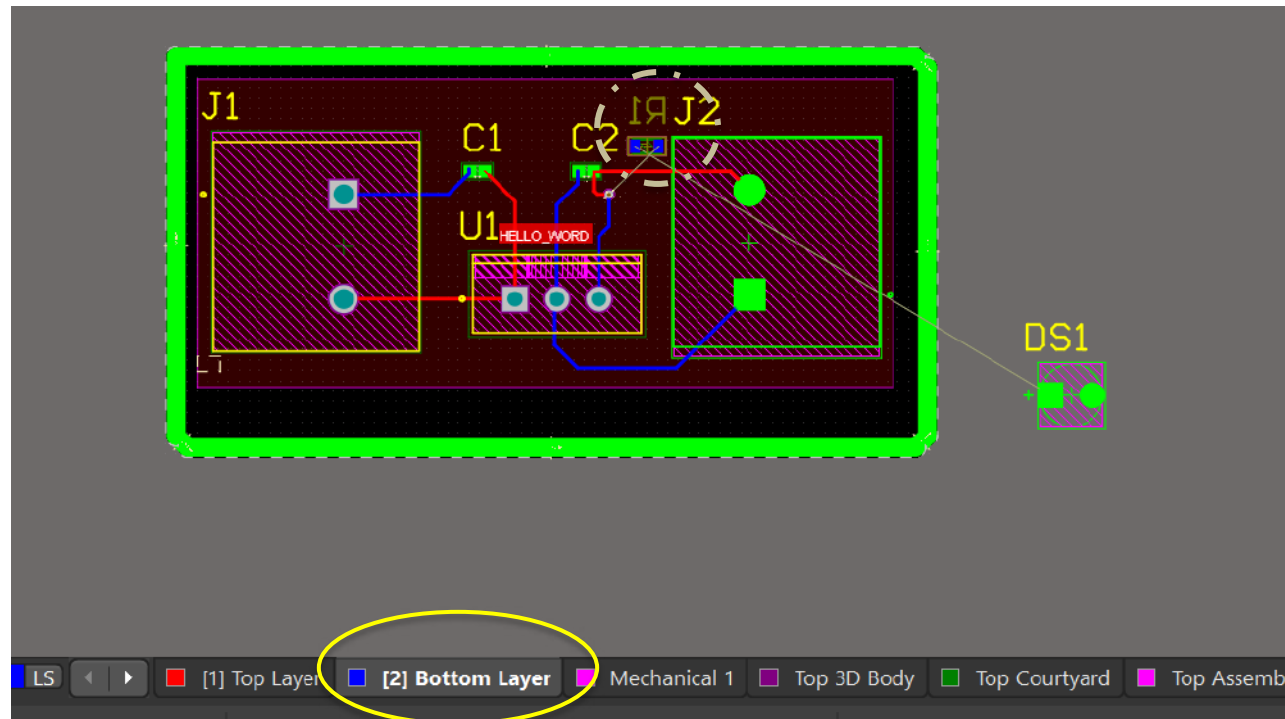
My first Layout

- New components have been imported to our PCB Layout.



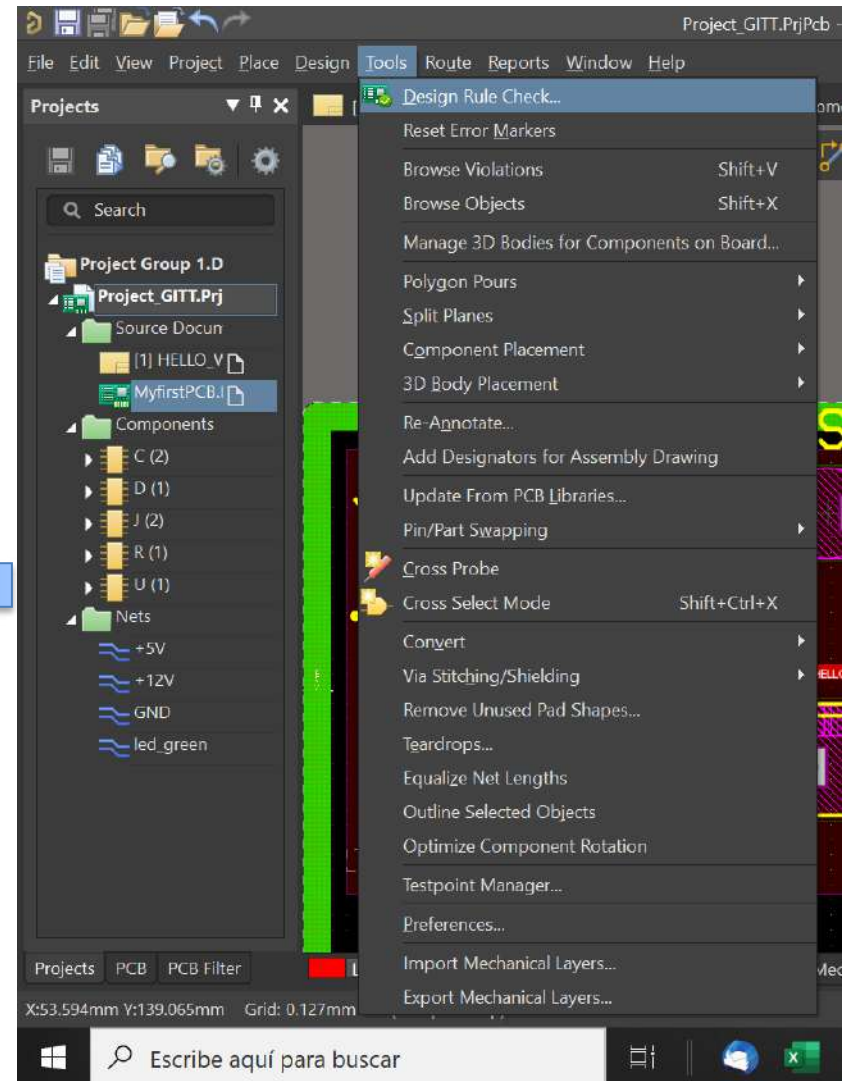
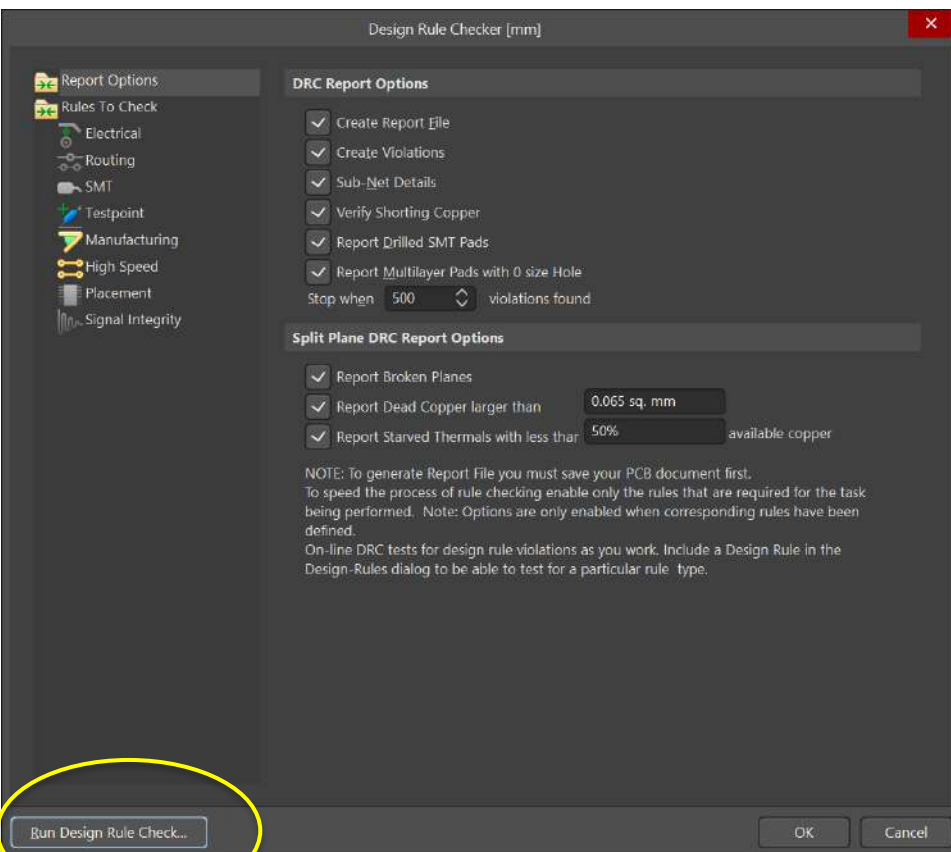
My first Layout

- To place a component in the Bottom Layer:
 1. Choose Bottom Layer
 2. Select the component
 3. Press 'L' Key.



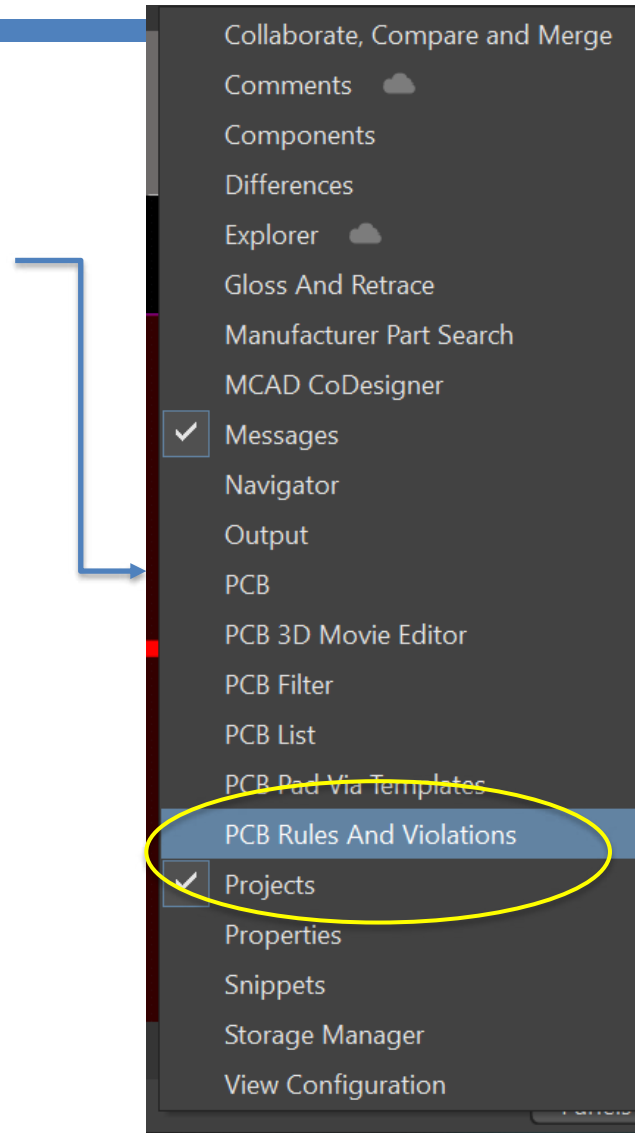
My first Layout

□ Checking the PCB →
Detecting violations

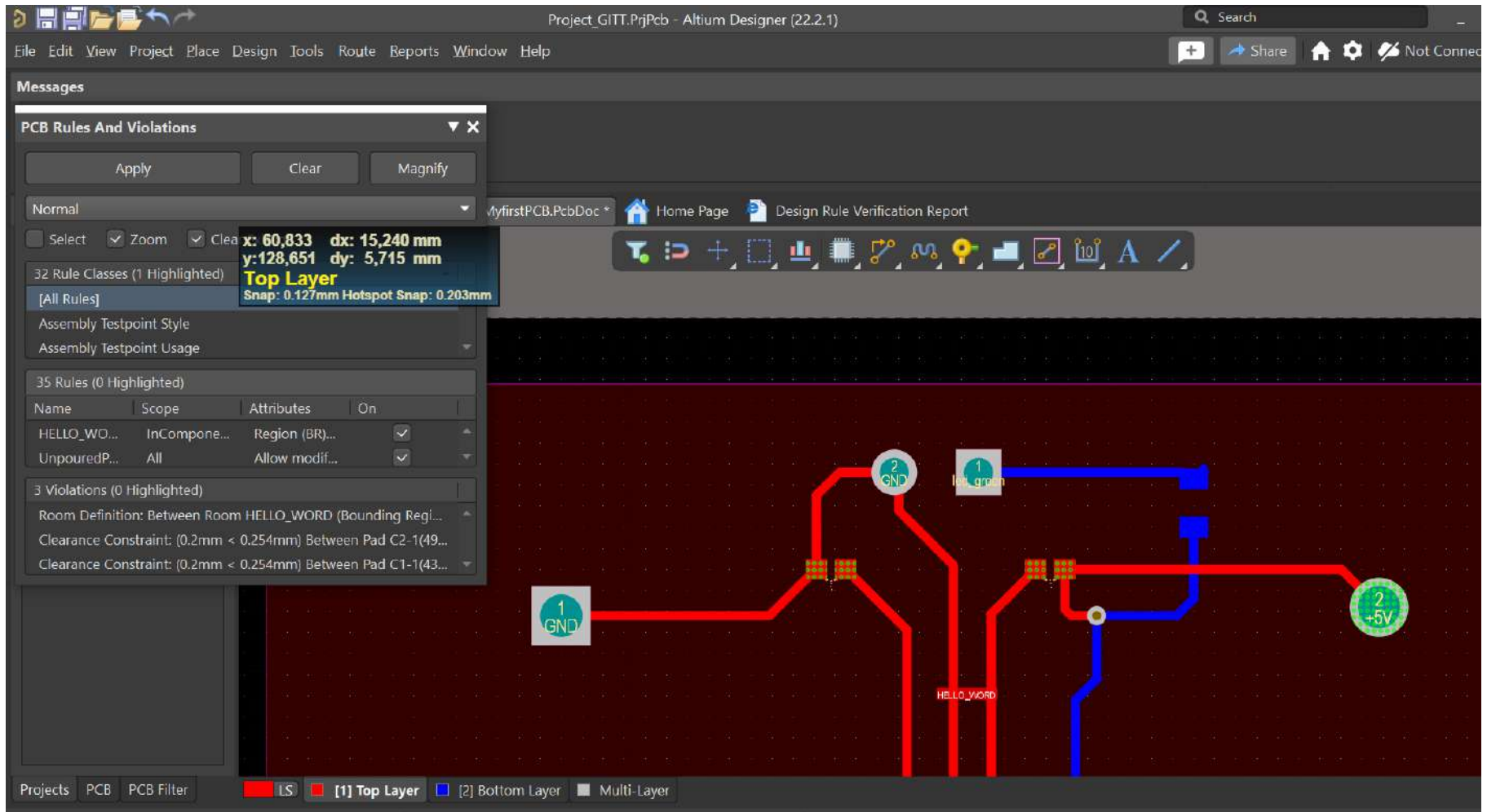


My first Layout

- ❑ To display *the violations* in a suitable mode:

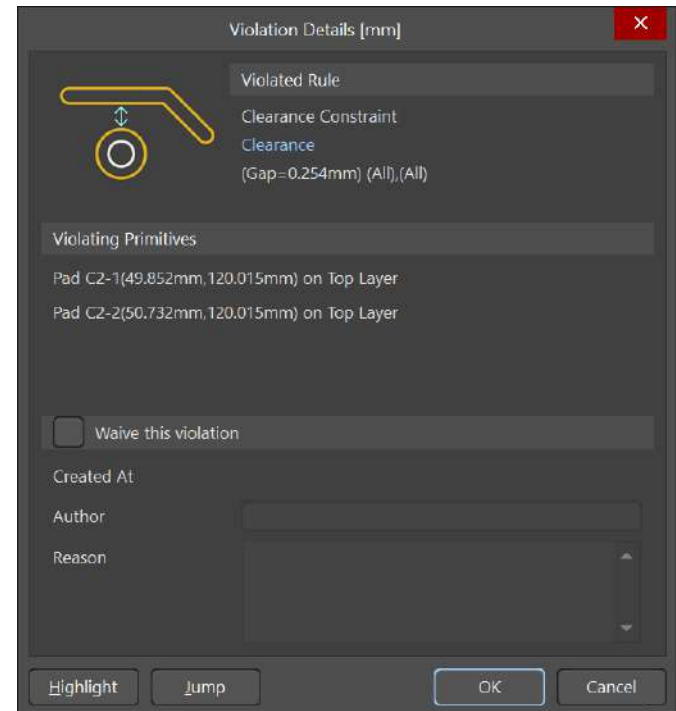


My first Layout



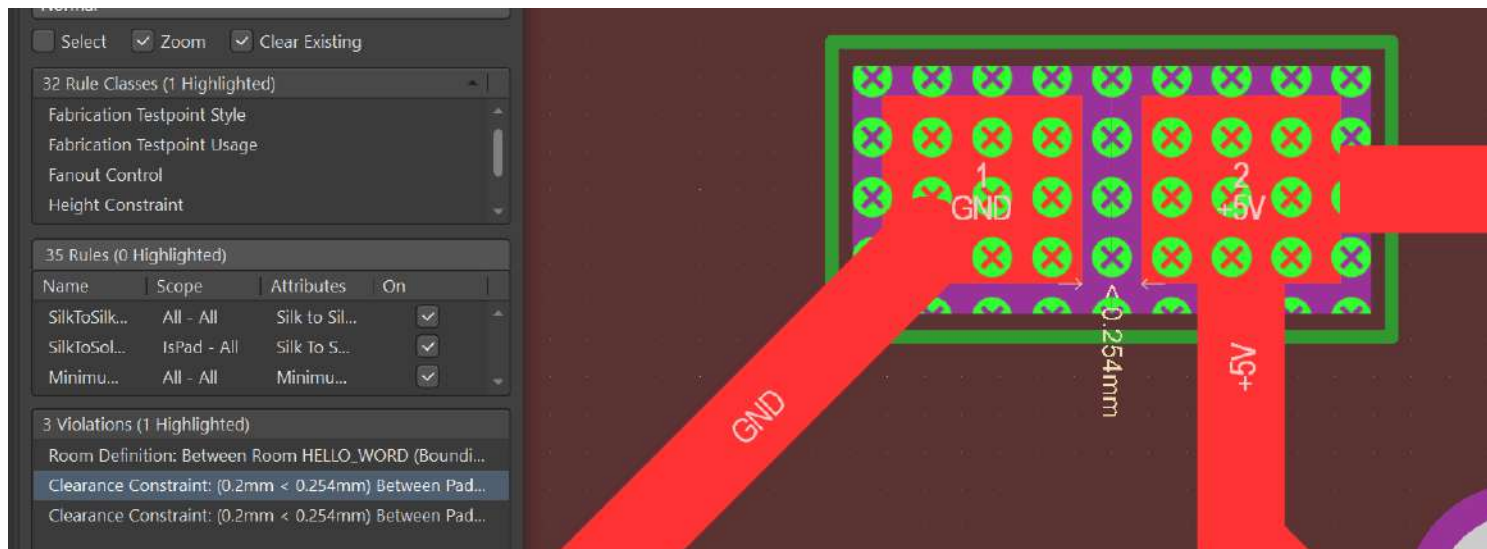
My first Layout

- ❑ Remember that there is a set of electrical rules defined by default.
- ❑ Some of them could not adjust to standard footprints



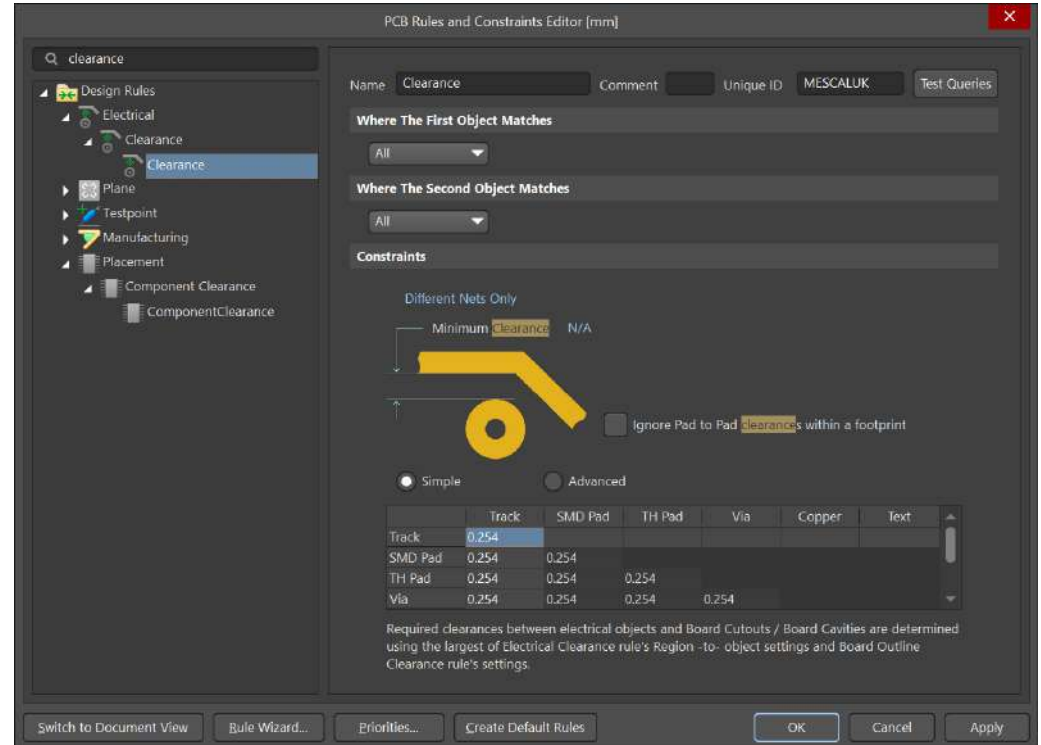
My first Layout

- This example displays how minimum distance is not fulfilled. However, the footprint is taken from manufacturer.



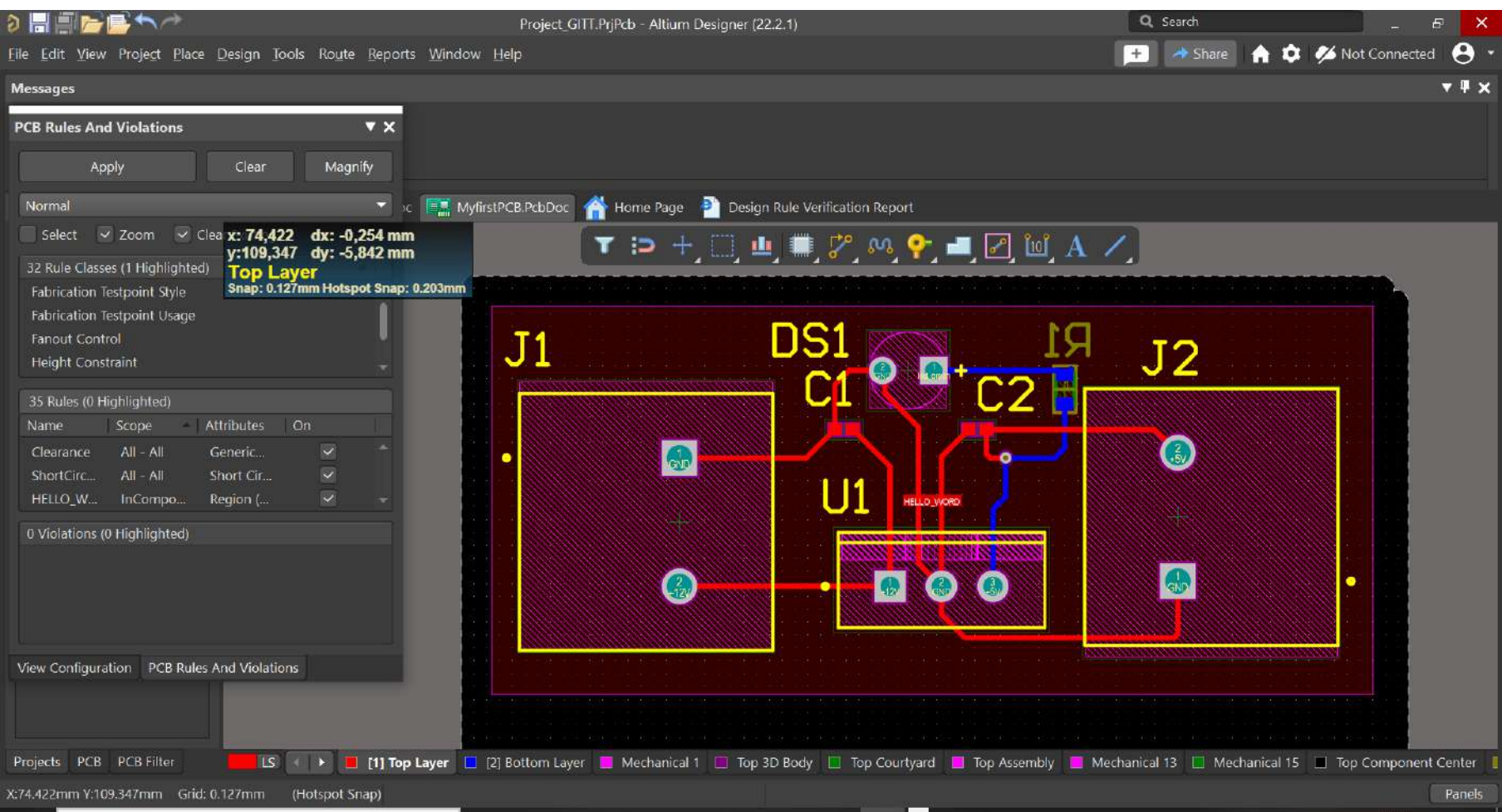
My first Layout

- ❑ To avoid any footprint modification, we are going to modify the electrical rule.
- ❑ Design → Rules → Search Clearance.
- ❑ Now, change it!!

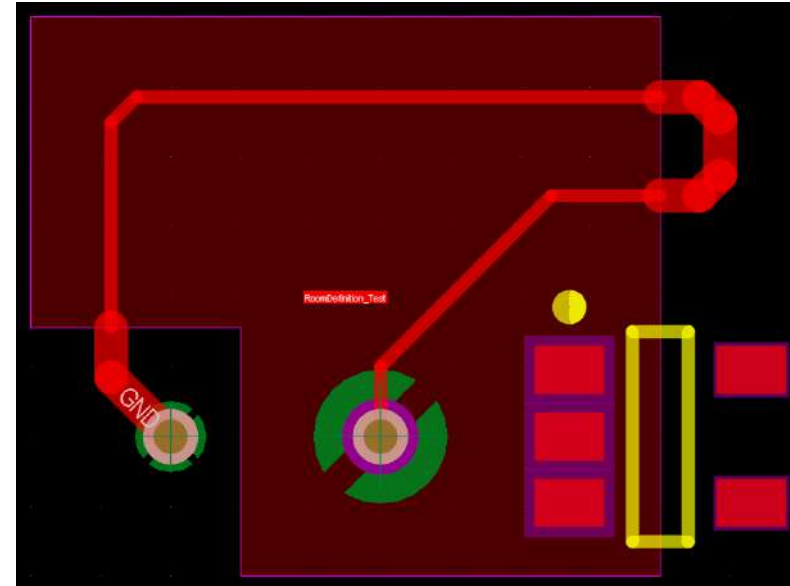


My first Layout

□ Final View

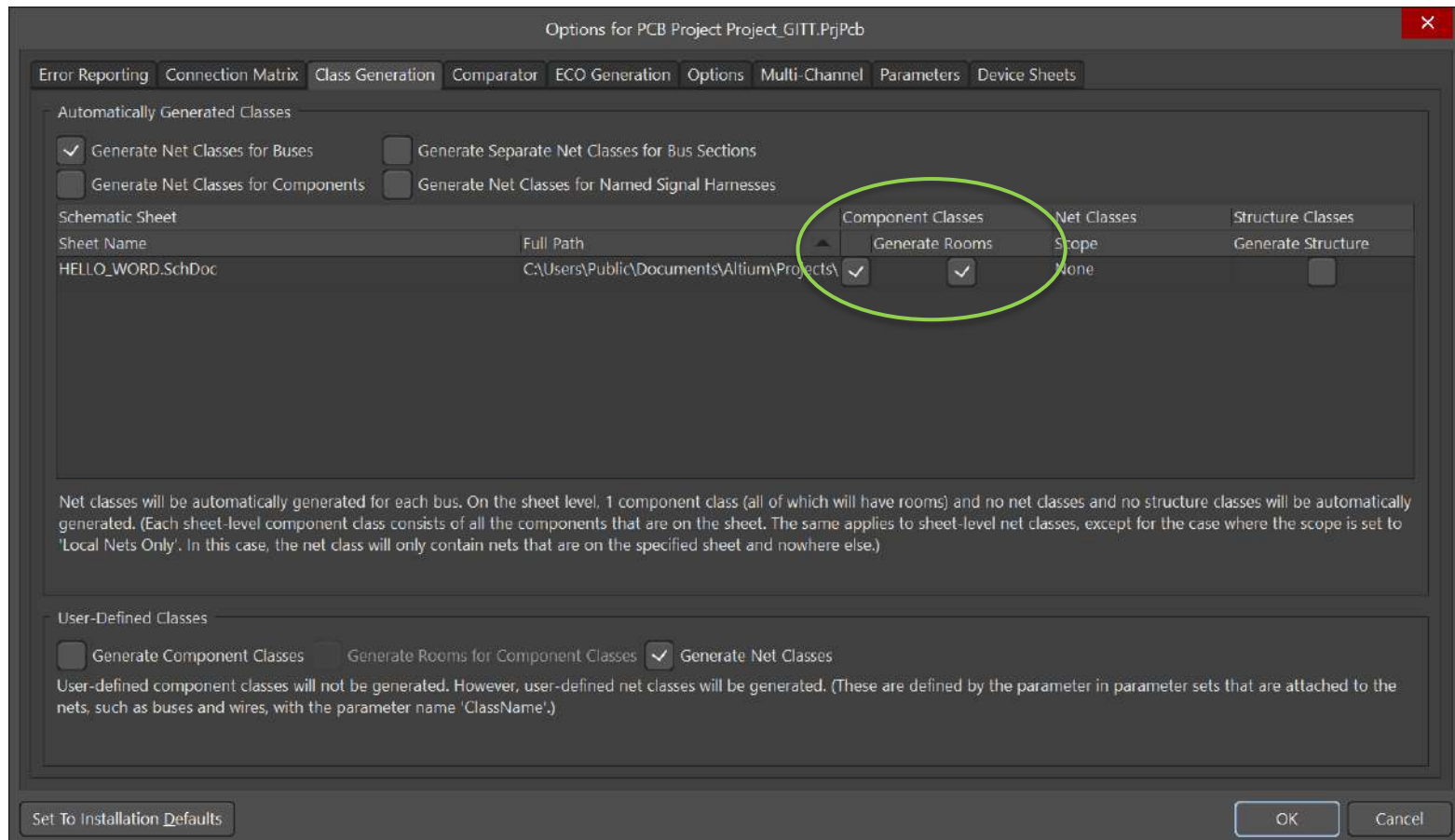


- ❑ Room Concept: It can be defined as a virtual area for PCB Layout where the components belonging to that region inherit a common set of specifications (rule designs, track width, same region, etc.)
- ❑ By default a room area is set by the system per sheet. It means that a hierarchical design contains as rooms as symbol sheets are defined.



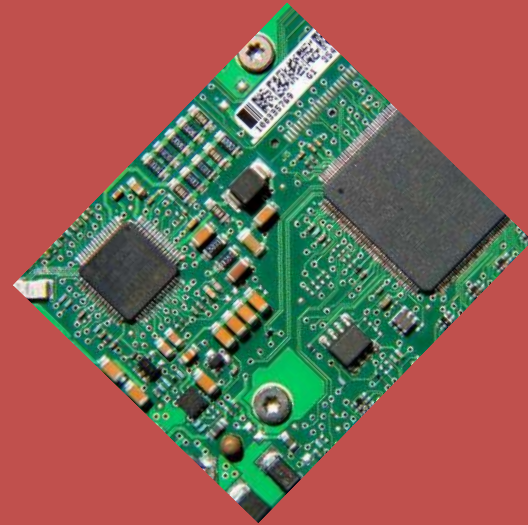
Room

□ Unchecking room option: Project → Project Options



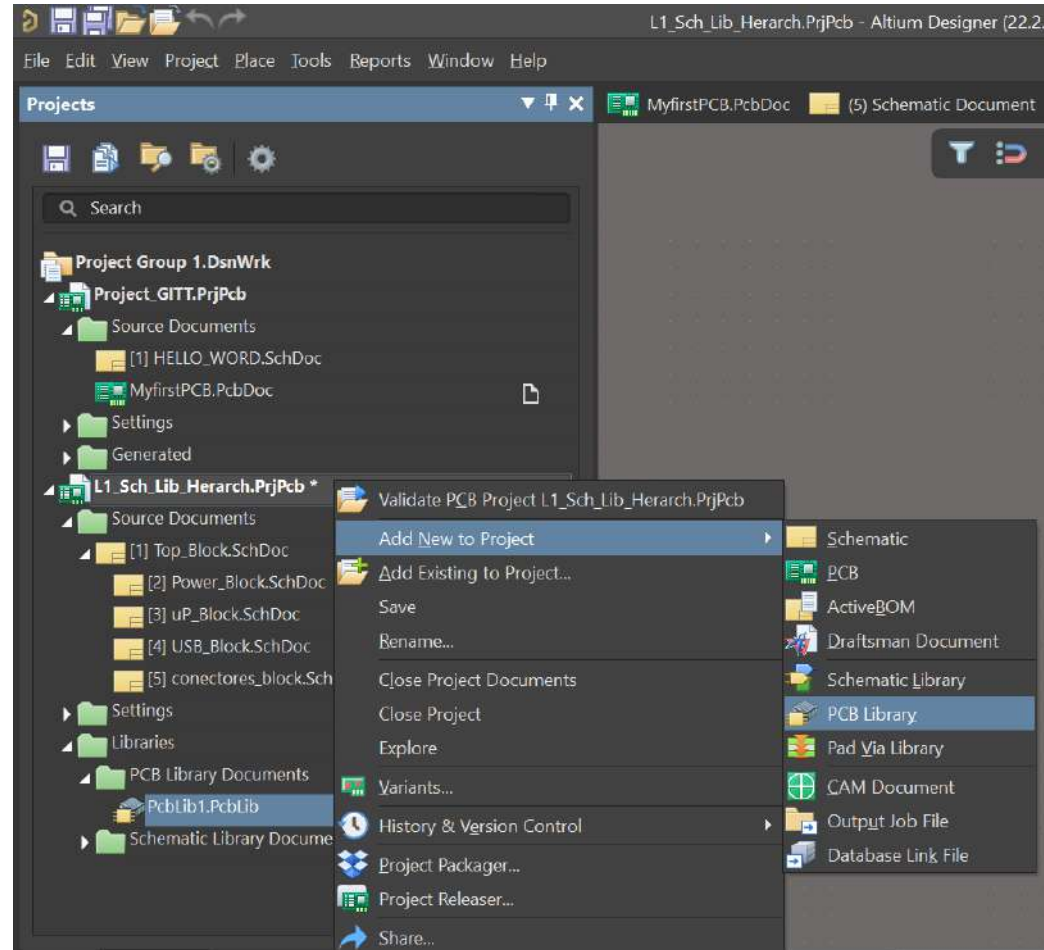
Introduction to Printed Circuit Board (PCB) design with ALTIUM

5.- PCB Library



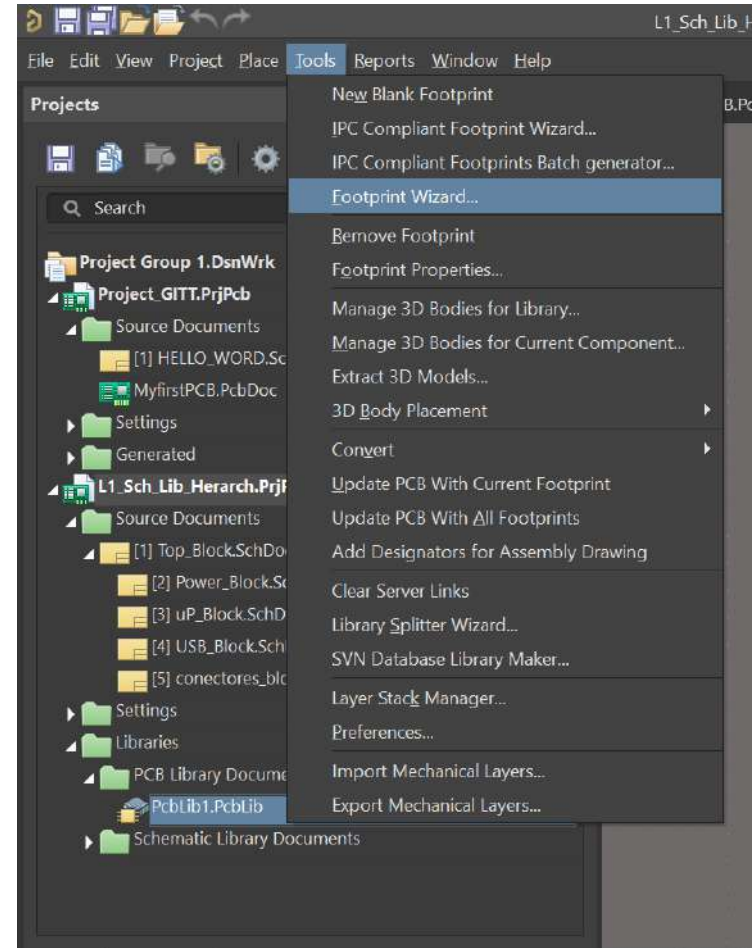
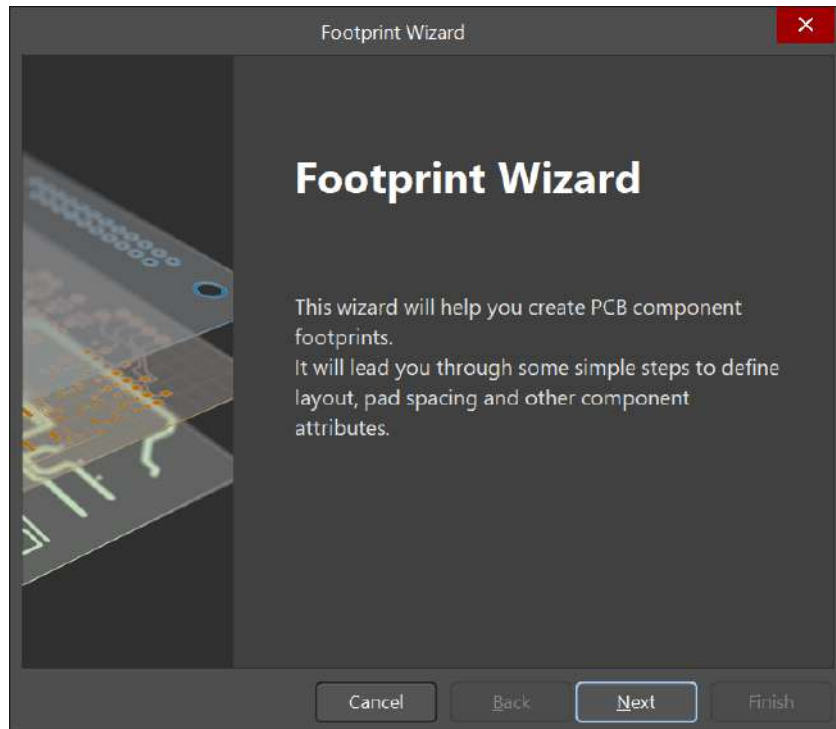
Creating a new PCB_Lib

- ❑ We are going to produce a Library for PCB tool in a similar way than Schematic Library.
- ❑ It is useful when we are not able to find the expected device within Manufacturer Part Search or Component Tab.
- ❑ We will create a new Library in the same way that New Sch, PCB or SCH_LIB.



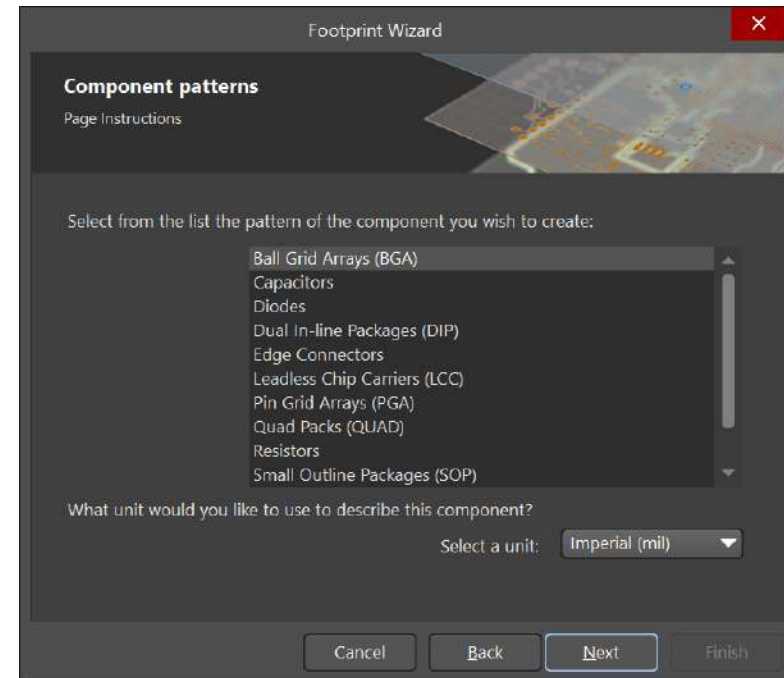
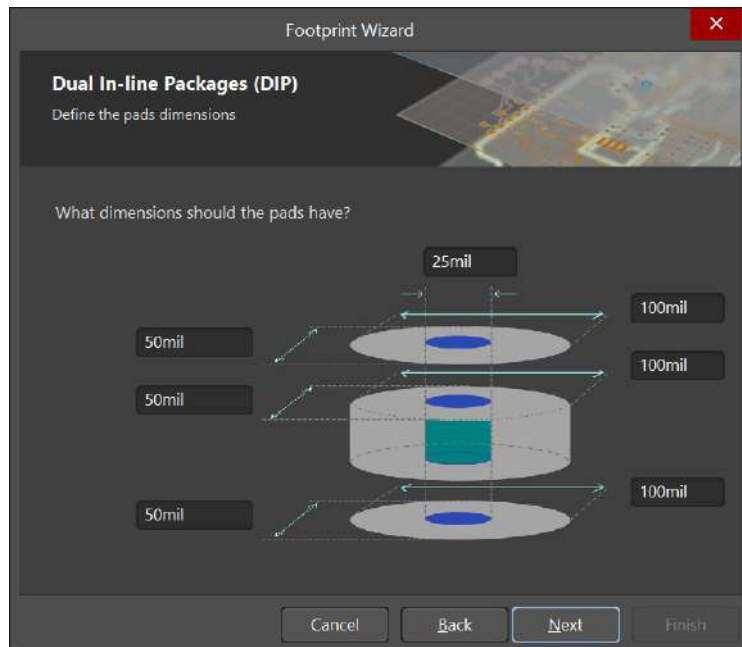
Creating a new PCB_Lib

- We will use a Wizard to make our footprint.



Creating a new Footprint

- A menu with a set of typical footprints are showed to choose one.



- We select a DIP (e.g.) and a new window with information about the pin features pops-up.

Creating a new Footprint

Footprint Wizard

Dual In-line Packages (DIP)
Set number of the pads

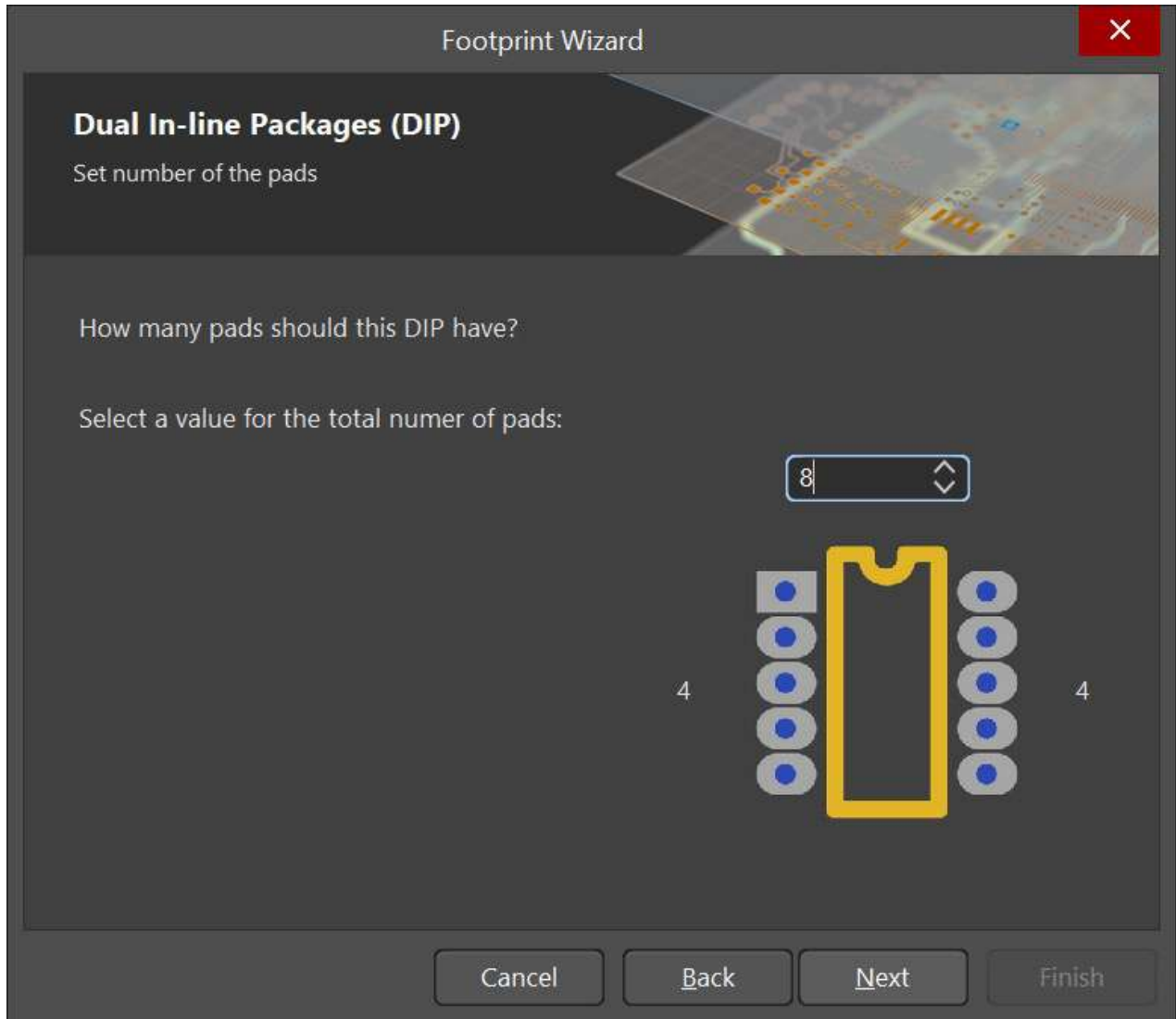
How many pads should this DIP have?

Select a value for the total number of pads:

8

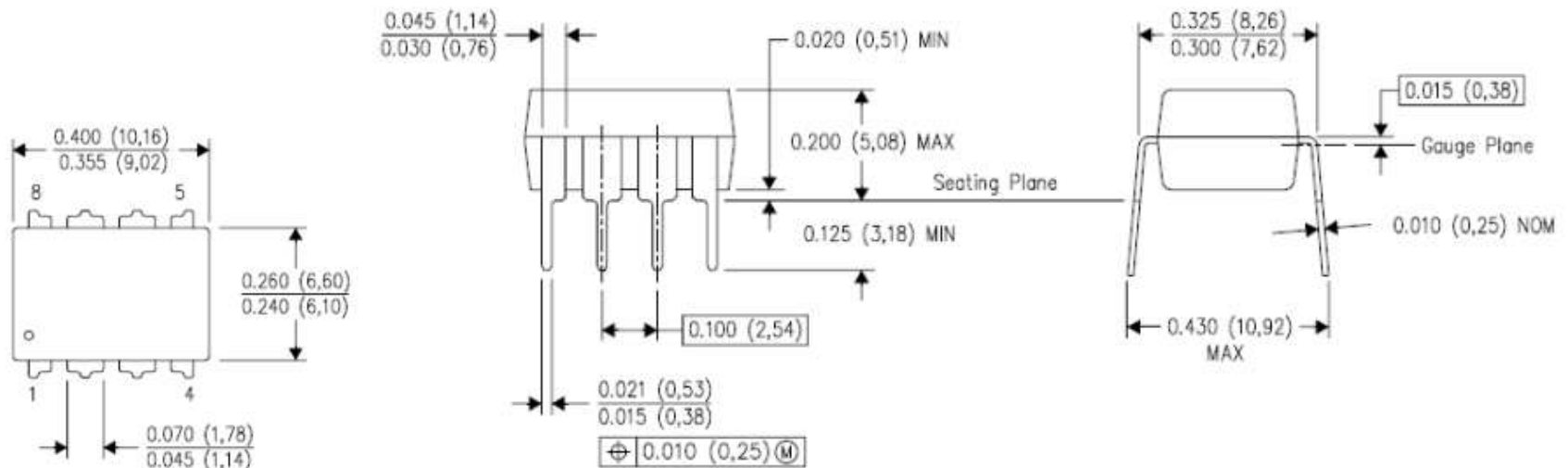
4 4

Cancel Back Next Finish



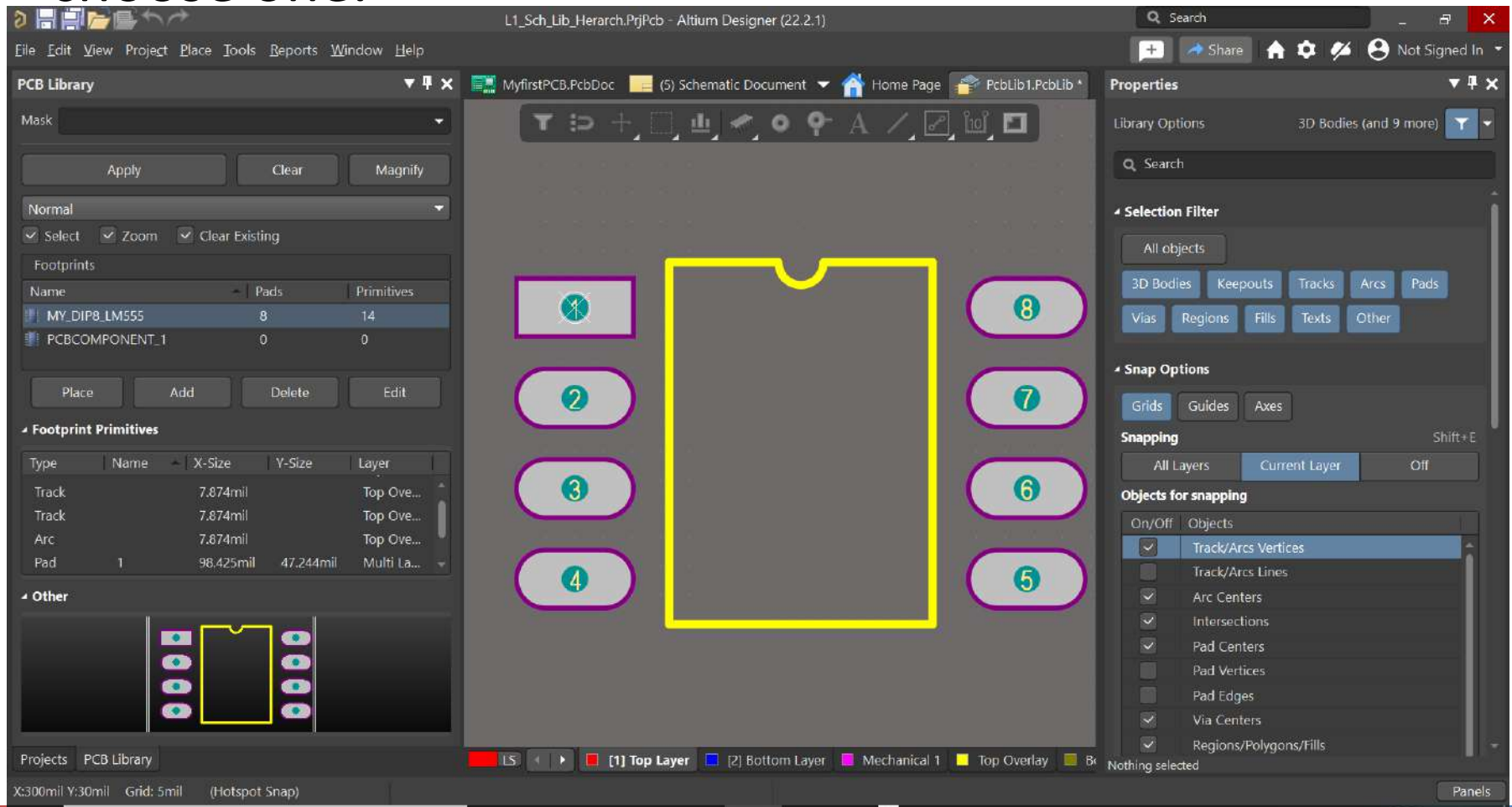
Creating a new Footprint

- The physical dimension of the device must be checked through the datasheet.



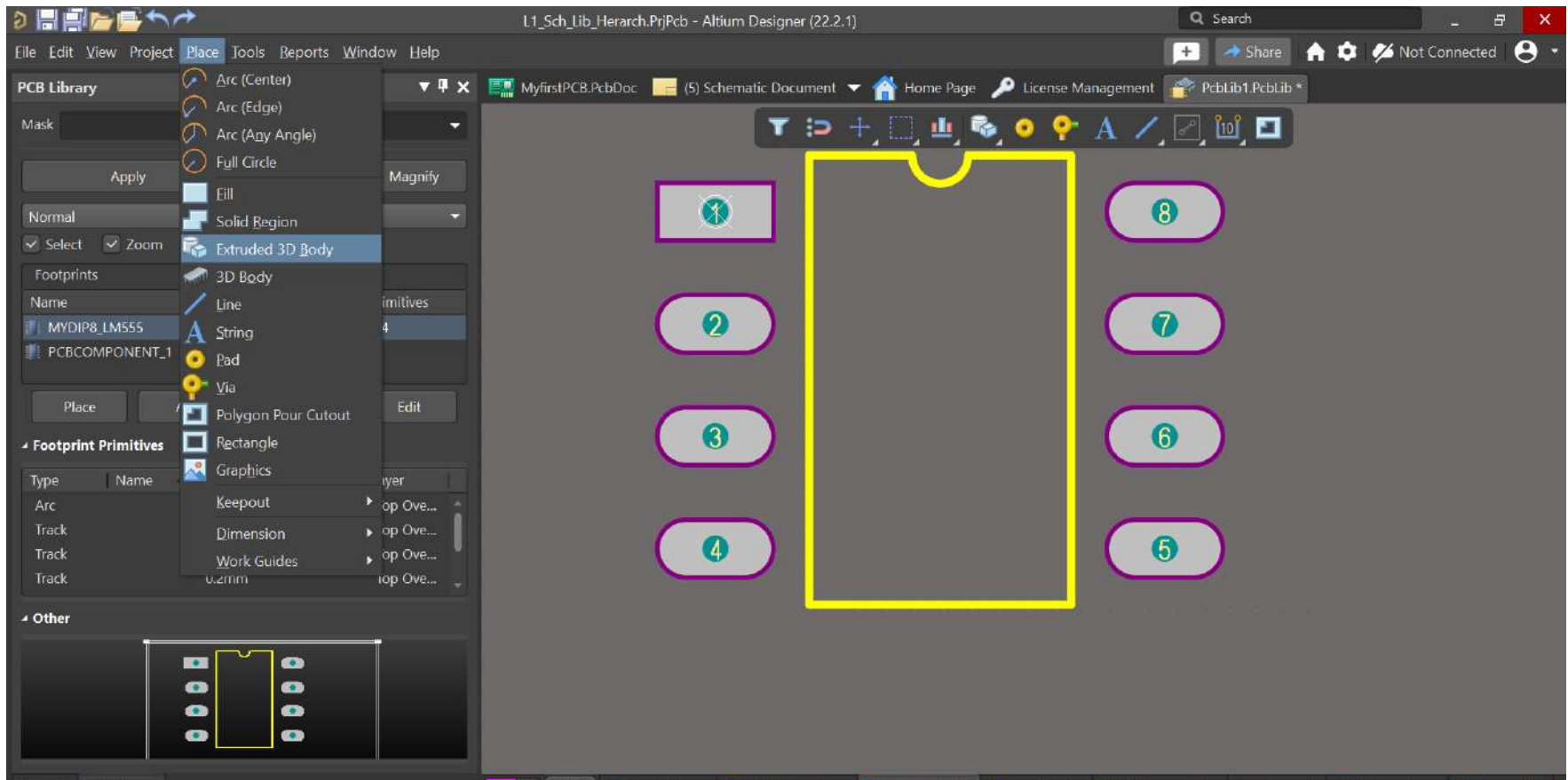
Creating a new Footprint

- ❑ A menu with a set of typical footprints are showed to choose one.

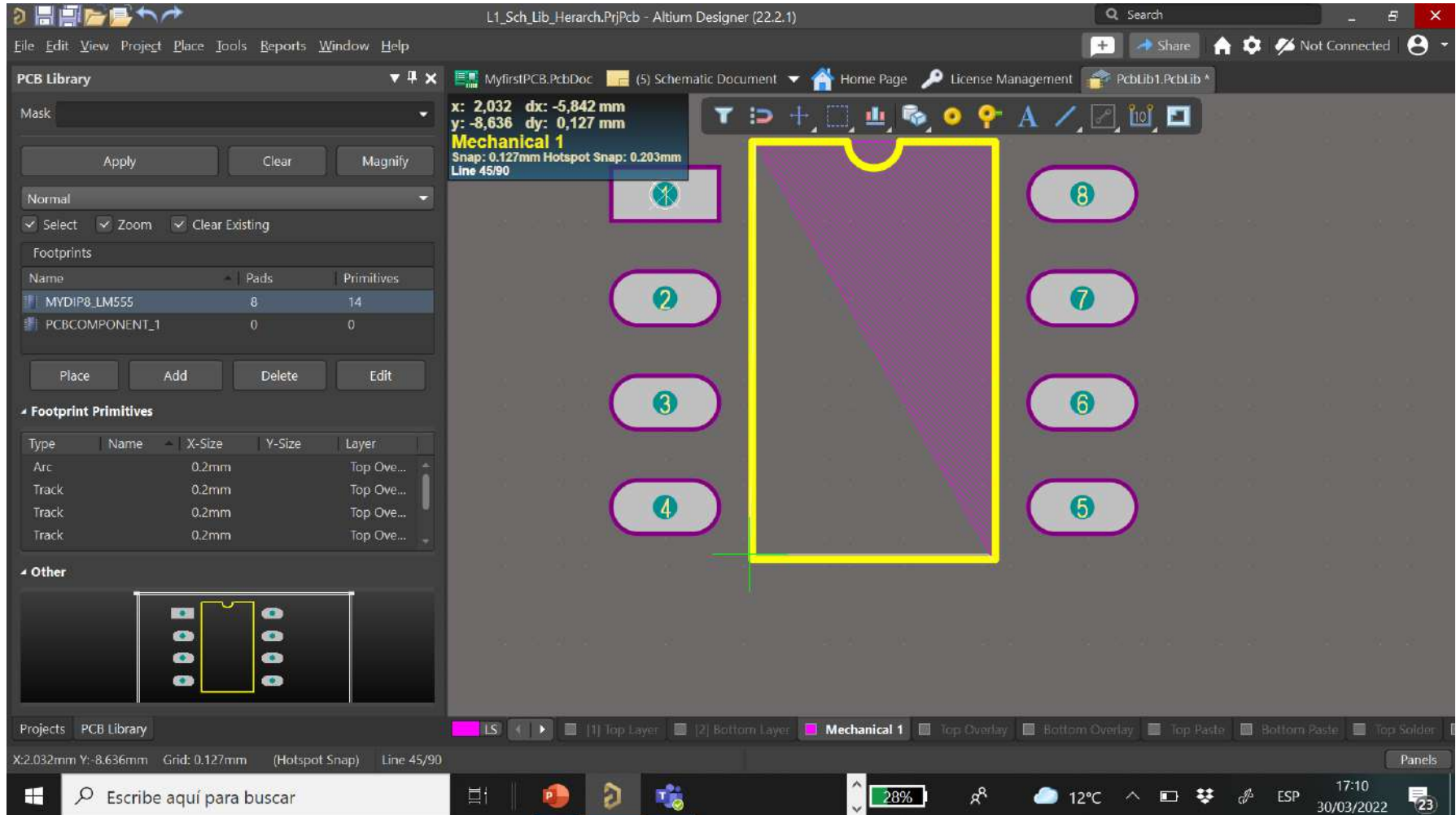


Creating a new Footprint

- Additionally, we could add a 3D model to our footprint. It is useful for the final view.

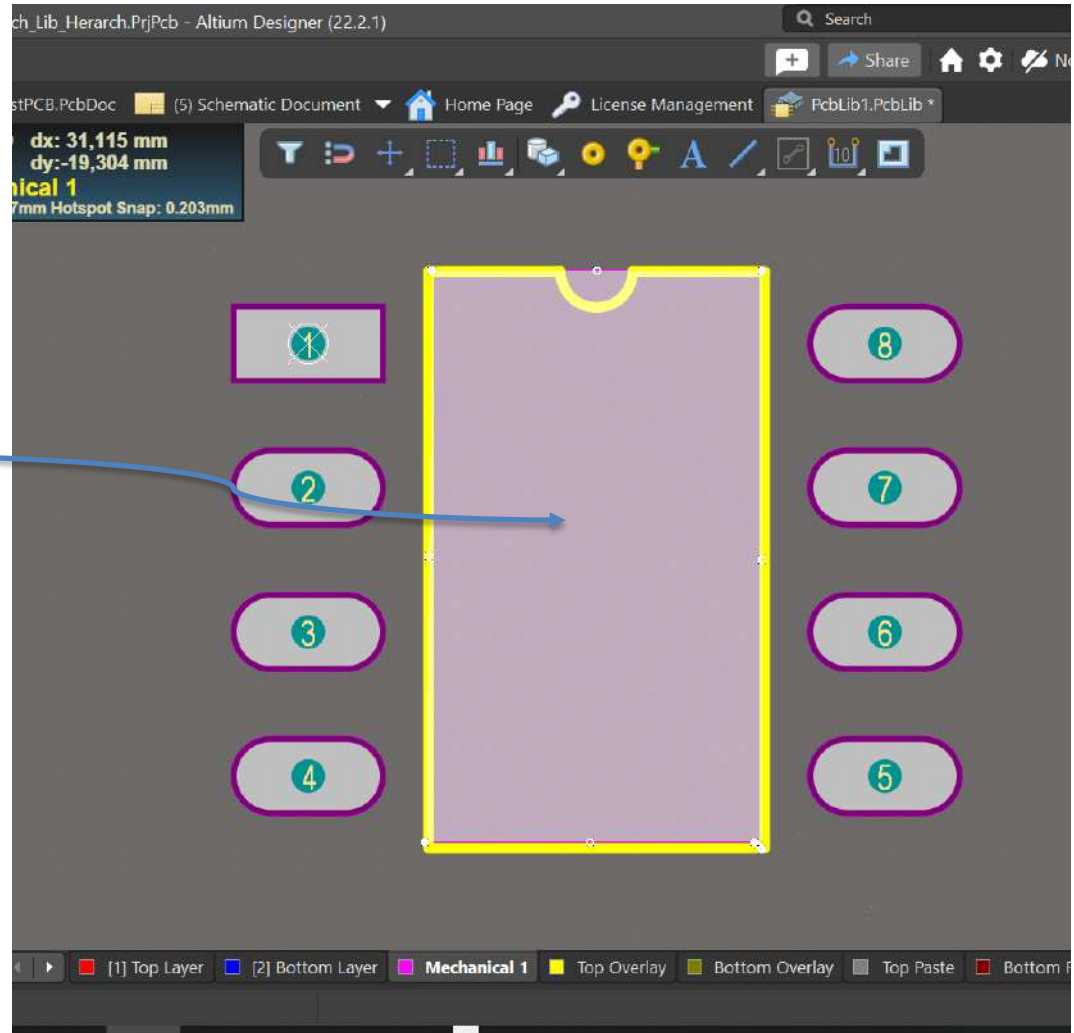


Creating a new Footprint



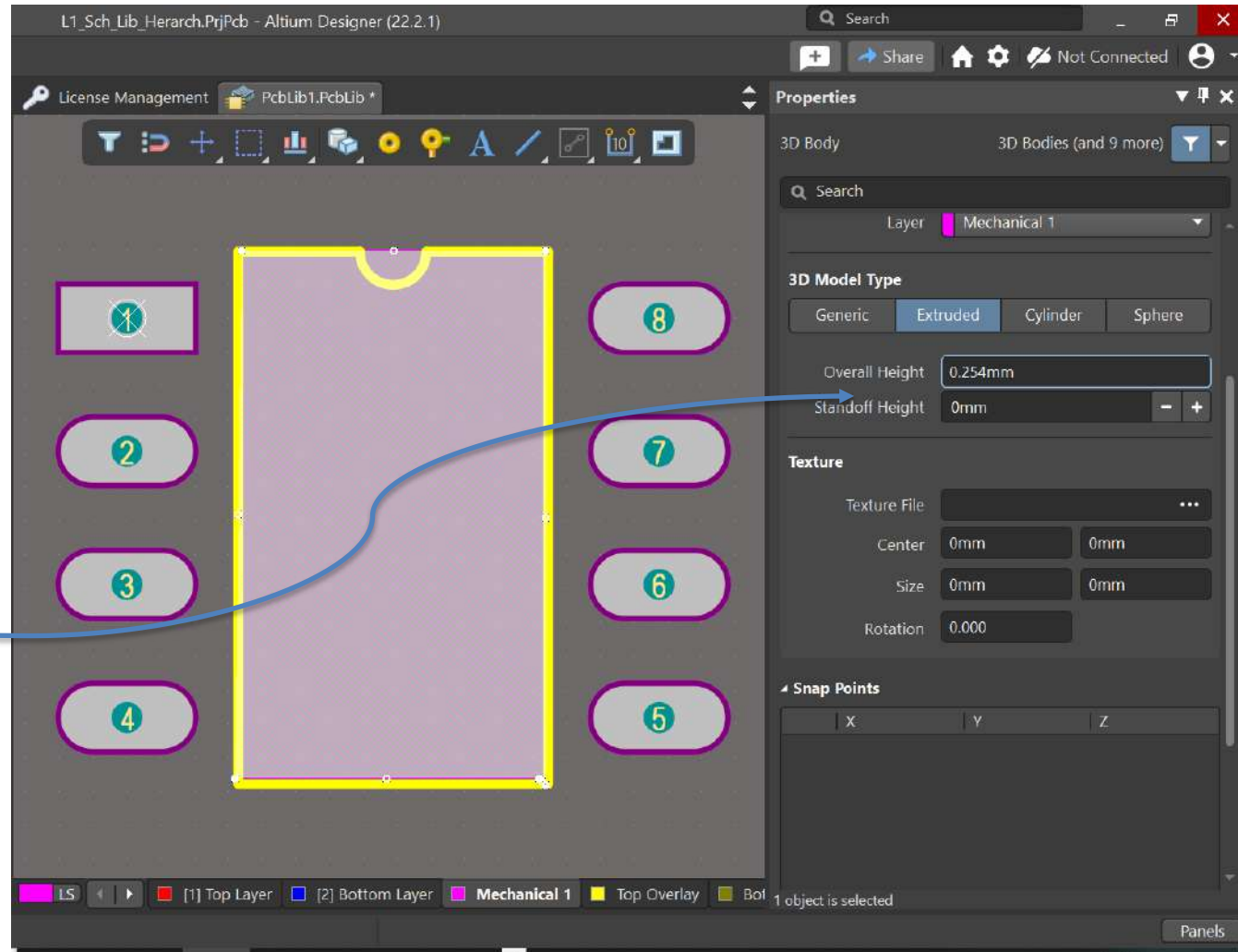
Creating a new Footprint

- This area is linked with the 3D model



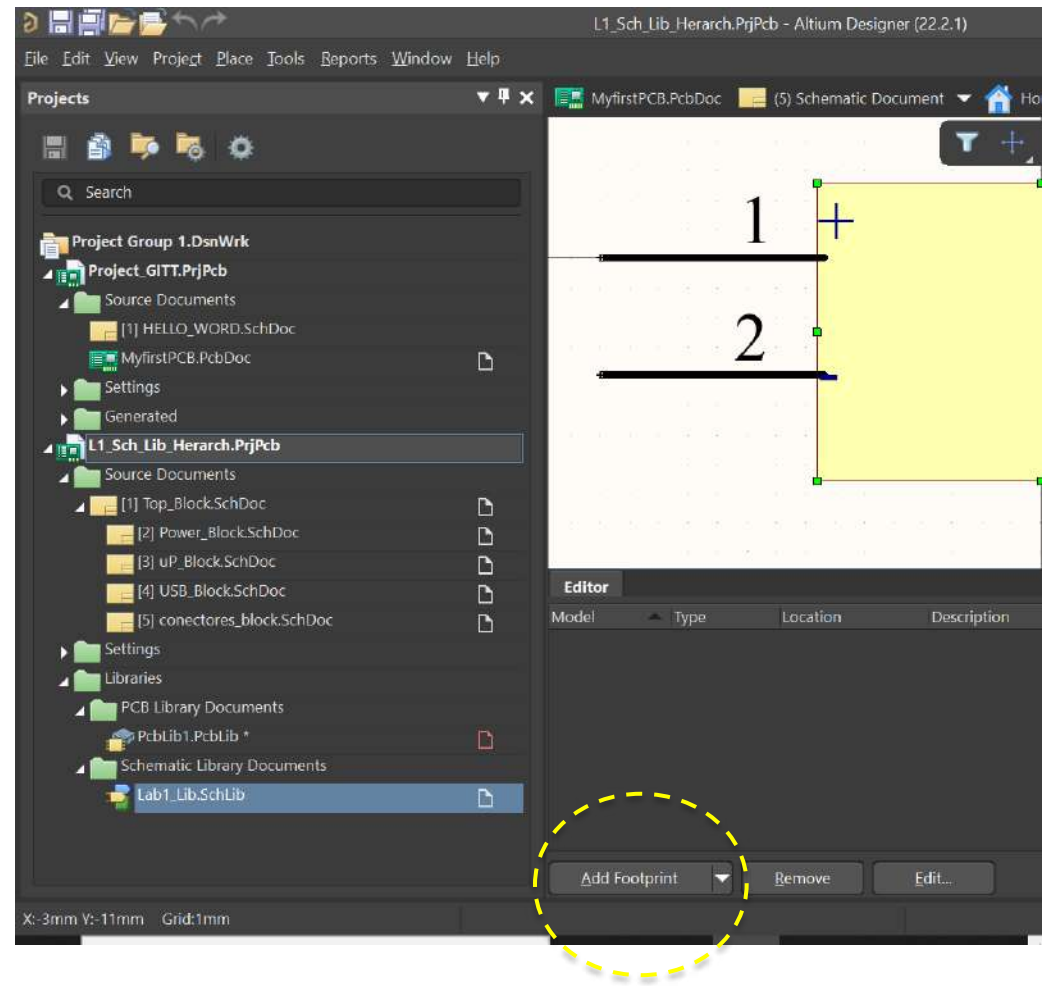
Creating a new Footprint

- We should include a height based on its datasheet



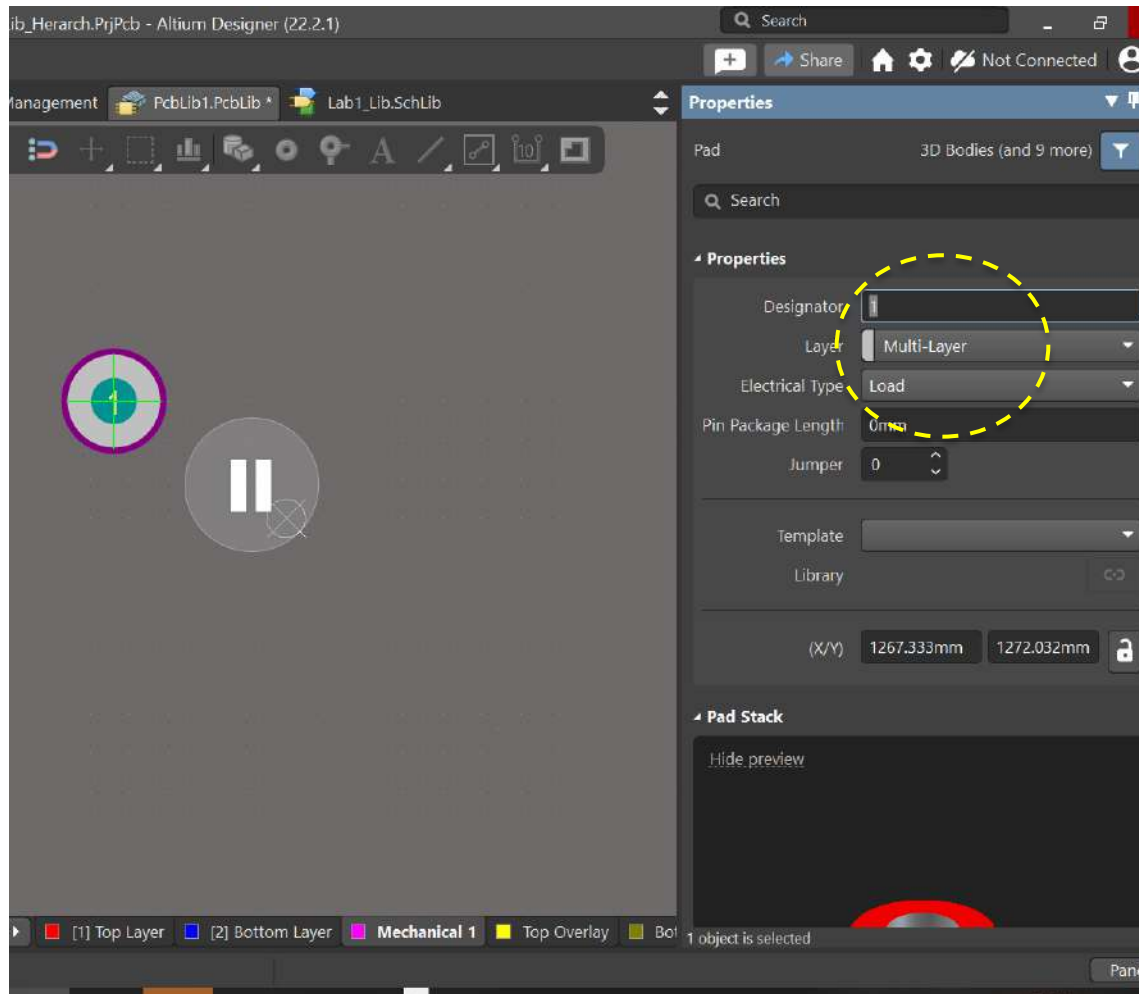
Linking the footprint with the symbol

- ❑ Footprint is ready to link with the symbol.
- ❑ We must connect the footprint with the symbol within SCHLIB.



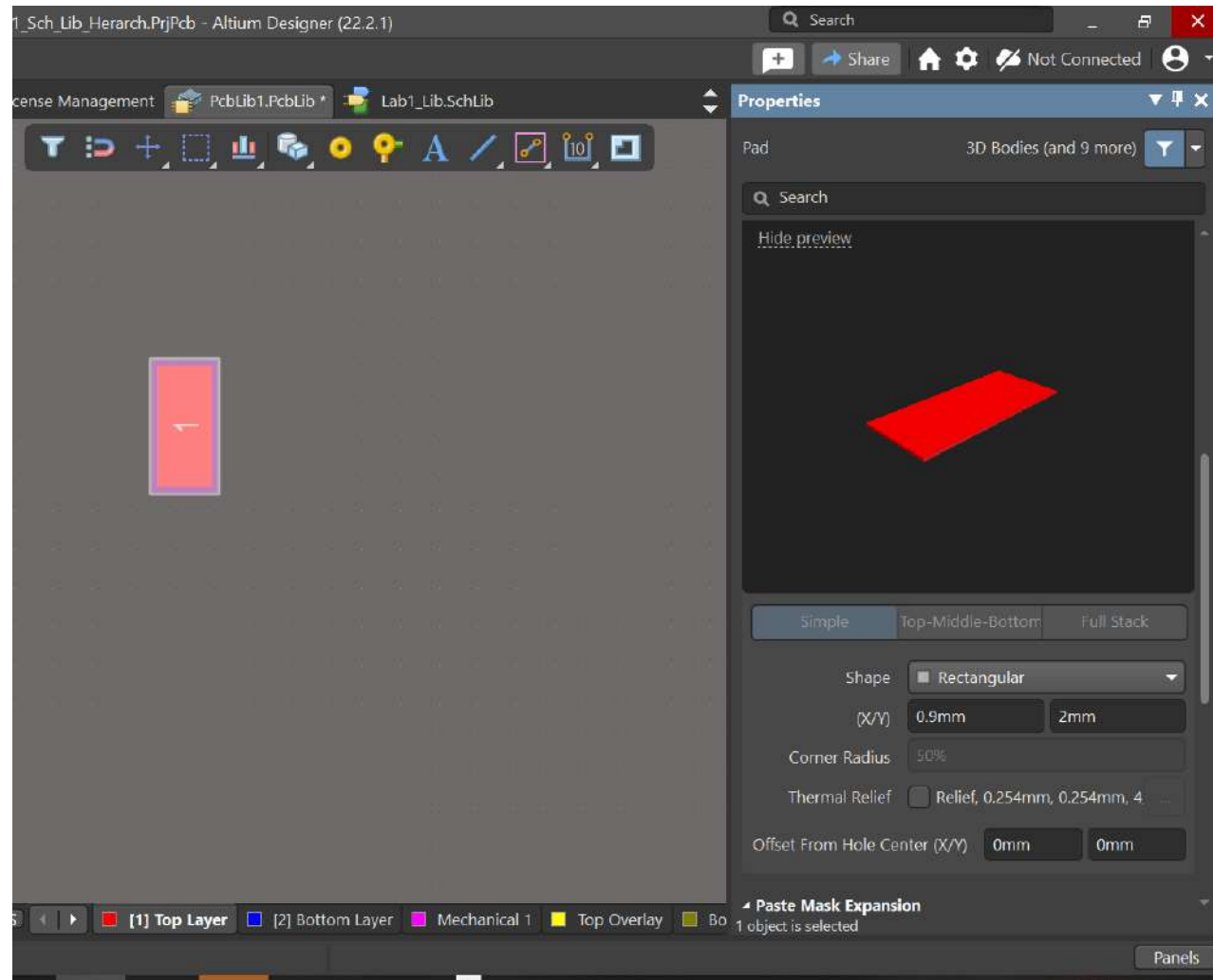
Doing a footprint without wizard

- ❑ We can create a footprint from zero without support.
- ❑ We use the Place menu to set the PADs
- ❑ Properties of the PAD allows to use through-hole or SMD types



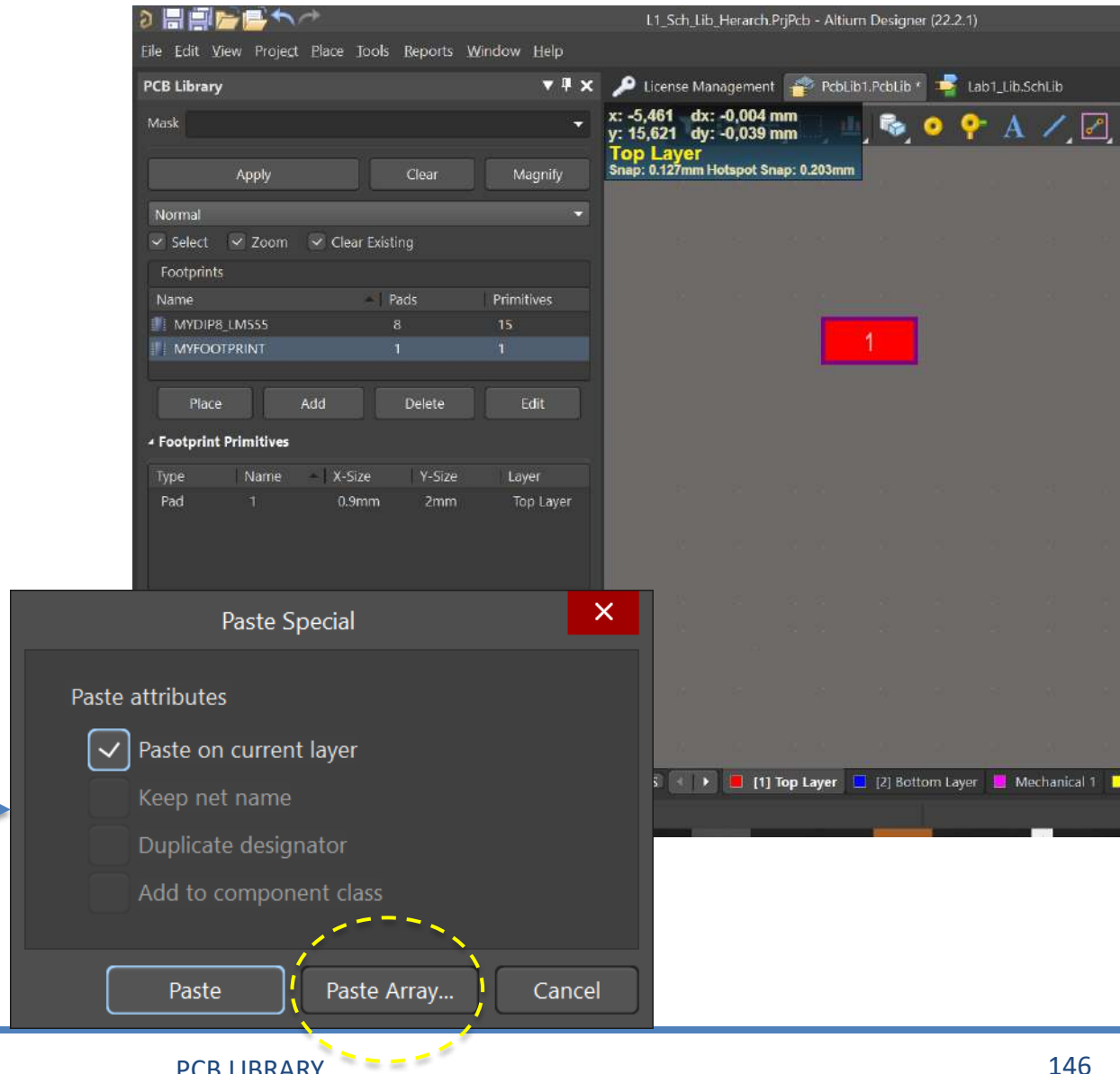
Doing a footprint without wizard

- Also, Properties of the PAD allows to define size and features of the PAD according to the datasheet



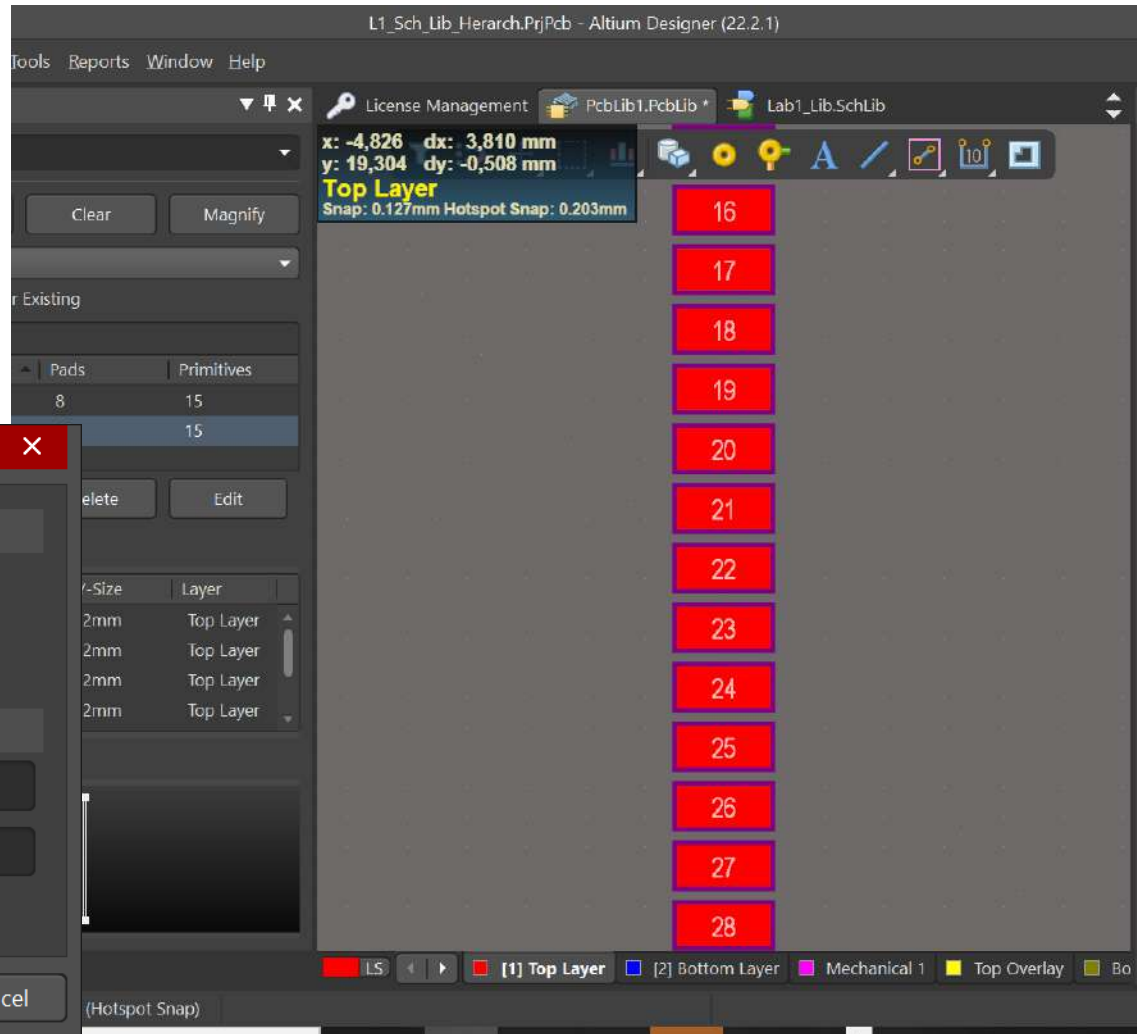
Doing a footprint without wizard

- ❑ If we want to create an array of pads:
- ❑ Edit->Copy. We match the green cursor in the PAD
- ❑ Then, Edit->Paste Special



Doing a footprint without wizard

- Here, we define the number of pads



Doing our footprint21

- ☐ You must create a footprint for a LM555 with DIP8 shape.
- ☐ Take the datasheets and create this footprint in a PCB Library

Doing our footprint2

- Trying to do the ESP32 footprint.
- You should take care with the lengths and distances.

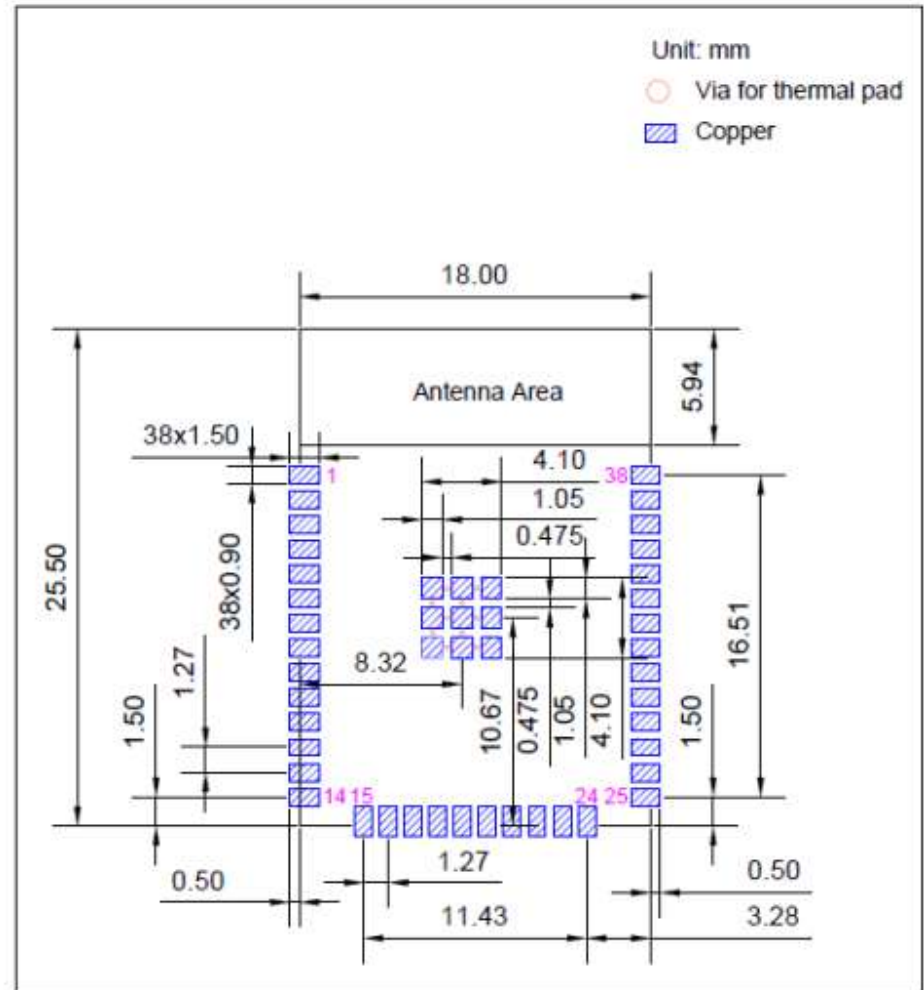
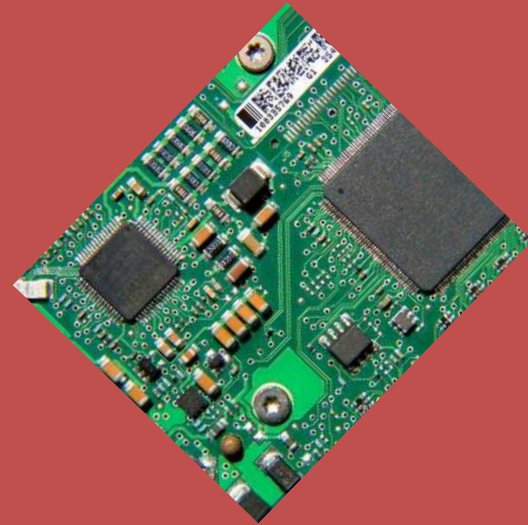


Figure 6: Recommended PCB Land Pattern

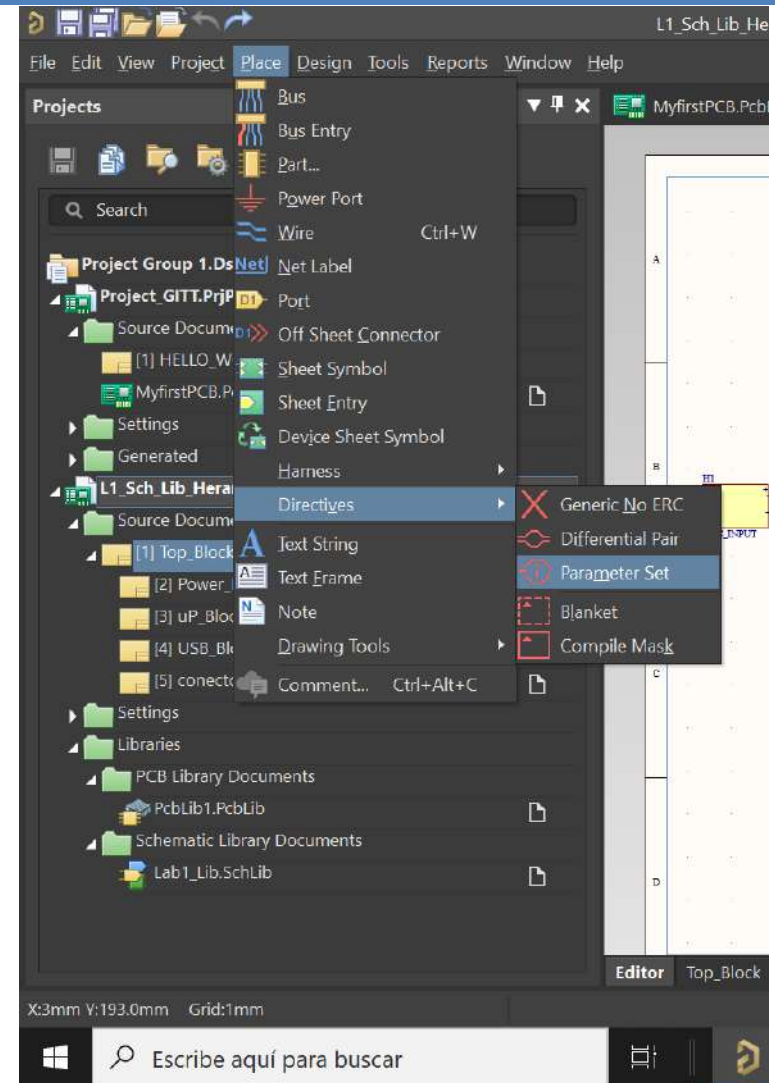
Introduction to Printed Circuit Board (PCB) design with ALTIUM

6.- PCB Advanced



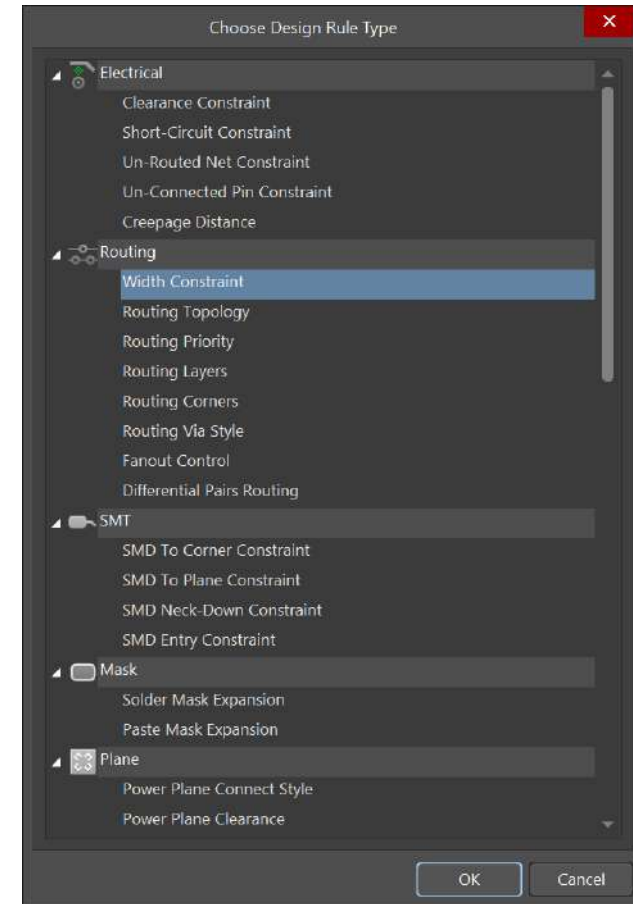
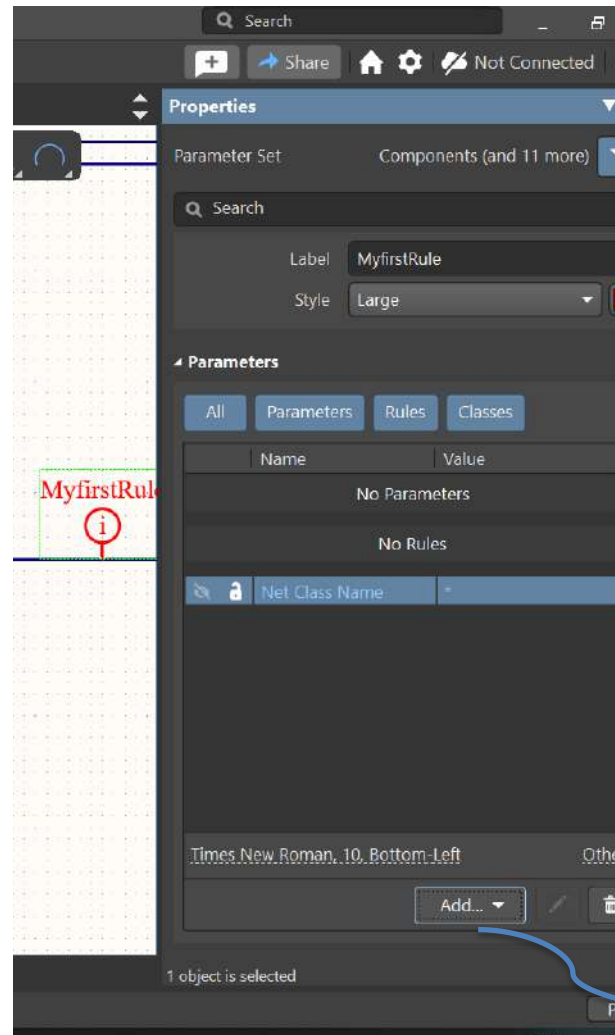
Rules

- ❑ The definition of rules for a set of nets can simplify our PCB design.
- ❑ It must be asserted in SCH Sheets.
- ❑ We will use the Place menu -> Directives-> Parameter Set



Rules

- ❑ We change the rule net.
- ❑ Also, we can create a class and we link to the class a Rule.
- ❑ We can assign the width of the PCB Track.



Electrical rules

- We can check the maximum current (SATURN PCB)

The screenshot shows the 'Conductor Properties' tab in the SATURN PCB Design, Inc. software. The interface is divided into several sections:

- Conductor Characteristics:** Includes 'Solve For' (Amperage selected), 'Plane Present?' (No selected), 'Conductor Width' (0,1 mm), 'Conductor Length' (25,4 mm), 'PCB Thickness' (1,5748 mm), 'Frequency' (1 MHz), and 'Parallel Conductors?' (No selected).
- Options:** Includes 'Base Copper Weight' (9um selected), 'Units' (Imperial selected), 'Substrate Options' (FR-4 STD selected), 'Plating Thickness' (Bare PCB selected), 'Plane Thickness' (35um selected), and 'Conductor Layer' (Internal Layer selected).
- Results Table:** A table with 3 columns and 4 rows showing calculated values for Skin Depth, Power Dissipation, Conductor DC Resistance, Skin Depth Percentage, Power Dissipation in dBm, Conductor Cross Section, Voltage Drop, and Conductor Current.

IPC-2152 with modifiers mode	Etch Factor: 1:1	
Skin Depth	Power Dissipation	Conductor DC Resistance
66.00620 um	0.06829 Watts	0.22172 Ohms
Skin Depth Percentage	Power Dissipation in dBm	Conductor Cross Section
100%	18.3433 dBm	0.0025 Sq.mm
	Voltage Drop	Conductor Current
	0.1230 Volts	0.5550 Amps

Autrouting

- ❑ **AutoRoute** is an automatic tool for routing.
- ❑ We can route the whole PCB, the rules, classes, etc.
- ❑ Is it magical? No it could be convenience when we are not able to find solutions in a specific area.

